

UNIVERSITÀ DI PISA
DEPARTMENT OF PHYSICS
MASTER'S DEGREE IN PHYSICS

Anomalous Transport in Soaring Flights

Collective Strategies and Intermittent Search Models

Candidate

Matteo Di Sante

Supervisors

Prof. Michael Benzaquen

Dr Alexandre Darmon

Internal tutor (Pisa)

Prof. Riccardo Mannella

Academic year 2025–2026

Working draft — last modified September 18, 2026.

Sources and code: <https://github.com/matteodisante/soaring-anomalous-transport>

Abstract

Soaring flight couples persistent horizontal motion to the search for atmospheric lift and the management of altitude. This thesis investigates which stochastic descriptions can account for that motion using paraglider and hang-glider records from the French free-flight federation. It combines empirical observables, probabilistic flight-phase segmentation and a programme of mathematical modelling and Monte Carlo validation.

This volume contains the introduction, the data chapter and an analysis of horizontal transport. The latter compares circuit and launch-altitude classes and tests displacement scaling with a common flight population, quantiles, moment spectra and kurtosis. Appendices 2.A and 2.B provide the spectral conventions and geodetic transformations used in Chapter 2.

Contents

1	Introduction	1
1.1	A flight is a transport process with an energy budget	1
1.2	The questions and the evidence needed to answer them	2
1.2.1	Which stochastic models survive the observations?	2
1.2.2	How do the flight phases generate global transport?	2
1.2.3	Does information from other pilots change the motion?	2
1.3	From empirical constraints to a physical model	3
1.4	Structure and current scope	3
1.5	Chapter summary	4
2	The dataset	5
2.1	Data acquisition	5
2.1.1	Source: the FFVL Coupe Fédérale de Distance	5
2.1.2	Collected data	5
2.1.3	Acquisition and verification	5
2.1.4	Further data sources	6
2.2	The IGC tracklog format	6
2.3	Flight metadata and the catalog	6
2.4	Glider categories	7
2.5	Coverage and statistics	7
2.6	Data quality and caveats	9
2.7	Trajectory pre-processing	9
2.7.1	Choice of altitude channel	11
2.7.2	Fix-level cleaning	13
2.7.3	Trimming of ground phases	23
2.7.4	Flight-level filtering	24
2.7.5	From geographic to local Cartesian coordinates	25
2.7.6	Uniform sampling within a flight	28
2.7.7	Smoothing and differentiation	31
2.7.8	Notation	35
2.8	Pipeline summary	35
2.9	Preliminary characterization	39
2.10	Trajectories for the transport analysis	45
3	Horizontal transport and displacement scaling	46
3.1	Mean-square displacement	46
3.1.1	Estimator and temporal resolution	46
3.1.2	General transport and uncertainty	47
3.1.3	Fixed population and the effect of duration selection	53
3.2	Transport by circuit type and initial altitude	55
3.3	Anisotropy and principal directions	57
3.4	Scaling of displacement distributions	57
3.4.1	Self-similarity and its observable consequences	57
3.4.2	A common empirical distribution	58
3.4.3	Quantiles, growth exponents and relative spread	58
3.4.4	Moment spectrum and finite-range scaling	61

3.4.5	Excess kurtosis of the signed components	62
3.4.6	Assessment of self-similarity	63
Appendices		68
2.A Power spectral density and Welch's method		69
2.A.1	From covariance to a spectrum	69
2.A.2	The estimator used here	69
2.B Geodetic transforms: from (φ, λ, h) to ECEF to ENU		71
2.B.1	Geodetic coordinates to ECEF	71
2.B.2	ECEF to local ENU	73
2.B.3	Size of the tangent-plane error	73
2.B.4	Size of the spherical-approximation error	74

Chapter 1

Introduction

1.1 A flight is a transport process with an energy budget

A soaring pilot travels by repeatedly exchanging height for distance and recovering height in rising air. The atmosphere supplies the energy, but the pilot must locate that supply and decide how to use it. A long displacement is therefore the outcome of several coupled processes: motion through the air, encounters with lift, decisions made with incomplete information, and constraints imposed by wind and terrain. Because the recorded trajectory contains all these contributions, inferring a stochastic description requires more than fitting its geometry: the description must also preserve the physical mechanisms that allow the pilot to remain airborne.

Two elementary relations explain why neither an unconstrained random walk nor a purely geometrical description is sufficient. Writing $\mathbf{r}$ for horizontal position and z for altitude, a local kinematic description is

$$\dot{\mathbf{r}}(t) = \mathbf{v}_{\text{air}}(t) + \mathbf{w}_h(\mathbf{r}(t), z(t), t), \quad (1.1)$$

$$\dot{z}(t) \simeq w_z(\mathbf{r}(t), z(t), t) - s(|\mathbf{v}_{\text{air}}(t)|, \phi(t)). \quad (1.2)$$

Here $\mathbf{w}_h$ and w_z are the horizontal and vertical air velocities, $s > 0$ is the sink rate relative to the air, and ϕ is bank angle. The second relation is a quasi-steady approximation: it makes explicit the competition between lift and aerodynamic sink without attempting to model every manoeuvre. Altitude is therefore a finite resource, and a transport model must account for its use and replenishment if its trajectories are to represent sustained flight.

This physical picture suggests three descriptions of behaviour: directed *transition*, exploratory *search*, and *climb* in lift. They are hypotheses about the organisation of motion, rather than labels directly supplied by a recorder. A pilot can find lift during a transition, climb without completing a circle, or search while losing height. Determining which distinctions the data support is part of the scientific problem.

Soaring control has been studied through the problem of locating and exploiting thermals, including adaptive strategies based on local information [1]. The present work asks a complementary question: what statistical laws emerge over entire flights, and which mechanisms can account for those laws? A direct empirical precedent is the analysis of soaring transport and flight phases by Vilpellet et al. [2]. This thesis revisits that connection using the FFVL archive, explicit data-quality controls, and a separation between measured properties and assumptions used to interpret them.

1.2 The questions and the evidence needed to answer them

1.2.1 Which stochastic models survive the observations?

A first observable is the squared displacement over an interval τ ,

$$M_2(\tau) = \left\langle |\mathbf{r}(t + \tau) - \mathbf{r}(t)|^2 \right\rangle. \quad (1.3)$$

Ordinary diffusion without drift has $M_2 \propto \tau$; directed motion can produce a ballistic law $M_2 \propto \tau^2$. An intermediate slope is informative, but it does not identify a mechanism, since correlated Gaussian motion, persistent motion observed during a crossover, and mixtures of flight behaviours can produce similar second moments. Random-walk frameworks offer competing explanations of anomalous transport [3]. The purpose here is to discriminate between such explanations using more than one curve.

The empirical programme therefore calls for measurements of the MSD, displacement quantiles and directional anisotropy, and how each depends on route, duration and equipment. Quantile ratios test whether the distribution changes shape as the lag increases. Open and closed tasks help expose the effect of the flight's intended geometry. These measurements, together with the spectrum of moments, the shape of the increment distribution and velocity correlations, must be compared with explicit model predictions: higher moments compare the growth of ordinary and large displacements, while velocity correlations resolve persistence in time. A model is challenged where its predictions disagree with these observations under a comparable measurement protocol.

This logic also sets the limit of an exclusion. A discrepancy with a homogeneous Gaussian model for pooled flights need not exclude Gaussian fluctuations within a flight: different conditions can create a non-Gaussian mixture. Similarly, evidence against a particular constant-speed renewal Lévy walk is not a theorem against every process with finite-speed legs [4]. Each comparison must specify the hypotheses being tested and leave unsupported exclusions open.

1.2.2 How do the flight phases generate global transport?

Although an unsegmented flight supplies constraints on the whole process, it does not identify which behaviour generates a long displacement, a change of heading or a period of persistence. A probabilistic segmentation into transition, search and climb can address this missing link. It requires an explicit mathematical model, readable trajectory examples, and a distinction between internal consistency and accuracy against independent annotations.

The next step will be a chapter devoted to observables conditioned on phase: episode durations, distance and height changes, speed and turning distributions, displacement scaling, and dependencies between successive episodes. A useful model must reproduce both these conditional statistics and the global ones, a stronger requirement than generating a visually plausible sequence of coloured flight segments. Windows that cross phase boundaries must also be retained when reconstructing global observables.

1.2.3 Does information from other pilots change the motion?

A nearby pilot can reveal lift before a follower reaches it. This creates a possible information channel that is absent from a model of independent searchers. A later chapter will compare solitary and group flight using both global and phase-conditioned observables. The operational definition of a group will concern simultaneous proximity and plausible visibility, not membership of a competition. A competition can contain isolated pilots, and pilots outside a competition can travel together.

The comparison will ask where differences occur: in time spent searching, climb selection, transition direction, episode persistence, or the efficiency of converting altitude into distance.

Site, weather, wing and pilot characteristics must be controlled because they can produce the same differences without social interaction. Incomplete recording of nearby pilots will limit what can be called solitary flight. These are conditions for an interpretable comparison, not effects assumed in advance.

1.3 From empirical constraints to a physical model

The thesis combines empirical analysis, mathematical modelling and simulation, with a distinct role for each. Data analysis establishes the constraints that a candidate process must satisfy; mathematics then makes that process precise and derives its predictions, which Monte Carlo simulation tests under finite flight durations, heterogeneous environments and the actual observation protocol.

Model construction will follow the global analysis and the analysis conditioned on phase. Possible ingredients include finite-speed transitions, lift encounters, distributions of episode duration, and dependencies between successive directions or episodes. Renewal independence is one hypothesis to test; correlated random walks provide alternatives when that independence fails [5, 6]. The choice among these ingredients must be justified by the observations and an explicit physical mechanism, rather than fixed by the outline of the thesis. Parameters fitted on one set of flights will be checked on separate flights or days, using several observables jointly.

A reference model without wind and terrain may help isolate mechanisms, but obtaining it requires an inference about the environment: mean ground velocity alone does not determine wind, and rotating a trajectory into its principal axes does not remove orography. Artificially making a covariance matrix isotropic changes the measured object; it does not recreate a flight in still air. The analysis consequently preserves the motion in geographical coordinates and studies environmental strata before asking which properties could survive in a common reference model.

A further part of the programme concerns the spatial organisation of thermals and route choice. It will first estimate the two-dimensional distribution of *visited* lift regions, accounting for where pilots have searched. Recovering the distribution of available thermals requires an observation model: unvisited terrain is not evidence of absent lift. A mathematical description of thermal availability and strength could then support separate optimisation problems, such as minimum-time travel between two locations, a closed route, or maximum distance within a specified flight budget. Wind and, subsequently, possible geometrical constraints from terrain will be introduced in stages. A spatial density alone cannot determine an optimal path without assumptions about thermal strength, spatial dependence, altitude and the pilot's information.

1.4 Structure and current scope

Figure 1.1 shows how the measurements support the later modelling. The present document contains the introduction, the data chapter and the analysis of horizontal transport and displacement scaling, together with the two appendices cited in Chapter 2. Flight-phase segmentation, conditional observables, social flight, stochastic modelling and route optimisation form the subsequent research programme.

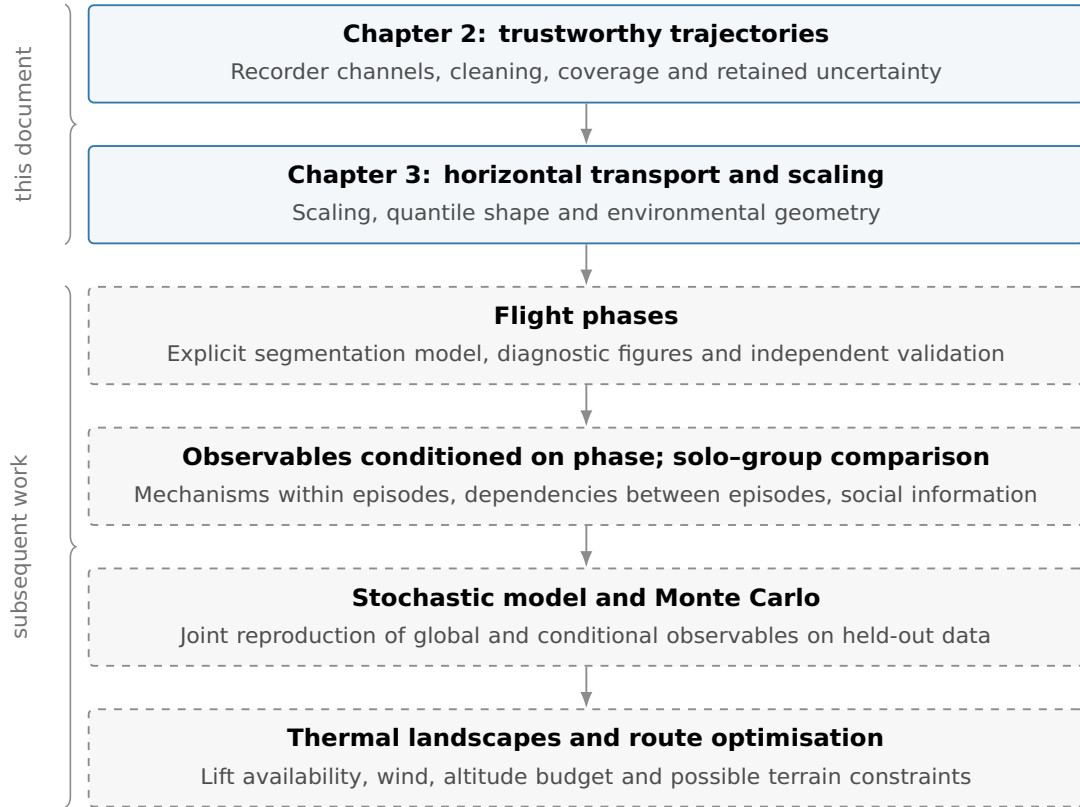


Figure 1.1. Logical structure of the thesis. The solid boxes are developed in this document; dashed boxes describe subsequent work. Model comparison must return to the measured observables, not stop at a fitted trajectory or a single exponent.

Chapter 2 follows the records from acquisition to retained trajectories, including the treatment of missing or inconsistent channels. The two appendices state the spectral conventions and the geodetic transformation used in that chapter. Chapter 3 measures horizontal transport and distributional scaling with fixed flight populations, and resolves circuit and launch-altitude differences.

1.5 Chapter summary

The thesis asks which stochastic description can account for soaring trajectories while respecting the physical roles of lift, travel and environmental constraints. The route from observations to a model begins with reliable trajectories and global transport statistics, then uses validated flight phases to identify the mechanisms mixed by those statistics. Comparisons between solitary and interacting flights, followed by simulation and thermal-landscape modelling, extend this programme. Their conclusions remain future work: the present document establishes the cleaned trajectories and the global transport statistics that those comparisons must explain.

Chapter 2

The dataset

The FFVL archive provides flight metadata and recorded trajectories for paragliders and hang gliders. We describe its coverage, then the cleaning, coordinate conversion and resampling that produce the trajectories used in the transport analysis.

2.1 Data acquisition

2.1.1 Source: the FFVL Coupe Fédérale de Distance

The *Coupe Fédérale de Distance* (CFD) is the cross-country competition of the French free-flight federation (FFVL). Its two archives separate paragliders (`parapente.ffvl.fr`) from hang gliders (`delta.ffvl.fr`). Each entry contains flight metadata and, usually, an IGC trajectory.

Seasons are named by their starting year: 1999 means 1999–2000. The paraglider archive begins in that season; the hang-glider archive begins in 2001–2002.

2.1.2 Collected data

For every season the pipeline retrieves two things:

- the **season XML export**, archived verbatim as provenance (`raw_xml/<year>.xml`); it is the authoritative list of flights and links;
- the **IGC tracklog files** referenced by the XML, one file per flight that has a track.

From these, a derived **catalog** is built (`catalog.csv`, one row per flight, its columns detailed in Sec. 2.3) together with a per-season index (`seasons_index.csv`). The raw data (tens of gigabytes) live on an external disk and are never committed to the repository. The catalog is itself rebuildable from the archived XMLs and likewise uncommitted (Sec. 2.3).

2.1.3 Acquisition and verification

Two command-line tools, one per discipline, share the same acquisition code. Downloads are resumable and rate-limited; each response must contain a readable IGC file with an A header and at least one B fix record (Sec. 2.2). The archived XMLs preserve the source lists, and the output location is configurable.

An independent check of the files on disk verified all 186,052 paraglider and 6,716 hang-glider tracklogs against the same content requirements. Section 2.5 compares these downloads with the full list of declared flights.

2.1.4 Further data sources

The present analysis uses the FFVL archive. WeGlide could add sailplanes, and XContest could broaden paraglider and hang-glider coverage. Both would feed the same catalog format.

Research-access requests were pending at both portals on 20 July 2026. At that date, WeGlide’s default quota of roughly sixty API requests per day precluded bulk acquisition. These additional sources are not included here.

2.2 The IGC tracklog format

An *IGC file* is a plain-text flight record with fixed-position fields, defined by the FAI/IGC specification [7]. The first character identifies the record type:

Record	Content
A	Recorder identifier.
H	Headers: date, pilot, glider, datum, and related metadata.
I	Declaration of optional extension fields.
B	UTC time, WGS84 latitude and longitude, GNSS validity flag, pressure altitude and GNSS altitude (both in metres).

The B-record flag describes the GNSS solution: **A** means a three-dimensional fix; **V** means a two-dimensional fix or no GNSS data. We invalidate GNSS altitude at **V** fixes and retain their horizontal coordinates provisionally, pending the checks in Sec. 2.7.2. A pressure value alone does not establish a working barometer.

The parser rejects malformed records: wrong length, illegal times, coordinates or hemisphere letters. Physically implausible but legally encoded fixes pass to cleaning. The resulting sequence of fixes has a device-dependent cadence, typically a few seconds, which need not be uniform.

2.3 Flight metadata and the catalog

Every flight in the season XML carries a set of metadata as record attributes, which the acquisition preserves. Beyond the identifiers and links, the available fields are collected in Table 2.1.

Table 2.1. Flight metadata preserved from the season XML (raw attribute names), beyond the identifiers and links.

Field(s)	Content / caveat
date	flight date
pilot, club	pilot and club
take-off, landing, department	the two sites and the French department
aile	wing model name (raw XML field; French for “wing”)
aile_class	wing class, one system per discipline (Sec. 2.4)
flight type	the <i>scoring</i> category: free distance, out-and-return, flat or FAI triangle, hike-and-fly, ... (<i>caveat</i> : a scoring label, not a glider or manoeuvre type)
distance, points	scored distance and points
duration, speed	duration and average speed, where present

The derived **catalog** (`catalog.csv`) has one row per flight: the metadata above, a `local_path` and a `downloaded` flag. The **season index** (`seasons_index.csv`) counts declared flights, track links and downloads. Both can be rebuilt from the archived XMLs.

2.4 Glider categories

Aircraft design changes speed, glide performance and turning geometry, motivating separate transport analyses [2]. Paragliders have a flexible canopy, hang gliders a framed flexible or rigid wing, and sailplanes a rigid airframe. The FFVL source identifies the first two families; sailplanes are absent from this dataset.

Within each family, `ail_e_class` defines the equipment groups in Table 2.2. Paraglider EN and legacy LTF codes, such as “C ou 2”, are harmonized as EN A–D; CCC and tandem/non-certified entries remain distinct. Blank and placeholder entries stay unclassified.

EN labels describe safety-test responses and recovery demands, rather than measured speed or glide ratio [8]. The descriptions here use the consulted 2012 final draft; the 2013 standard and later amendments were not compared. FAI hang-glider classes distinguish design and control systems [9].

Two operational equipment proxies are defined:

beginners : EN A/B, experts : EN C/D/CCC.

Neither measures individual experience or assumes equally spaced performance. The harmonization does not reassess certificates, and a tandem label alone does not establish EN certification.

2.5 Coverage and statistics

The statistics below describe the FFVL CFD dataset currently in hand; further sources may be added later (Sec. 2.1). It comprises 212,602 flights in total, of which 192,971 carry a GPS tracklog and 192,768 have been downloaded and verified, split across two disciplines.

The *paraglider* collection spans 27 seasons, from 1999–2000 to 2025–2026: 203,343 flights, of which 186,225 (91.6 %) carry a tracklog and 186,052 (99.9 %) are downloaded and verified (Table 2.3). The *hang-glider* collection spans 25 seasons, from 2001–2002 to 2025–2026: 9,259 flights, of which 6,746 (72.9 %) carry a tracklog and 6,716 (99.6 %) are downloaded and verified (Table 2.4). The hang-glider archive is roughly twenty times smaller, reflecting the smaller active population of the discipline.

In both disciplines the number of flights carrying a track grows by orders of magnitude over the period, reflecting the adoption of GPS loggers and online scoring: the early seasons contribute a handful of tracks each, while recent seasons contribute thousands of tracks for paragliders and several hundred for hang gliders.

Table 2.3. FFVL CFD *paraglider* flights per season (`parapente.ffvl.fr`): total declared, with a GPS track, and downloaded.

Season	Flights	With GPS	Downloaded	Coverage (%)
1999–2000	10	10	10	100.0
2000–2001	832	1	1	100.0

Season	Flights	With GPS	Downloaded	Coverage (%)
2001–2002	1,150	3	3	100.0
2002–2003	1,996	123	123	100.0
2003–2004	1,470	337	337	100.0
2004–2005	1,765	574	574	100.0
2005–2006	2,162	894	894	100.0
2006–2007	2,386	1,016	1,016	100.0
2007–2008	3,081	1,667	1,667	100.0
2008–2009	3,974	2,614	2,613	100.0
2009–2010	4,171	3,059	3,059	100.0
2010–2011	4,872	4,185	4,169	99.6
2011–2012	6,744	6,075	6,065	99.8
2012–2013	7,873	7,106	7,075	99.6
2013–2014	9,804	9,157	9,127	99.7
2014–2015	13,793	13,293	13,254	99.7
2015–2016	13,982	13,703	13,696	99.9
2016–2017	15,760	15,534	15,528	100.0
2017–2018	14,917	14,682	14,676	100.0
2018–2019	16,702	16,517	16,508	99.9
2019–2020	10,966	10,868	10,863	100.0
2020–2021	9,500	9,483	9,480	100.0
2021–2022	14,552	14,507	14,502	100.0
2022–2023	11,590	11,561	11,560	100.0
2023–2024	10,372	10,362	10,360	100.0
2024–2025	10,906	10,885	10,884	100.0
2025–2026	8,013	8,009	8,008	100.0
Total	203,343	186,225	186,052	99.9

Table 2.4. FFVL CFD *hang-glider* flights per season (`delta.ffvl.fr`): total declared, with a GPS track, and downloaded.

Season	Flights	With GPS	Downloaded	Coverage (%)
2001–2002	248	0	0	0.0
2002–2003	281	6	6	100.0
2003–2004	256	28	25	89.3
2004–2005	244	35	25	71.4
2005–2006	365	80	76	95.0
2006–2007	337	100	96	96.0
2007–2008	388	136	132	97.1
2008–2009	344	155	155	100.0
2009–2010	324	136	136	100.0
2010–2011	425	147	147	100.0
2011–2012	117	103	103	100.0
2012–2013	428	406	406	100.0
2013–2014	423	409	409	100.0
2014–2015	628	595	591	99.3
2015–2016	417	413	413	100.0
2016–2017	468	462	462	100.0

Season	Flights	With GPS	Downloaded	Coverage (%)
2017–2018	342	331	331	100.0
2018–2019	604	594	594	100.0
2019–2020	378	373	373	100.0
2020–2021	328	328	328	100.0
2021–2022	570	568	568	100.0
2022–2023	376	374	374	100.0
2023–2024	437	437	436	99.8
2024–2025	353	352	352	100.0
2025–2026	178	178	178	100.0
Total	9,259	6,746	6,716	99.6

2.6 Data quality and caveats

A few limitations are already known and will shape the analysis:

- **Flights without a track.** About 8 % of paraglider entries and about 27 % of hang-glider entries have no IGC file (older or manually declared flights); they carry metadata only. These are the flights outside the “With GPS” column of Tables 2.3 and 2.4.
- **A small residue of unrecoverable files.** A few flights advertise a track whose download fails or returns something that is not valid IGC content (broken or placeholder links); these are recorded as failures and excluded.
- **Heterogeneous logging.** Sampling rate, availability of the two altitude channels, and per-fix positioning noise depend on the device; trajectories must be resampled and quality-filtered before any ensemble statistic is computed.
- **Altitude channel.** A sizeable minority of loggers record no pressure altitude – almost a third of paraglider and roughly a sixth of hang-glider flights. GNSS altitude is used for every admitted flight, with the channel recorded per flight (Sec. 2.7.1).
- **Glider metadata.** Flights are paragliders or hang gliders, tagged by source (Section 2.4); the class field `aile_class` must be normalised before it is used to stratify.
- **Incomplete dates.** Some historical entries carry the placeholder date 0000-00-00, written as a sentinel at acquisition. Such entries cannot support a calendar-conditioned comparison until a date is recovered independently; the relative-time preprocessing does not itself enforce this metadata exclusion. The count is small: 8 of 203,343 paraglider entries carry the sentinel (0.0039 %), 0 of 9,259 hang-glider ones, and the largest within-season fraction is only 0.17 % of its entries (2001-2002).

These caveats motivate the pre-processing described in the remainder of this chapter.

2.7 Trajectory pre-processing

Each IGC file is parsed into a sequence of fixes $(t_i, \varphi_i, \lambda_i, h_i)$ – UTC time, geodetic latitude and longitude, and altitude. The pipeline then proceeds in a fixed order:

- (i) choose the altitude channel (Sec. 2.7.1);

Table 2.2. Class labels recorded in `aile_class`, one system per discipline. EN labels summarize test-based recovery characteristics; FAI classes distinguish control and design. They are used as equipment groups, without assigning pilot ability. Sources: the FAI Sporting Code, Common Section 7 [9] (§1.4.1) for the FAI classes and the sub-class ceilings; the EN class descriptions are condensed from Table 1 of the consulted 2012 final draft [8].

Class	Notes
Paragliders — harmonized EN/LTF catalogue groups	
EN A	maximum passive safety, extremely forgiving; any level of training
EN B	good passive safety, forgiving; ceiling of the FAI <i>Standard</i> sub-class
EN C	moderate passive safety, dynamic reactions; recovery needs precise input; ceiling of the FAI <i>Sport</i> sub-class
EN D	demanding, potentially violent reactions; for pilots practised in recovery
CCC	CIVL Competition Class racing wings
tandem / non-certified	two-seater and uncertified wings, pooled in this field
Hang gliders — FAI Class O (paragliders are Class 3 of the same scheme)	
Delta Class 1	weight-shift as the sole method of control
Delta Class Sport	sub-class of Class 1: type-certified production wing with a king post
Rigide Class 2	movable aerodynamic surfaces as the primary control
Rigide Class 5	as Class 2, but in the roll axis only; no fairings

- (ii) clean individual fixes (Sec. 2.7.2);
- (iii) trim the ground phases (Sec. 2.7.3);
- (iv) filter out unfit whole flights (Sec. 2.7.4);
- (v) convert to a local Cartesian frame (Sec. 2.7.5);
- (vi) enforce a uniform sampling interval within each flight (Sec. 2.7.6);
- (vii) smooth and differentiate the trajectory (Sec. 2.7.7).

The output consists of fix, segment and attempted-flight tables, accompanied by a table of suspect intervals retained for later sensitivity analysis. What each file holds and how the disk is organised is set out in Sec. 2.8.

Steps (i)–(iv) use raw geographic coordinates: great-circle distances give horizontal speeds, and altitude is read as logged. Step (v) places the cleaned, trimmed trajectory in a local East–North–Up frame, anchored at its first surviving fix. Resampling and differentiation then act on Cartesian components.

Table 2.8 collects the working thresholds. They are specified independently of the transport exponents; Table 2.7 records their effect on retention.

Status of the numerical snapshot. The census, transport measurements and phase analysis use the archive regenerated with pipeline version 2.3.0. It applies the local paired-barometer requirement, the 10 ms^{-1} vertical-speed threshold and the propagation of reconstructed altitude through derivative-estimation windows. Both inter-fix gaps and intervals between finite altitudes obey the same bridge-or-split bound; longer altitude gaps split the complete trajectory. The checks establish agreement with these numerical rules, not the absence of

every recorder error. The witness audit below concerns the isolated version-2.0.1 correction. The targeted comparison reran the legacy and corrected rules on 2,388 candidates: 189 flights changed. Frozen-run counts changed in 188 cases; interior-ground counts changed in 2 cases (the categories can overlap). The corrected witness retained 22,468 additional horizontal fixes; 0 flight admission decisions changed. These counts describe the targeted set, not detector error rates. They do not replace regeneration of the archive and its observables.

2.7.1 Choice of altitude channel

The analysis uses *logged GNSS altitude* throughout: $z = h_{\text{GNSS}}$. This keeps one channel across flights but leaves the recorder’s vertical datum unresolved:

Recorder specification	GNSS height convention
IGC-approved	Ellipsoidal height [7].
CIVL hang-/paragliding	Geoid-referenced height in the consulted document, which contains draft annotations [10].

Compliance has not been established across this historical archive, and the parser preserves the numerical field. Absolute-height comparisons therefore need a datum audit.

A constant height offset cancels from increments; spatially varying geoid corrections and changing sensor biases do not. Pressure altitude also depends on atmospheric pressure and temperature. Using GNSS consistently does not make it uniformly more accurate.

Comparing the barometric and GNSS channels. From 3,000 sampled raw flights per discipline, we retain those with at least 95% pressure-altitude presence and median cadence within 0.25 s of the pooled mode (median intervals rounded to seconds).

Each pair of spectra uses the same longest block of finite, nonzero readings with valid GNSS flags. Duplicate/backward timestamps and gaps above 1.5 modal intervals break the block; ties in fix count select the earliest block. Linear resampling at the modal interval requires at least 256 raw and 256 grid fixes.

Figure 2.1 shows a time-varying channel difference: a constant offset does not align the 30 min traces. The high-frequency median spectra nearly coincide, with broad variation between flights. These comparisons exclude missing-channel runs and do not represent every sampled recorder. Appendix 2.A defines the spectral estimator.

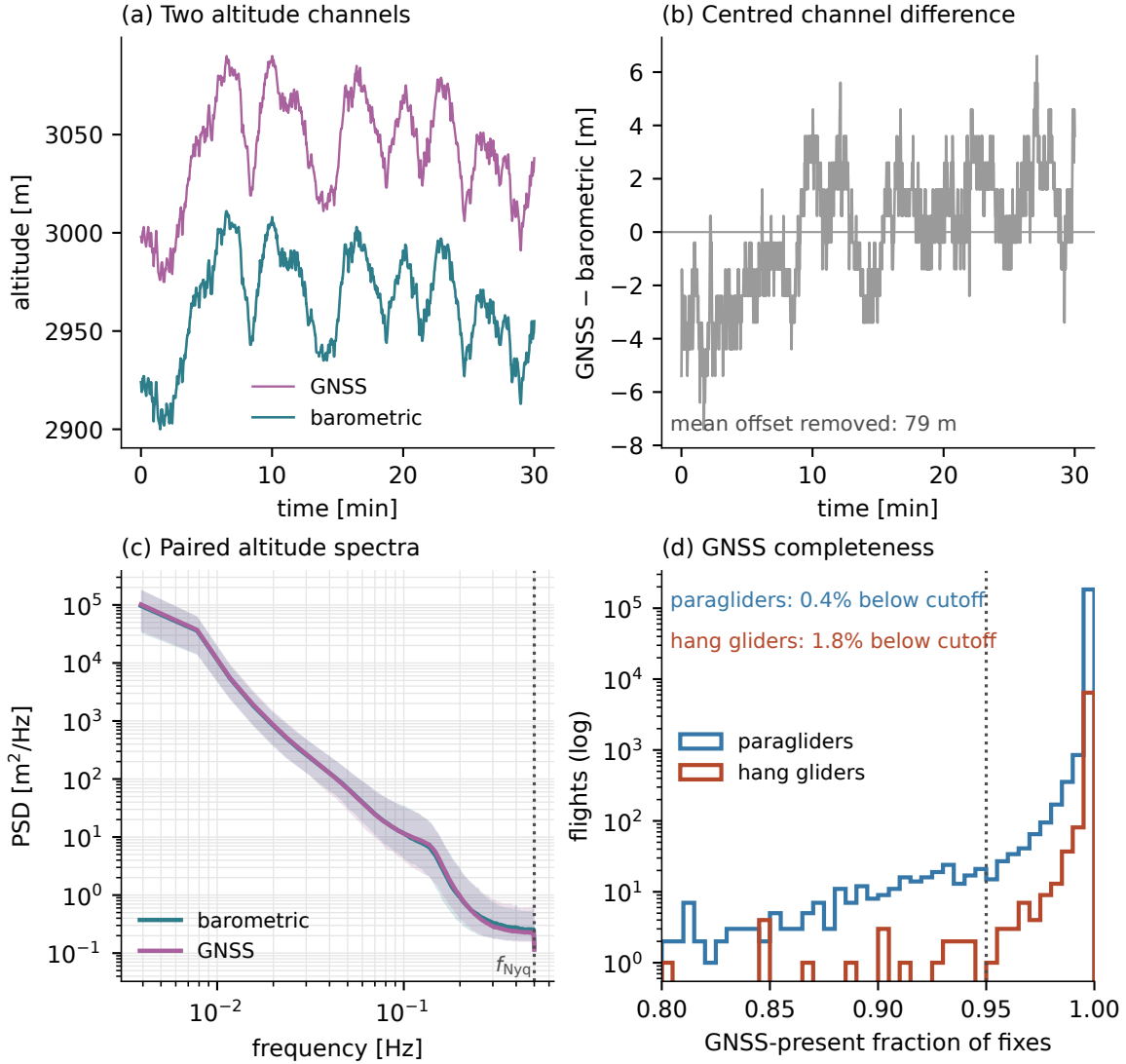


Figure 2.1. Barometric and GNSS altitude. (a) Paired raw channels over 30 min; (b) their demeaned difference. (c) Per-frequency median Welch PSD with the 10–90% range across paired flights, pooled over both disciplines (Appendix 2.A). The band describes flight variability; the dotted line marks $f_{\text{Nyq}} = 1/(2\Delta t)$. (d) GNSS-presence fractions by discipline, shown over 0.80–1.00 with logarithmic counts. The dotted line is the 95% admission threshold.

Variation between flights. Similar medians can conceal different upper tails. We therefore measure each flight’s median PSD in the 0.35 Hz to 0.50 Hz band and compare it with the median of these values across barometric records. Of the 2,008 paired flights contributing to the spectrum, 6.5% carry a GNSS floor above three times the barometric channel’s typical one, and 2.1% above ten times it. Multipath is one possible GNSS error source [11]; this comparison does not attribute the measured excess to a unique physical mechanism.

Sensitivity to the observation interval. A same-flight comparison separates the selection of valid intervals from subsequent trajectory smoothing. Using the same decoder and Welch estimator, we compare a whole-record rule, which can retain missing-value codes and interpolate across recording gaps, with the paired valid-block rule above. On the same 2,008 flights, the 90th percentile of the per-flight GNSS spectral floor decreases from 3.758 to 0.590 m² Hz^{−1}; the corresponding barometric values change from 0.545 to 0.508 m² Hz^{−1}. Here the floor is

each flight’s median PSD over 0.35 Hz to 0.50 Hz; its percentile is distinct from a per-frequency percentile in Fig. 2.1.

Neither comparison uses the pipeline’s Savitzky–Golay output: both use logged altitudes, linear resampling and Welch detrending. The smaller GNSS upper tail therefore reflects the changed observation interval in this comparison. It does not demonstrate lower physical sensor noise. The experiment changes the combined validity, missing-value and gap-selection rule, without identifying the contribution of each criterion separately. These spectra remain conditional on recorders and intervals with adequate paired-channel support.

Availability. A flight needs finite, nonzero GNSS altitude at 95 % of its fixes and an altitude range of at least 30 m. These criteria reject incomplete or nearly constant channels, including genuine flights with too little vertical excursion. No pressure altitude is substituted.

The presence criterion excludes 0.4 % of paraglider and 1.8 % of hang-glider flights. Both channels fall below their presence thresholds in 0.14 % and 1.13 %, respectively. Within admitted flights, missing GNSS altitude initially leaves the horizontal fix intact. At resampling, short altitude holes are reconstructed and flagged; long holes split the full trajectory (Sec. 2.7.6). Missing readings affect 6.1 % of paraglider and 10 % of hang-glider flights, with a median of 5 missing altitudes among affected paraglider flights.

Local barometric support. The pressure channel has a separate role: it can corroborate a suspected frozen GNSS position and an interior ground interval (Secs. 2.7.2 and 2.7.3). Its flight-level eligibility requires at least 95 % nonzero finite readings and a range of at least 30 m. The corresponding archive fractions, 72.1 % and 84.7 %, describe eligibility, not coverage of every candidate interval.

At fix i , define $b_i = 1$ when a finite, nonzero pressure altitude accompanies the horizontal position in the same decoded record, and $b_i = 0$ otherwise. For a candidate interval I , the barometric test is available only if

$$B_I = B_{\text{flight}} \prod_{i \in I} b_i = 1. \quad (2.1)$$

Zero pressure values are treated as missing before duplicate timestamps are merged: the pair $\{0, 1000\}$ therefore supplies one 1,000 m reading. No temporal interpolation supplies a witness. Equation (2.1) requires pressure and position in every record of I , even when GNSS altitude is invalid.

An eligible barometer with incomplete local coverage causes the barometric test to abstain. Flights without an eligible barometer remain in the analysis, subject to the weaker recorder-declaration rule below. Restricting a sensitivity analysis to $B_I = 1$ would measure the effect of this difference in evidence.

2.7.2 Fix-level cleaning

Fix-level cleaning distinguishes three cases (Fig. 2.2):

- An *off-the-trend defect*, such as a horizontal spike, may receive a Hampel flag. The separate impossibility gate decides deletion.
- An *on-the-trend defect*, such as a duplicate timestamp or frozen position, requires timestamp or recorder checks.

- A *vertical defect* invalidates altitude while provisionally retaining the horizontal fix.

These local tests leave slowly varying and plausible-looking errors unresolved. Deletions, boundaries and reconstruction flags record what the pipeline changed.

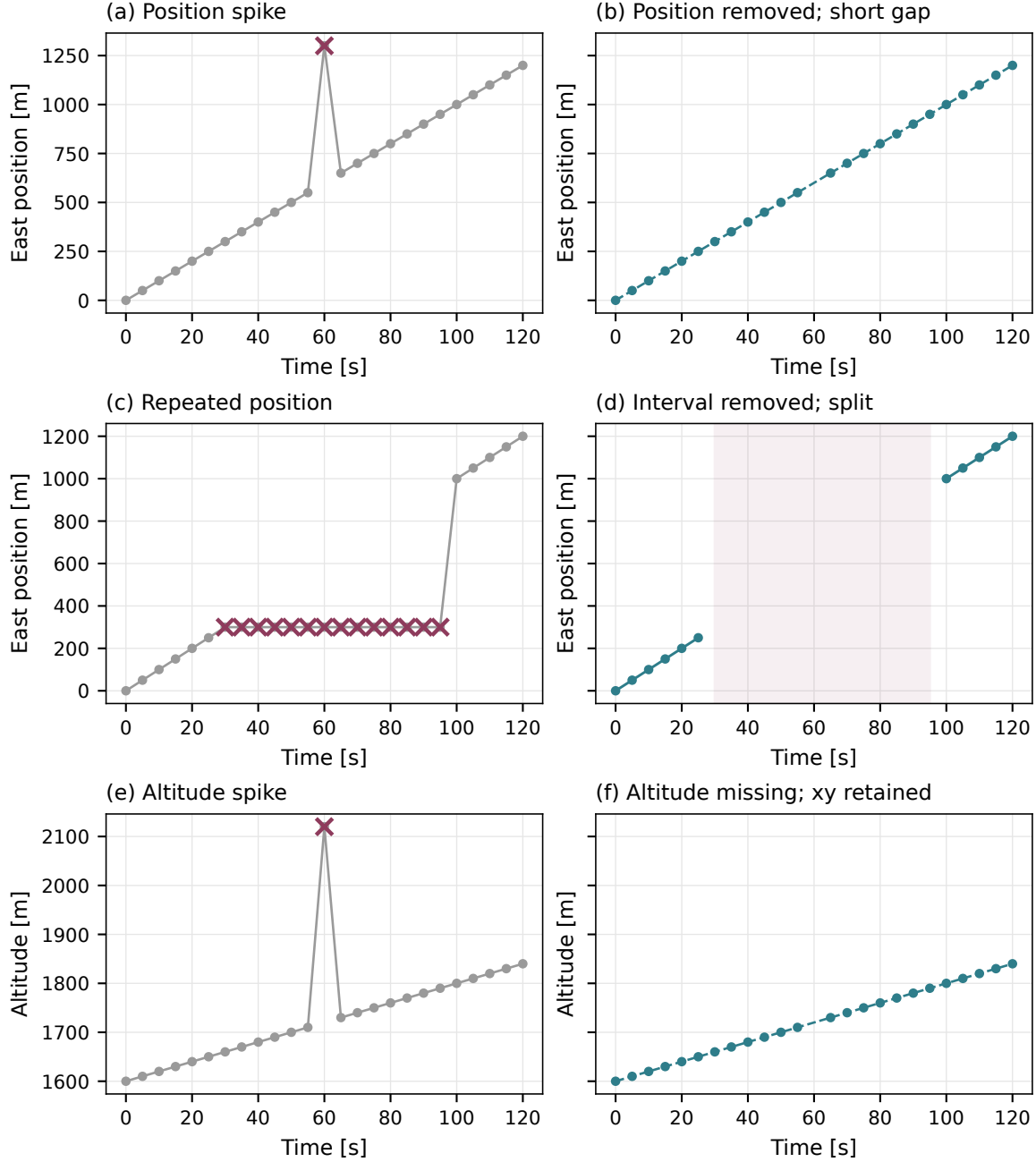


Figure 2.2. Synthetic cleaning examples. Crosses mark suspect readings and dots retained information. A horizontal spike is deleted after the speed/rejoin test; a frozen-position run is removed and split when duration and witness conditions hold; an altitude spike leaves horizontal fixes intact. Dashed paths indicate possible reconstruction at resampling.

Geometry on the raw coordinates. The cleaning runs on the raw geodetic coordinates (φ_i, λ_i) , before the Cartesian conversion (Sec. 2.7.5), so distances between fixes are great-circle

(haversine) distances [12] on a sphere of mean radius $R_\oplus = 6,371$ km:

$$\Delta s_i = 2R_\oplus \arcsin \sqrt{\sin^2 \frac{\varphi_{i+1} - \varphi_i}{2} + \cos \varphi_i \cos \varphi_{i+1} \sin^2 \frac{\lambda_{i+1} - \lambda_i}{2}}, \quad (2.2)$$

The same distances give step velocities, total path length and extent:

$$\begin{aligned} v_{xy,i} &= \frac{\Delta s_i}{t_{i+1} - t_i}, & v_{z,i} &= \frac{h_{i+1} - h_i}{t_{i+1} - t_i}, \\ L &= \sum_i \Delta s_i, & D &= \max_i \Delta s(\text{fix}_0, \text{fix}_i). \end{aligned} \quad (2.3)$$

Near 45° latitude, the spherical distance error is at most about 0.3%: 3 cm on a 10 m step, below metre-scale GNSS noise. Appendix 2.B derives the ellipsoidal correction and its latitude dependence.

The Hampel identifier. Let the window of fix k be the fixes within $\pm w$ of t_k , and $\tilde{\varphi}_k, \tilde{\lambda}_k$ the component-wise window medians. The residual is the great-circle distance from that median point,

$$r_k = \Delta s((\varphi_k, \lambda_k), (\tilde{\varphi}_k, \tilde{\lambda}_k)), \quad (2.4)$$

with local scale $\text{MAD}_k = \text{med}\{r_j : |t_j - t_k| \leq w\}$, and the fix is flagged when

$$r_k > \max(k \sigma_k, \varepsilon_{\min}), \quad \sigma_k = 1.4826 \text{MAD}_k, \quad (2.5)$$

σ_k a robust local scale [13]. The factor 1.4826 is the usual one-dimensional Gaussian MAD normalization; for these two-dimensional radial residuals it is a scale convention, not an exact standard deviation or a calibrated false-alarm probability.

The working values are $k = 5$, $w = 20$ s and $\varepsilon_{\min} = 15$ m, with at least 5 fixes per window.

Flags contribute to the anomaly count; they never gate deletion. A thermalling circle can depart from its local median while remaining reachable at the absolute speed bound.

Reach, rejoin, or cut. Write a for the last fix the scan accepted, the *anchor*, and measure every candidate against the chord from it,

$$V(a, i) = \frac{\Delta s((\varphi_a, \lambda_a), (\varphi_i, \lambda_i))}{t_i - t_a}. \quad (2.6)$$

An unreachable fix opens a pending block. Its anchor a stays fixed while elapsed time grows. The scan then applies three actions:

Condition	Action
Reachable fix, no pending block	Keep it and advance the anchor.
First rejoin within w	Delete the pending predecessors; keep the rejoin as the new anchor.
No rejoin within w	Keep the block, split at its first fix and re-anchor on the current fix.

The cap starts at the pending block's first fix. It reuses the Hampel half-window without a shared calibration. Gaps left by deletion are bridged or split at resampling (Sec. 2.7.6).

Figure 2.3 compares a spike, a moderate offset and a large offset. The moderate offset becomes reachable as time grows and can cost genuine fixes; the large offset remains unreachable and is split. This gate cannot identify which branch contains the true positions.

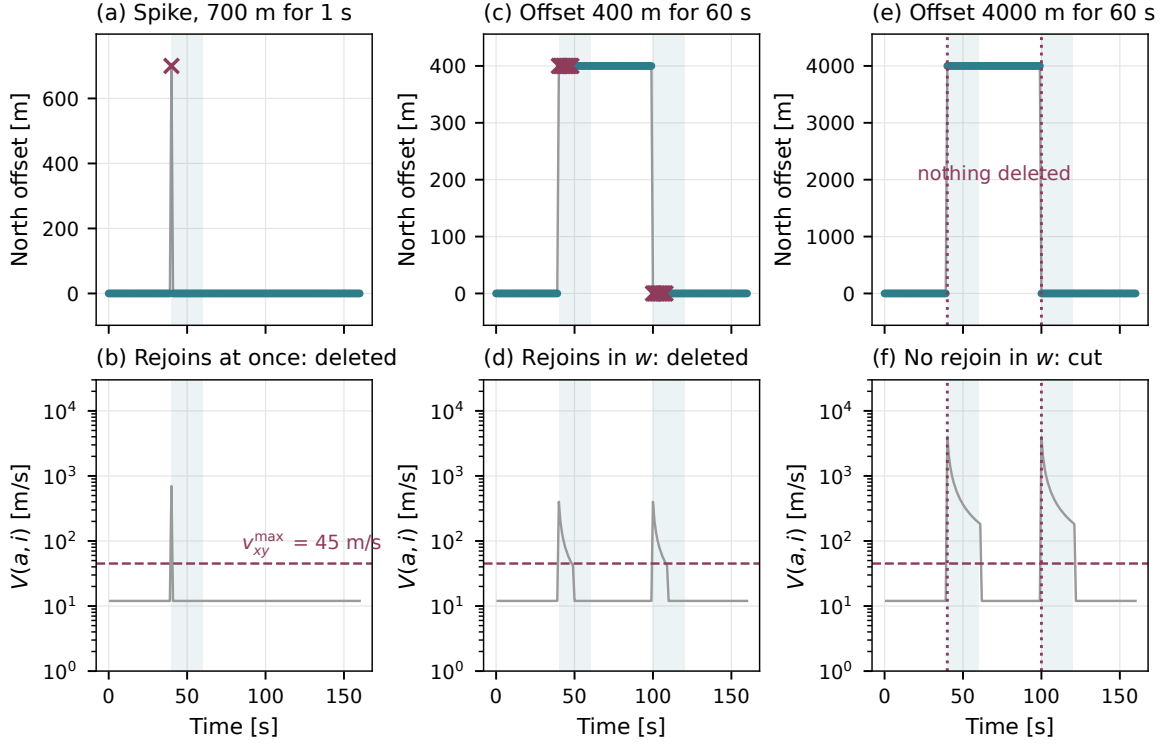


Figure 2.3. Implemented gate on synthetic tracks cruising east at 12 m s^{-1} , sampled every second. Northward offsets are 700 m for one fix (a,b), 400 m for 60 s (c,d), and 4,000 m for 60 s (e,f). Crosses mark deletions, dotted lines boundaries, and shading the block cap w . Lower panels show chord speed from the current anchor and the bound $v_{xy}^{\max}$. The moderate offset removes nine fixes at each end, including genuine ones; the large offset is kept as a separate segment.

Vertical-speed checks. Write the observed altitude as $h_i = z_i + \epsilon_i$. Even a smooth true trajectory produces noisy finite differences,

$$v_{z,i}^{\text{obs}} = \frac{h_{i+1} - h_i}{t_{i+1} - t_i} = \bar{v}_{z,i}^{\text{true}} + \frac{\epsilon_{i+1} - \epsilon_i}{t_{i+1} - t_i}. \quad (2.7)$$

For independent errors of variance σ_z^2 , the differenced noise has variance $2\sigma_z^2/\Delta t^2$. Short cadences therefore amplify altitude error. This calculation illustrates the effect; real GNSS errors may be correlated.

The sustained-excess statistic is

$$\tilde{v}_j = \text{med}\{|v_{z,i}^{\text{obs}}| : |m_i - m_j| \leq w_v, v_{z,i}^{\text{obs}} \text{ finite}\}, \quad m_i = \frac{1}{2}(t_i + t_{i+1}), \quad (2.8)$$

with $w_v = 5 \text{ s}$. The statistic measures local speed magnitude. Each of the $N - 1$ steps is assigned to its midpoint m_i . Figure 2.4 illustrates window selection and the median: for $2r + 1$ finite values, at least $r + 1$ must exceed the bound for the median to do so. For an even count, the two central values are averaged.

With fewer than 5 finite steps, the implementation does not use the median. Writing n_j for that count, the quantity used by the cleaning is

$$u_j = \begin{cases} \tilde{v}_j, & n_j \geq 5, \\ |v_{z,j}^{\text{obs}}|, & n_j < 5. \end{cases}$$

The second branch is the sparse-window fallback. It can occur at a slow recording cadence, near missing altitudes, or at the ends of a record. The sustained-excess rule marks the altitude

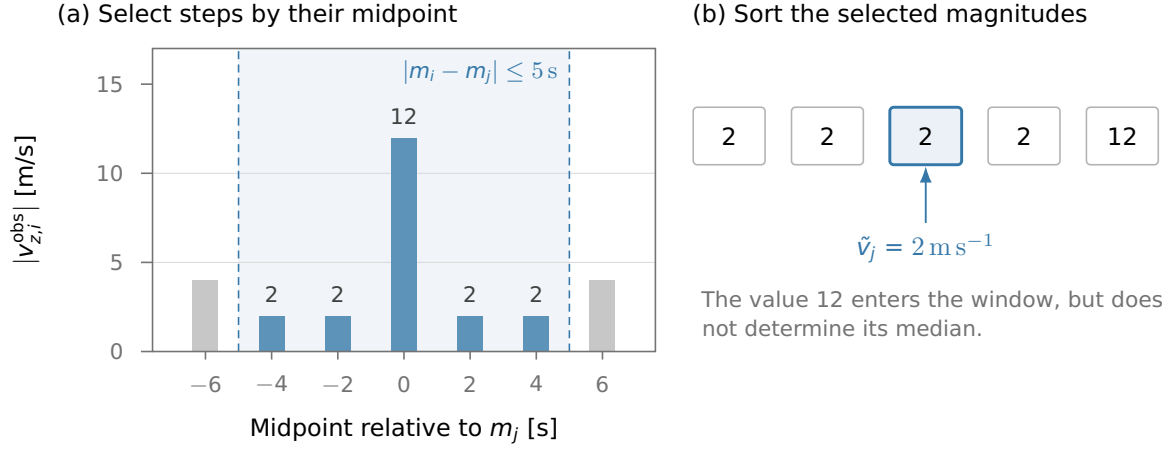


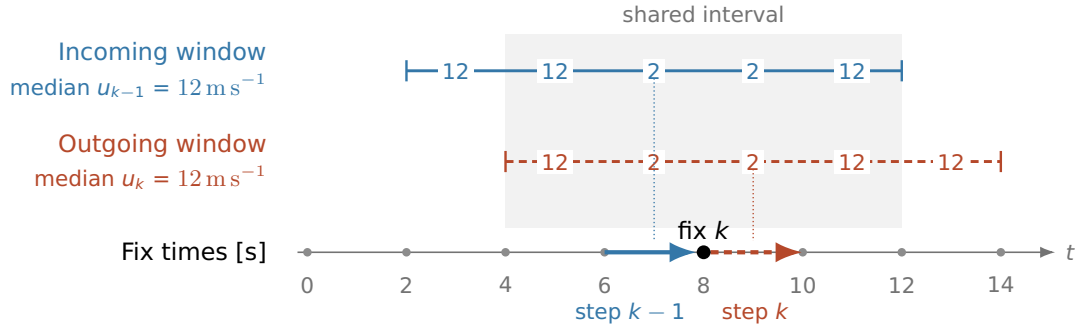
Figure 2.4. How Eq. (2.8) is evaluated. This schematic uses fixes every 2 s and a half-window of 5 s: five step midpoints lie inside it. Grey bars are outside the window and are excluded, regardless of their height. Missing or non-finite speeds are also excluded. Taking absolute values makes an ascent and a descent of equal magnitude contribute equally. These are illustrative values, not flight data.

of an interior fix k as missing when

$$u_{k-1} > v_z^{\max} \quad \text{and} \quad u_k > v_z^{\max}, \quad v_z^{\max} = 10 \text{ m s}^{-1}. \quad (2.9)$$

Both the incoming and outgoing window statistics must exceed the bound strictly. With populated windows, the rule can invalidate altitude even when the two steps touching fix k are below the bound (Fig. 2.5). The overlapping windows locate a neighbourhood of excessive speeds; they do not identify its erroneous altitude independently.

(a) Two windows meet at one decision



(b) Apply the AND condition

u_{k-1} [m/s]	u_k [m/s]	Both $> 10 \text{ m s}^{-1}$?	Result of this rule
2	2	No	Keep altitude
2	12	No	Keep altitude
12	2	No	Keep altitude
12	12	Yes	Mark altitude missing

Figure 2.5. Two overlapping windows determine the altitude decision at fix k . Fixes are 2 s apart and the half-window is 5 s; numbers along the bars are speed magnitudes in m/s. Both medians equal 12 m s^{-1} , although the steps touching k have magnitude 2 m s^{-1} . Equation (2.9) therefore invalidates its altitude at the 10 m s^{-1} bound. The table illustrates the AND condition. Timestamp and horizontal position remain intact.

An isolated spike on a low-speed background leaves a well-supported median low (Fig. 2.6).

A separate out-and-back rule invalidates it when both adjacent steps exceed $v_z^{\max}$ with opposite signs. Sustained excess instead raises the median, whether caused by recorder error or exceptionally strong physical motion.

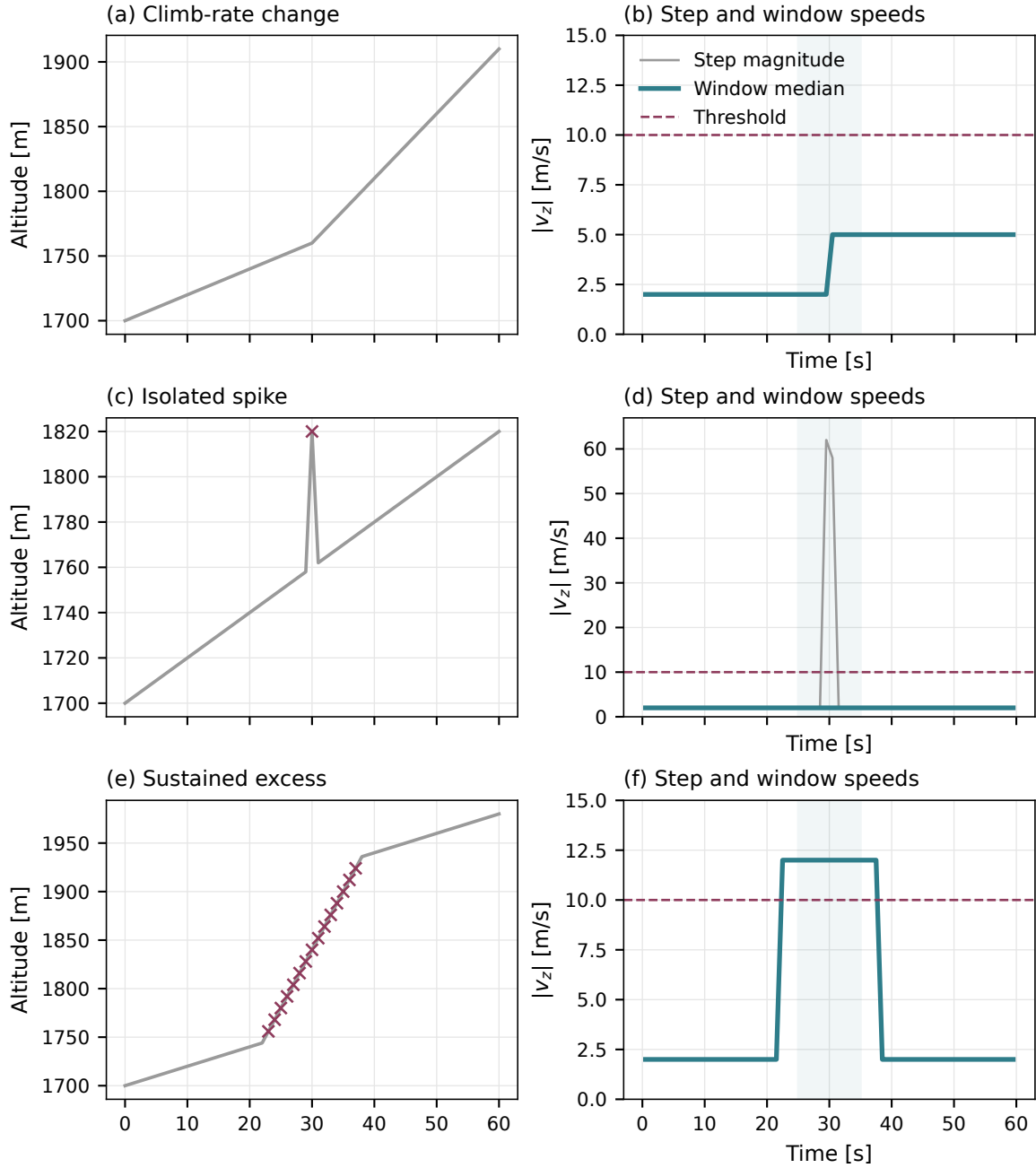


Figure 2.6. The median test at a one-second cadence, using the configured window and speed threshold. Left: synthetic observed altitude. Right: absolute per-step speed (grey), window median (solid), and the bound (dashed). A change between plausible climb rates remains below the bound; an isolated spike barely affects the median and is handled by the separate spike rule; a sustained excessive interval raises the median. Crosses mark altitudes invalidated by the actual implementation.

Unreturned vertical steps. An isolated jump without a prompt return can escape both checks. Define

$$e_i = 1\{|v_{z,i}^{\text{obs}}| > v_z^{\text{max}}\}, \quad J_i = |h_{i+1} - h_i|, \quad (2.10)$$

$$d_i^{\text{return}} = \min_{\substack{j=i+1, \dots, i+5 \\ h_j \text{ finite}}} |h_j - h_i|. \quad (2.11)$$

The minimum uses the next five fix positions, fewer at the record end, and equals $+\infty$ if none has finite altitude. Set $e_i = 0$ outside the observed steps. The counter `n_alt_level_shift` records candidates satisfying

$$e_i = 1, \quad e_{i-1} = e_{i+1} = 0, \quad d_i^{\text{return}} > 0.3J_i. \quad (2.12)$$

This short return check does not establish a permanent offset. It runs before trimming and segment selection, and can overlap other altitude rules.

Among retained-flight metadata, 47,440 paraglider and 1,350 hang-glider records contain at least one candidate. Reprocessing eight previously selected examples with the current pipeline reproduces their counters. In each discipline these include two high-count records and two seeded positive-count records. All eight still have candidate times near usable phase-feature centres. This is temporal association, not proof that an uncorrected sensor defect enters a feature. These counts describe candidates before later cuts, rather than surviving confirmed defects. Automatic repair is not applied: a level shift and a real large increment can have similar signatures. Their effect on phase features still requires a sensitivity analysis.

Figure 2.7b motivates the working bound 10 m s^{-1} : around 9 m s^{-1} to 10 m s^{-1} , the density begins to decay more slowly and the paraglider curve crosses the hang-glider curve. The transition is gradual. The bound is an empirical cleaning choice, so extreme genuine descents can be invalidated too. In the full diagnostic sample, 0.069 % of window medians exceed it. This exceedance fraction differs from the removal rate, which also depends on neighbouring windows and the spike rule.

Deletion and mandatory boundaries. Altitude rules retain timestamp and horizontal coordinates; position and timestamp rules may remove the whole fix. Reconstruction is deferred to resampling. After all cleaning checks, every surviving step above the horizontal speed bound becomes a mandatory segment boundary, counted as `n_boundaried`. This also catches an impossible return from a block that the forward scan kept.

The forward scan does not re-evaluate its initial anchor, so an erroneous first fix can survive this local procedure and displace the origin of the later Cartesian frame. A separate flight-level test therefore rejects a track containing a fix beyond the distance reachable from its first retained position (Sec. 2.7.4). The reach test acts on spherical distances before projection and smoothing, with a 50 m tolerance. The written Cartesian trajectories undergo separate diagnostic checks (Sec. 2.9); those operations do not preserve the reach inequality as an exact invariant.

Frozen-position runs and the evidence for GNSS lock loss. A logger may repeat its last position after losing GNSS lock while timestamps continue. An unremoved run then resembles a period of spatial confinement. The detector uses three joint conditions,

$$D_I < \delta_{xy}, \quad t_{\text{max } I} - t_{\text{min } I} \geq \tau_{\text{freeze}}, \quad W_I = 1, \quad (2.13)$$

where D_I is the great-circle diagonal of the latitude–longitude bounding box of candidate run I , evaluated after longitude unwrapping. This is the implemented extent proxy, not an exact all-pairs diameter. The working values are $\delta_{xy} = 15 \text{ m}$ and $\tau_{\text{freeze}} = 60 \text{ s}$.

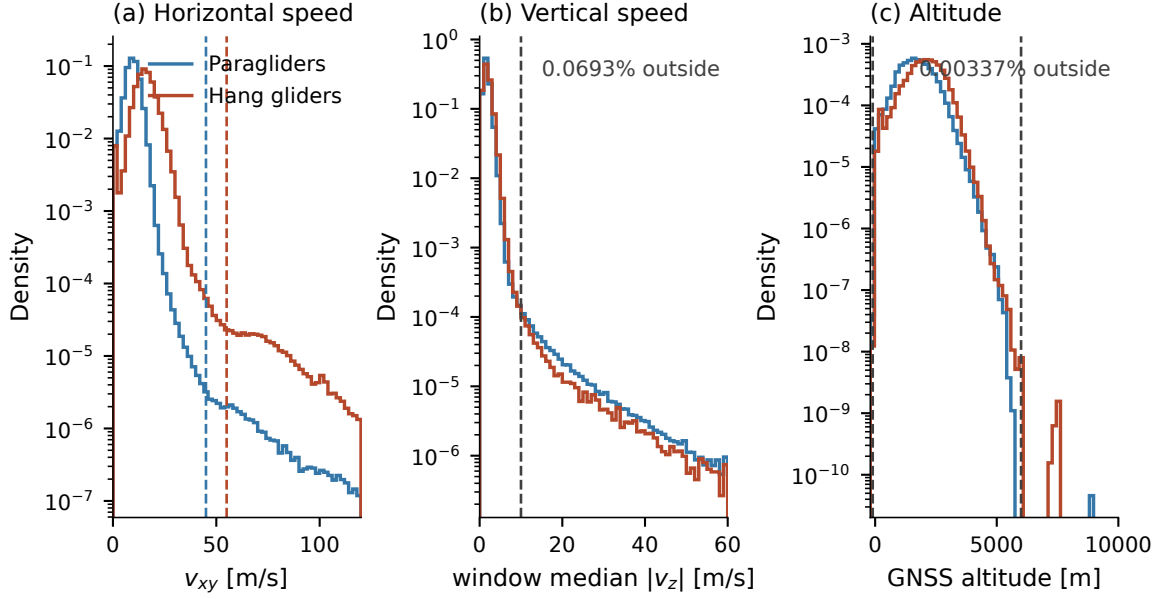


Figure 2.7. Raw diagnostics for the absolute bounds: (a) horizontal step speed, (b) windowed GNSS med $|v_z|$, and (c) GNSS altitude. Dashed lines mark the working thresholds. Densities pool a large paraglider subsample and the complete hang-glider population, using the full finite-sample denominator even where the displayed range cuts off a tail. Logarithmic ordinates expose rare values; abscissae are linear. Annotations count bound exceedances, while actual deletion and invalidation also depend on the rejoin and neighbouring-window rules.

Let E_I denote exact equality of both coordinate fields throughout the run, d_I the fraction of its fixes with a V flag or missing GNSS altitude, and Δb_I the median pressure altitude of the final quarter minus that of the first quarter. With B_I defined in Eq. (2.1), the witness is

$$W_I = E_I \vee \begin{cases} |\Delta b_I| < \delta_z, & B_I = 1, \\ \text{false}, & B_{\text{flight}} = 1, B_I = 0, \\ d_I > 1/2, & B_{\text{flight}} = 0, \end{cases} \quad \delta_z = 5 \text{ m}. \quad (2.14)$$

Removed frozen-lock runs always split the trajectory. Exact coordinate repetition is a separate witness; otherwise incomplete local pressure coverage forces abstention.

The signatures have limits. Quantized coordinates can repeat for a stationary receiver, and sparse fixes do not establish continuous stillness between them. Small net pressure change can hide an excursion; a barometer can also continue climbing during a horizontal GNSS freeze. Shared logger failures can affect both channels.

For scale, a coordinated turn in still air has

$$R = \frac{v_{\text{air}}^2}{g \tan \phi}, \quad T_{\text{turn}} = \frac{2\pi R}{v_{\text{air}}}. \quad (2.15)$$

At $v_{\text{air}} = 10 \text{ m s}^{-1}$ and bank angle $\phi = 35^\circ$, $2R \simeq 29 \text{ m}$ exceeds δ_{xy} , while a sustained 1 m s^{-1} climb changes pressure altitude far more than δ_z . These are reference scales, not universal minimum diameters or guarantees for every wind, bank angle and recorder cadence.

A genuine small manoeuvre can satisfy all signatures and be cut; a jittering freeze without a witness can survive. Runs shorter than τ_{freeze} are retained. Independent inspection or labelled injections are needed to estimate these error rates.

Defect	Detected by	Action
Off the trend — <i>horizontal position defects</i>		
Position spike, out-and-back jump	Unreachable block with a rejoin within w (Eq. 2.6). The Hampel flag [14] only records local disagreement.	delete block; apply gap rule at resampling
Position discontinuity (re-acquisition offset)	No reachable rejoin within the block cap w	keep fixes; split at the discontinuity
Impossible step still standing	any surviving step past the v_{xy} bound, whatever the scan resolved (postcondition)	make it a segment boundary
Corrupt block closing the record	a block still open when the track ends: nothing follows it to rejoin, and by construction it spans at most w	delete the block, instead of splitting
On the trend — <i>record structurally wrong, horizontal position agrees with its neighbours</i>		
Duplicate timestamp	two or more fixes in one UTC second	merge to one (centroid)
Backward timestamp	time decreases (not a midnight roll-over)	delete the fix
Frozen-lock run	Spatial collapse, minimum duration and witness (Eqs. 2.13–2.14)	remove run; split trajectory
Out of range — <i>horizontal fix intact; altitude unavailable or suspect</i>		
Missing / out-of-band altitude, sustained $ v_z $ excess, or $ v_z $ spike	Absent reading; outside $[-100, 6000]$ m; both adjacent window statistics above $v_z^{\max}$ (Eq. 2.9); or an opposite-sign spike pair above that bound	invalidate altitude; keep horizontal fix
Policy: altitude-only invalidation retains the timestamp and horizontal coordinates. Position and time rules may remove complete fixes. These are operational decisions, not known sensor truth. At resampling, short altitude holes are reconstructed and long ones exclude the full interval and split the trajectory (Sec. 2.7.6).		

Table 2.5. Fix-level decisions. Thresholds appear in Table 2.8; Fig. 2.7 shows the absolute-bound diagnostics. Horizontal flags, deletions and altitude invalidations remain separate operations.

Integrity and validation. The integrity gate rejects flights whose cleaning burden exceeds 10 %. Counts use only the airborne window selected by trimming (Sec. 2.7.3), so discarded ground recordings do not affect this decision. Let N_{del} count deleted fixes, N_{keep} surviving fixes and N_{zbad} survivors whose altitude was invalidated. The burden is

$$f_{\text{clean}} = \frac{N_{\text{del}} + N_{\text{zbad}}}{N_{\text{keep}} + N_{\text{del}}}. \quad (2.16)$$

Altitude invalidation includes out-of-band values, sustained excess, spikes and V declarations. Duplicate-second merges and later reconstruction are excluded. This gate removes 550 paraglider and 53 hang-glider flights. The larger loss occurs later when no segment survives resampling: 20,798 and 193 flights, respectively.

Removal fractions measure the amount changed; threshold sweeps measure sensitivity. Classification error rates require independent labels. Synthetic regressions and the real examples in Fig. 2.8 check specific failure modes, while a representative labelled archive audit remains to be done.

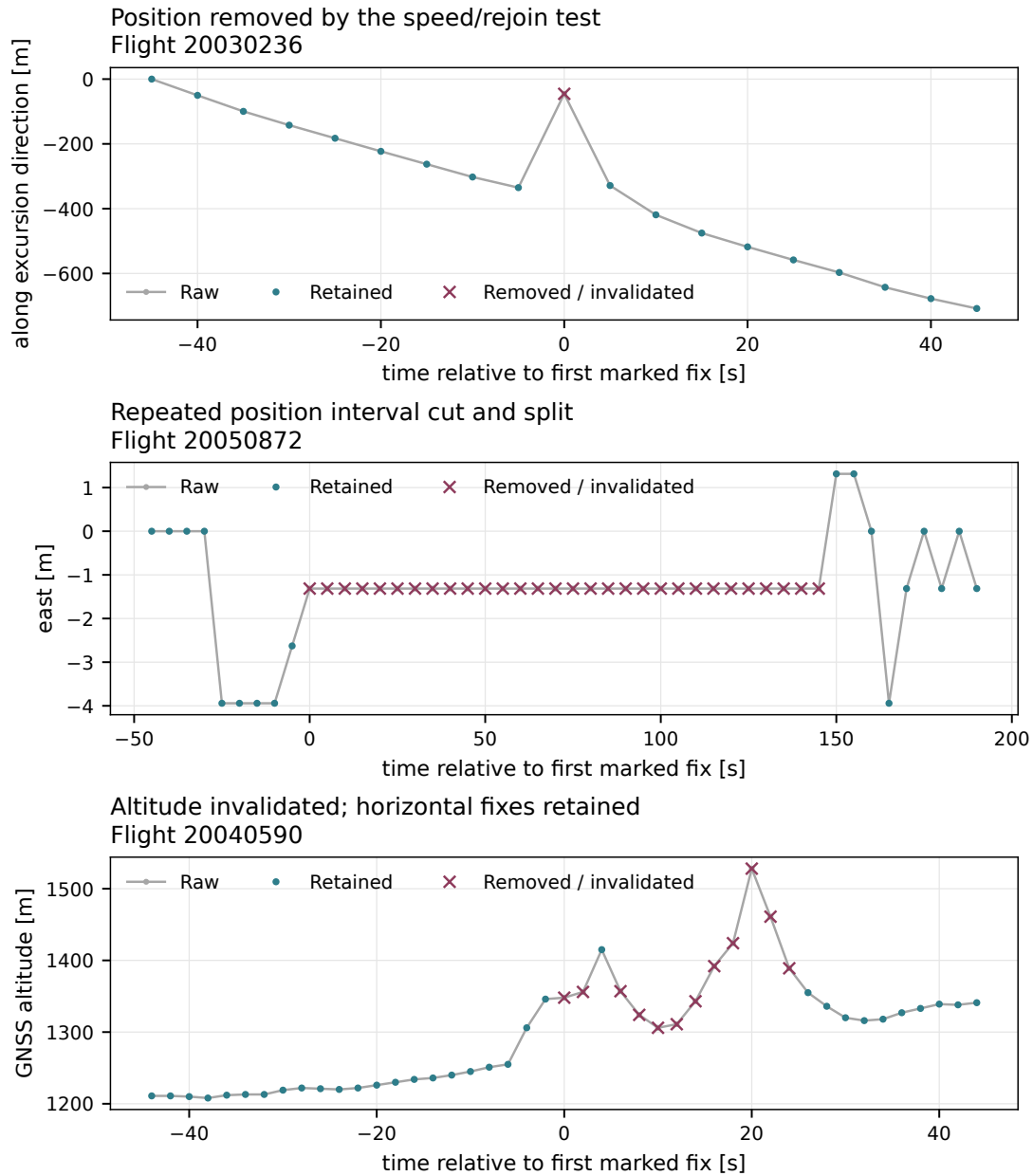


Figure 2.8. Short examples extracted from the interior of raw flight records and reprocessed with the current cleaning implementation. Top: the horizontal coordinate projected along the detected excursion direction, making the removed spike visible. Middle: a collapsed-position interval; crossed marks are removed and the surviving sides split. Bottom: GNSS altitude invalidation, with all horizontal fixes retained. Flight identifiers and plotted raw values are saved with the figure. The examples document detector decisions; no independent sensor truth is available for these intervals.

2.7.3 Trimming of ground phases

Outer ground phases (take-off and landing). The logger starts recording before take-off and stops after landing; these ground phases must be stripped before any kinematic analysis, using the horizontal speed v_{xy} alone. With $v_0 = 5 \text{ m s}^{-1}$ and $T_0 = 30 \text{ s}$ (Table 2.8), the retained airborne segment is $[t_{\text{on}}, t_{\text{off}}]$ with

$$t_{\text{on}} = \min\{t : v_{xy} > v_0 \text{ on } [t, t + T_0]\}, \quad t_{\text{off}} = \max\{t : v_{xy} > v_0 \text{ on } [t - T_0, t]\}, \quad (2.17)$$

the same sustained-speed rule read forward and time-reversed. Here v_{xy} is the speed inferred on each observed step, not a continuously measured velocity. Interior slow periods do not define these outer boundaries, although slow flight at either end can be clipped. A flight with no sustained stretch at all is discarded.

The persistence time T_0 reduces sensitivity to brief speed excursions. It does not bound take-off or landing error by T_0 : a sustained fast ground movement can be retained, while a long airborne interval below v_0 at either end can be removed. The trimmed fraction is recorded per flight, and such cases require direct inspection.

The median time removed before onset and after landing is 33 s and 71 s for paragliders, and 19 s and 62 s for hang gliders (Fig. 2.9). Counts cover 185,131 and 6,580 flights reaching a defined window. The stored `trimmed_fraction` combines the two ends. Longer post-landing intervals are consistent with continued logging after touchdown.

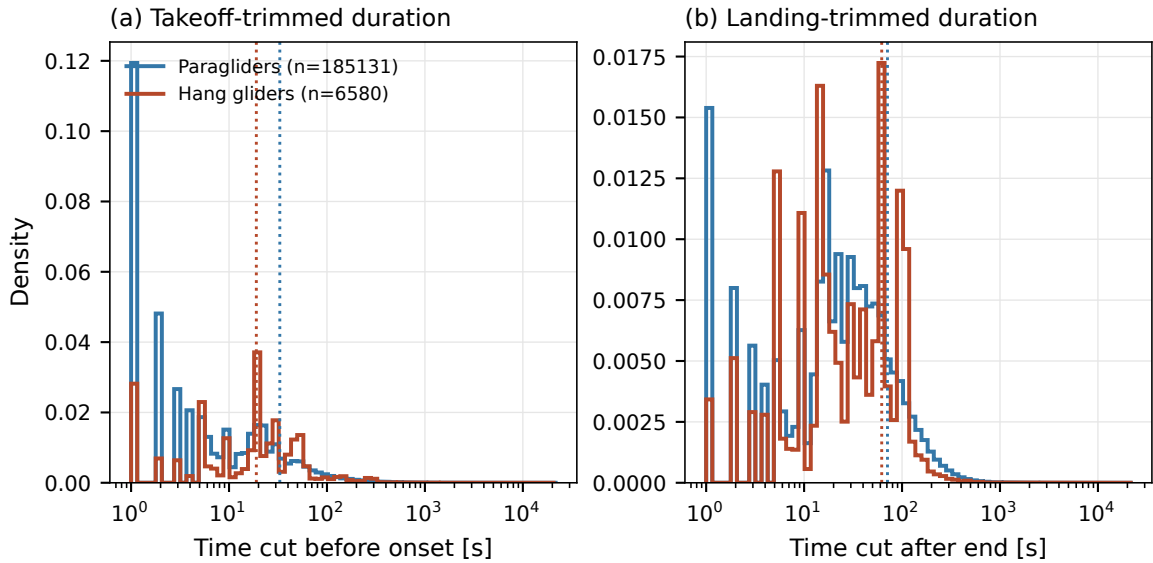


Figure 2.9. Time removed (a) before estimated onset and (b) after estimated landing, with one observation per flight per panel. Dotted lines mark discipline medians. The sustained-speed interval T_0 does not bound the time removed; the tails extend to hours.

Interior ground intervals. A landing and relaunch in one log can imitate a long waiting episode. The detector requires sustained low horizontal speed on an interval I containing at least three fixes, all with paired barometry. Fit $h_i^{\text{baro}} = \hat{\alpha} + \hat{\beta}t_i + r_i$ by least squares. The joint conditions are

$$\begin{aligned} v_{xy} < v_0 \text{ on } I, \quad |I| \geq T_{\text{ground}}, \quad B_I = 1, \\ Q_{.95}(r) - Q_{.05}(r) \leq \delta_z^{\text{gr}}, \quad |\hat{\beta}| \leq \delta_z / \tau_{\text{freeze}}, \end{aligned} \quad (2.18)$$

where $|I|$ denotes duration, $T_{\text{ground}} = 10 \text{ min}$ and $\delta_z^{\text{gr}} = 5 \text{ m}$. The residual span tolerates isolated extremes; the slope bound prevents detrending from erasing a steady climb. Without complete

eligible barometry the detector abstains, since slow ground-relative flight can otherwise resemble a landing.

Excised intervals split the trajectory. Subsequent flight-level cuts apply to the parent flight, summing duration and path length over its remaining blocks; they are not reapplied separately to each segment.

Interior excisions occur in 231 paraglider flights (0.12 %) and 3 hang-glider flights (0.05 %). Most affected flights have one excision; the maxima are 5 and 2 (Fig. 2.10).

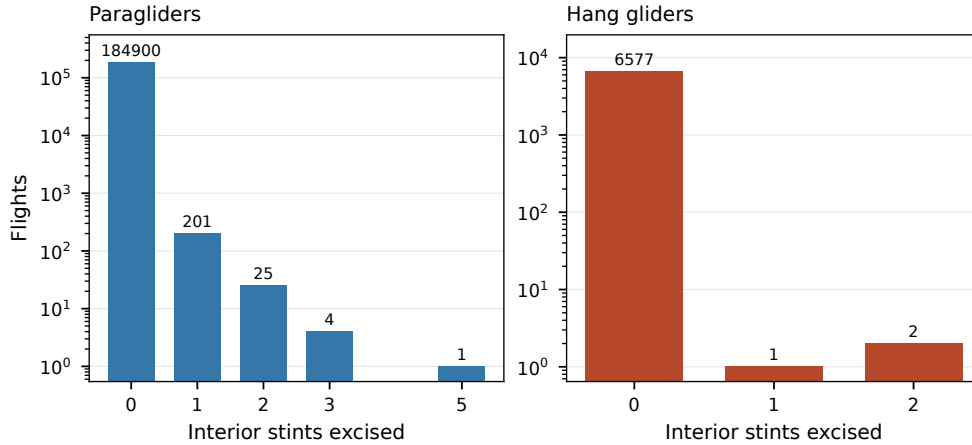


Figure 2.10. Interior intervals excised per flight among all flights reaching this stage. Logarithmic counts expose the rare nonzero cases. Empty bars denote counts absent from the archive.

Shorter intervals satisfying the same speed and barometric conditions are *flagged* when their duration reaches 60 s, the configured minimum passed to this stage. They remain in the trajectories. The stored flags support a future sensitivity analysis with and without these observations; that comparison has not yet been carried out.

2.7.4 Flight-level filtering

The cuts below define the analysed population. They act on the trimmed parent flight, with duration T_{rec} and path L accumulated only within contiguous blocks. Forced boundaries and gaps longer than g_{max} contribute to neither sum; elapsed span T_{span} includes them.

Quantity	Condition	Purpose
Recorded duration	$T_{\text{rec}} \geq 40 \text{ min}$	Exclude short records.
Elapsed span	$T_{\text{span}} \leq 16 \text{ h}$	Screen long recording sessions.
Path length	$L \geq 20 \text{ km}$	Require horizontal travel.
GNSS altitude range	$\max z - \min z \geq 600 \text{ m}$	Require vertical excursion.
Mean speed	$L/T_{\text{rec}} \leq v_{xy}^{\text{max}}$	Check consistency with step bounds.

These are sample restrictions, not a soaring classifier: a genuine flight with little vertical excursion can fail. Mean speed adds no independent evidence when all included steps already satisfy the bound. Figure 2.11 shows sample-size sensitivity; a plateau does not establish unbiased selection.

Raw records near the cuts. The raw census contains 9 paraglider records spanning 16 h to 166 h, and no overlong hang-glider records. Among raw records above the duration floor, the path cut excludes

	Below path floor	Above duration floor
Paragliders	236	184,541
Hang gliders	7	6,638

These raw counts are distinct from the sequential post-trimming removal counts.

Reach from the first fix. Before projection and smoothing, every fix must satisfy

$$\Delta s(\text{fix}_0, \text{fix}_i) \leq v_{xy}^{\max} t_i + \varepsilon_{\text{reach}}, \quad \varepsilon_{\text{reach}} = 50 \text{ m}, \quad \forall i, \quad (2.19)$$

with t_i measured from the first trimmed fix. Testing each elapsed time prevents a later long duration from excusing an early impossible displacement. It also catches a corrupt initial anchor whose first step is no longer summed into L .

This rejects 217 paraglider and 5 hang-glider flights in the stored snapshot. The written Cartesian trajectories receive separate checks with operational tolerances (Sec. 2.9).

Unclassified equipment remains in pooled analyses but is excluded from class-resolved comparisons (Sec. 2.4). Sampling regularity is tested after resampling (Sec. 2.7.6).

2.7.5 From geographic to local Cartesian coordinates

The cleaned, trimmed track is next mapped from geographic to local Cartesian coordinates: resampling and smoothing operate on the coordinates *componentwise* (Sec. 2.7), and latitudes and longitudes cannot play that role, being angles on a curved surface whose componentwise differences are not displacements. The mapping proceeds in two steps.

Step 1: geodetic to ECEF. Each fix is converted to Earth-Centred, Earth-Fixed (ECEF) coordinates on the WGS84 ellipsoid (semi-major axis $a = 6,378,137 \text{ m}$, flattening $f = 1/298.257223563$, $e^2 = f(2 - f)$) [15, 16],

$$N(\varphi) = \frac{a}{\sqrt{1 - e^2 \sin^2 \varphi}}, \quad \begin{aligned} X &= (N + h) \cos \varphi \cos \lambda, \\ Y &= (N + h) \cos \varphi \sin \lambda, \\ Z &= (N(1 - e^2) + h) \sin \varphi, \end{aligned} \quad (2.20)$$

$N(\varphi)$ the prime-vertical radius of curvature and h the ellipsoidal height along the normal, derived step by step in Appendix 2.B.

Geodetic versus geocentric latitude. WGS84 reports geodetic latitude φ , measured along the ellipsoid normal. Geocentric latitude ψ follows the radius from the Earth's centre (Fig. 2.12). Their difference is approximately

$$\varphi - \psi \simeq \frac{e^2}{2} \sin 2\varphi, \quad \max |\varphi - \psi| \simeq 0.19^\circ.$$

Confusing them can shift a position by about 20 km (Appendix 2.B). Equation (2.20) already takes the geodetic latitude recorded in IGC; no latitude conversion is needed.

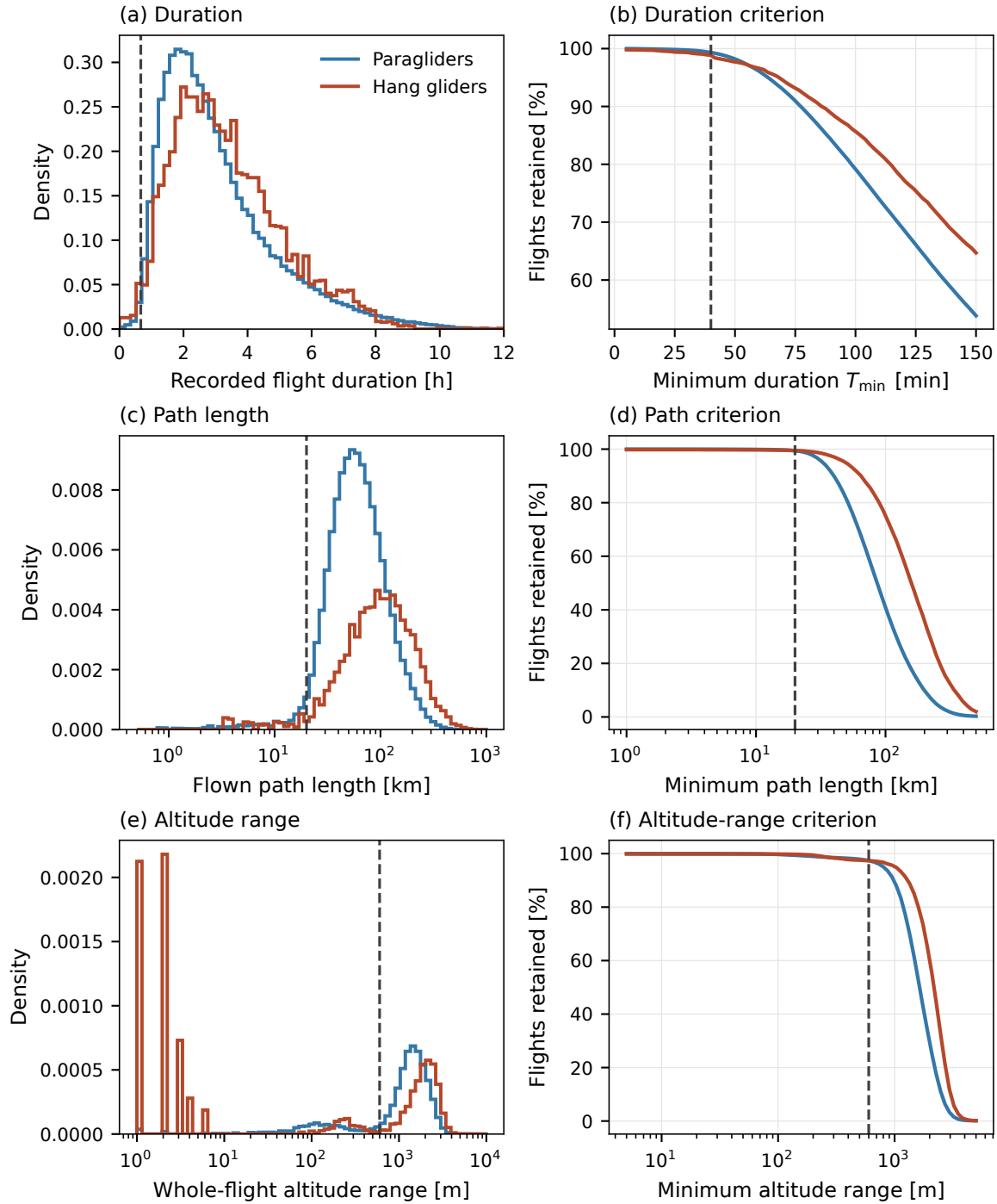


Figure 2.11. Raw flight-level diagnostics by discipline. Left column: (a) recorded duration, (c) flown path length and (e) GNSS altitude range. Right column: retention under the corresponding single criterion, evaluated separately on these raw summaries. Dashed lines mark the working thresholds ($T \geq 40$ min, $L \geq 20$ km and altitude range ≥ 600 m). The altitude panels require the configured minimum GNSS coverage. Histogram densities retain their full finite-sample denominators when tails extend outside the displayed range. These marginal raw diagnostics are not the retention fractions of the sequential cleaned-flight cuts.

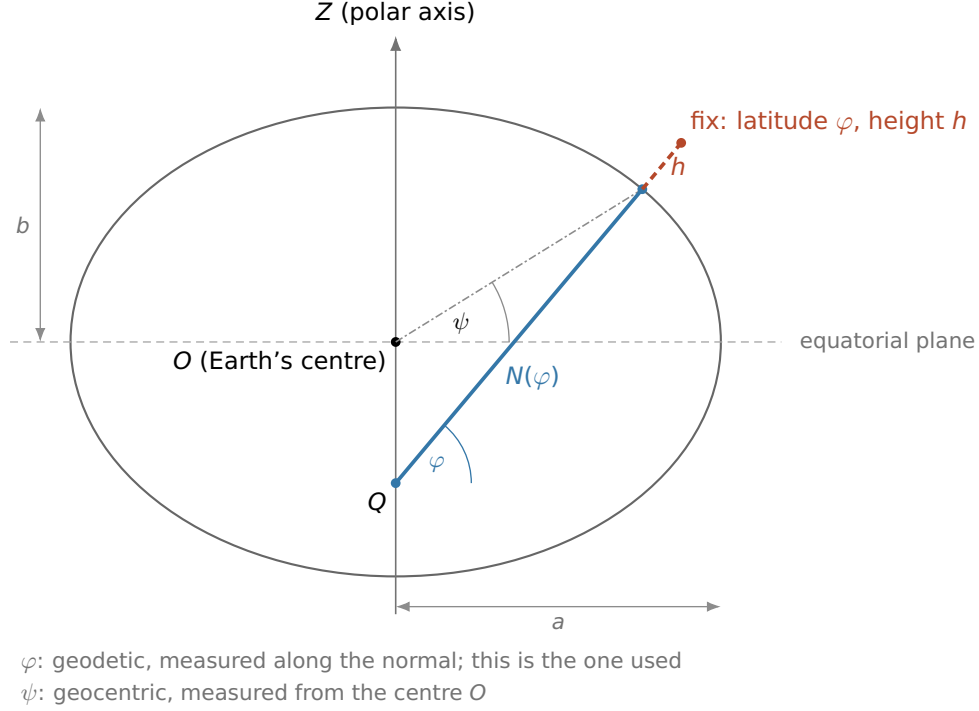


Figure 2.12. Geodetic latitude and height on the reference ellipsoid, in meridian cross-section (flattening exaggerated; the true WGS84 $f \approx 1/298$ would be indistinguishable from a circle): the semi-axes a and b ; the geodetic latitude φ , the angle between the ellipsoid normal and the equatorial plane; the height h along that normal; and $N(\varphi)$, the distance along the normal from the surface to the point Q on the polar axis. The normal does not pass through the centre O , so φ differs from the geocentric latitude ψ (dash-dotted). Longitude λ is out of the page here; it is shown in Figure 2.13.

Step 2: ECEF to local ENU. Taking the first fix $(\varphi_0, \lambda_0, h_0)$ of the *trimmed* track as the origin — close to but not the take-off point, since the ground phase has been removed (Sec. 2.7.3) — the ECEF displacement $\Delta \mathbf{X} = \mathbf{X} - \mathbf{X}_0$ is rotated into the local East–North–Up (ENU) frame tangent to the ellipsoid at that point,

$$\begin{pmatrix} E \\ N \\ U \end{pmatrix} = \begin{pmatrix} -\sin \lambda_0 & \cos \lambda_0 & 0 \\ -\sin \varphi_0 \cos \lambda_0 & -\sin \varphi_0 \sin \lambda_0 & \cos \varphi_0 \\ \cos \varphi_0 \cos \lambda_0 & \cos \varphi_0 \sin \lambda_0 & \sin \varphi_0 \end{pmatrix} \Delta \mathbf{X}, \quad (2.21)$$

The rotation is orthogonal with unit determinant: it preserves the three-dimensional ECEF chord and angles (Appendix 2.B). Keeping only (E, N) then projects the chord onto the tangent plane.

For a point at surface distance d from the origin on a sphere,

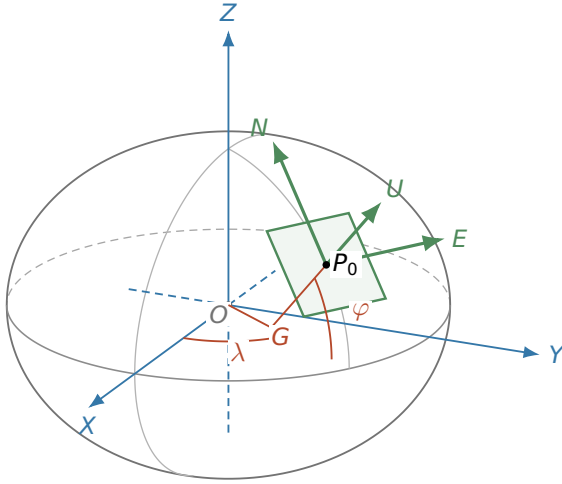
$$r_{EN} = R_{\oplus} \sin(d/R_{\oplus}), \quad d - r_{EN} = \frac{d^3}{6R_{\oplus}^2} + O(d^5/R_{\oplus}^4). \quad (2.22)$$

The horizontal shortening is about 0.03 m at 20 km and 0.51 m at 50 km; it reaches about 33 m at 200 km. The often-used $d^3/(24R_{\oplus}^2)$ expression instead compares an arc with its full three-dimensional chord. Neither estimate is a uniform error bound for every pair of fixes in a fixed frame.

The height input is the logged GNSS altitude, whose datum has not been harmonized. Its use as ellipsoidal height introduces a horizontal error of order

$$|\delta \mathbf{r}_{EN}| \sim |\delta h| \frac{d}{R_{\oplus}}.$$

(a) The global frame and the two angles



(b) The tangent frame, close up

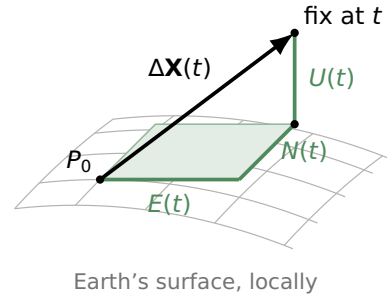


Figure 2.13. The local ENU frame. (a) The origin P_0 is the first fix of the trimmed track. Longitude λ and geodetic latitude φ orient its green unit vectors within the blue ECEF frame. Up follows the ellipsoid normal through G , with flattening exaggerated. (b) A later fix's displacement is resolved into E , N and U : the box edges project its space diagonal.

For example, a 100 m height error at 50 km contributes about 0.8 m. This illustrates the scale; it is not an archive-wide bound.

The vertical coordinate is $z(t) = h_{\text{GNSS}}(t)$, without subtracting the origin altitude. The rotated component U also contains Earth curvature and is not used for vertical dynamics. Velocities depend on altitude increments; absolute heights retain the datum limitation of Sec. 2.7.1.

2.7.6 Uniform sampling within a flight

The smoothing and derivative estimates in the next section require equally spaced observations. Each flight keeps its native interval, $\Delta t = \text{median}_i(t_{i+1} - t_i)$: a flight recorded every second stays at one second, and a flight recorded every five seconds stays at five seconds. Figure 2.14 shows these intervals in the archive.

Fill a short gap; split at a long one. For a gap $g = t_{i+1} - t_i$ between surviving fixes, interpolation is allowed when $g \leq g_{\text{max}}$, with

$$g_{\text{max}} = \min(10 \Delta t, \max(20 \text{ s}, 2\Delta t)). \quad (2.23)$$

At fast cadences the relative limit binds; at slower cadences the absolute cap binds, with a $2\Delta t$ floor allowing one missed fix. Thus the configured 20 s value is not an absolute ceiling for every logger: at $\Delta t = 15 \text{ s}$, the bound is 30 s. For example,

$$\Delta t = 1 \text{ s} \Rightarrow g_{\text{max}} = 10 \text{ s}, \quad \Delta t = 5 \text{ s} \Rightarrow g_{\text{max}} = 20 \text{ s}.$$

A longer gap starts a new segment. The same bound applies between consecutive finite altitude readings, even with horizontal fixes in between. Cleaning boundaries always force a split. The cap limits reconstruction but cannot ensure that manoeuvres inside short gaps are resolved.

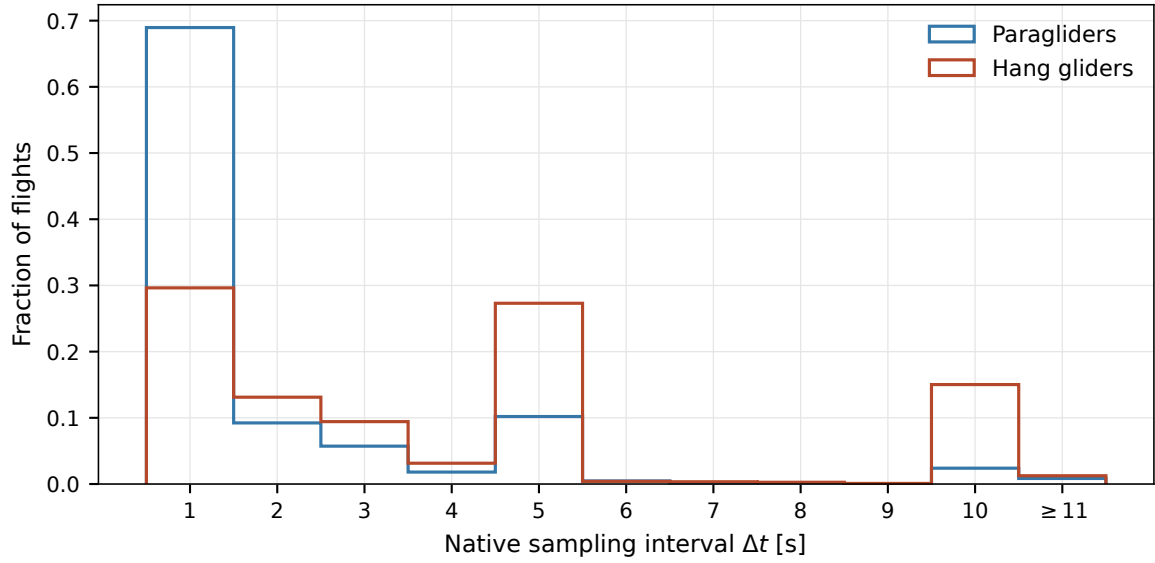


Figure 2.14. Native sampling interval per flight, measured as the median inter-fix time in the raw archive. Bars give the fraction of flights in one-second bins; the last bin also includes intervals beyond 11 s. Paragliders concentrate at 1 s, while hang gliders are spread across several cadences. Resampling preserves each flight’s native interval.

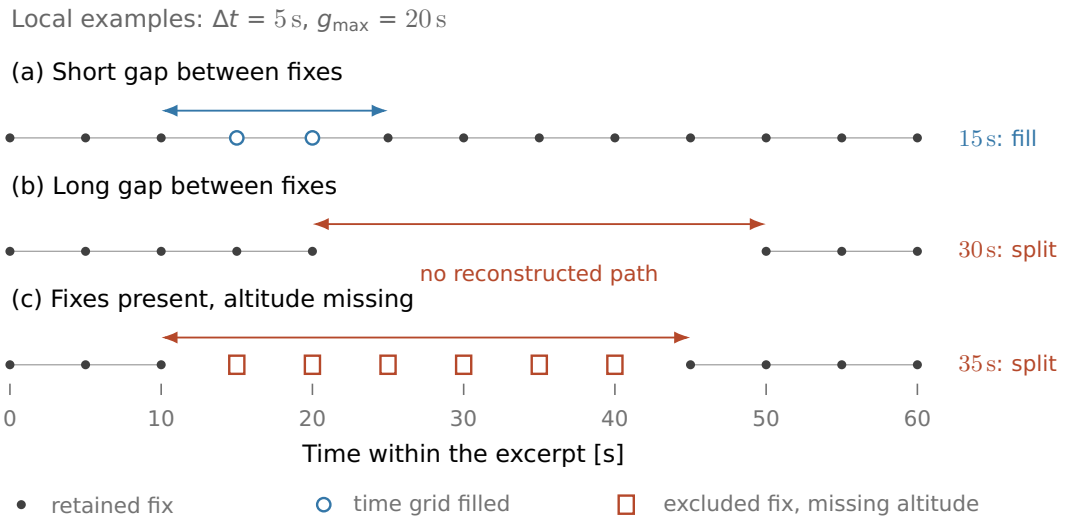


Figure 2.15. Three local examples of resampling. Circles mark new grid times in a short gap; a long gap leaves separate segments. In (c), squares mark existing horizontal fixes without valid altitude. The separation between valid altitude endpoints exceeds $g_{\max}$, so those fixes are excluded and the full trajectory is split.

Reconstructing the coordinates. Within segment s , the grid is $t_k^{(s)} = t_{\text{start}}^{(s)} + k\Delta t$, ending at or before its last fix. An already uniform, complete series is unchanged. Otherwise, E and N are evaluated with a shape-preserving cubic interpolant (PCHIP) [17], and z with linear interpolation. Linear filling avoids overshoot in altitude but can smooth away a climb–glide transition. Missing-altitude prefixes and suffixes are excluded before the grid is built; altitude is never extended beyond finite support.

For altitude missing at existing fixes, let g_z separate the surrounding finite altitude readings. The rule is

$$\begin{cases} g_z \leq g_{\text{max}} : & \text{fill altitude linearly,} \\ g_z > g_{\text{max}} : & \text{exclude intervening fixes and split the full trajectory.} \end{cases}$$

It applies to both missing raw readings and values invalidated by cleaning (Fig. 2.15c). Unsupported intervals remain recorded as rejected segment candidates. Sacrificing their horizontal coverage avoids constructing long unsupported three-dimensional paths.

Three flags record reconstruction for subsequent phase analysis:

Flag	Marked grid times
<code>interpolated</code>	No surviving fix within $\Delta t/2$.
<code>z_reconstructed</code>	The above, plus times whose nearest fix has no altitude.
<code>z_derivative_reconstructed</code>	Derivative windows touched by altitude reconstruction.

The flags identify grid times or derivative windows touched by reconstruction; numerical completeness alone does not establish accuracy. The altitude-gap bound concerns the separation of finite source readings, not the duration of a consecutive run of flags on the new grid. With irregular source timing, nearest-fix flagging can merge adjacent reconstructed portions into a longer flagged run even when every finite-altitude bracket satisfies the bound.

Which segments are retained. A segment must span at least 90 s, have at most 10 % of its grid times flagged as `interpolated`, and yield finite values in all three coordinates. The temporal-coverage fraction and the altitude-gap bound are separate criteria; the latter is applied before interpolation. A rejected segment leaves the other segments available; the parent’s flight-level cuts are not reapplied.

Segments retain the flight’s spatial origin and elapsed time. Time-lagged displacement pairs and analysis windows are formed within segments. Splitting therefore changes the available coverage and can truncate flight phases; it must be accounted for when comparing observables. Figure 2.16 describes gaps in the raw archive; observables that depend on the recording interval, such as fix-to-fix turning angles, require a common temporal scale or a comparison within cadence groups.

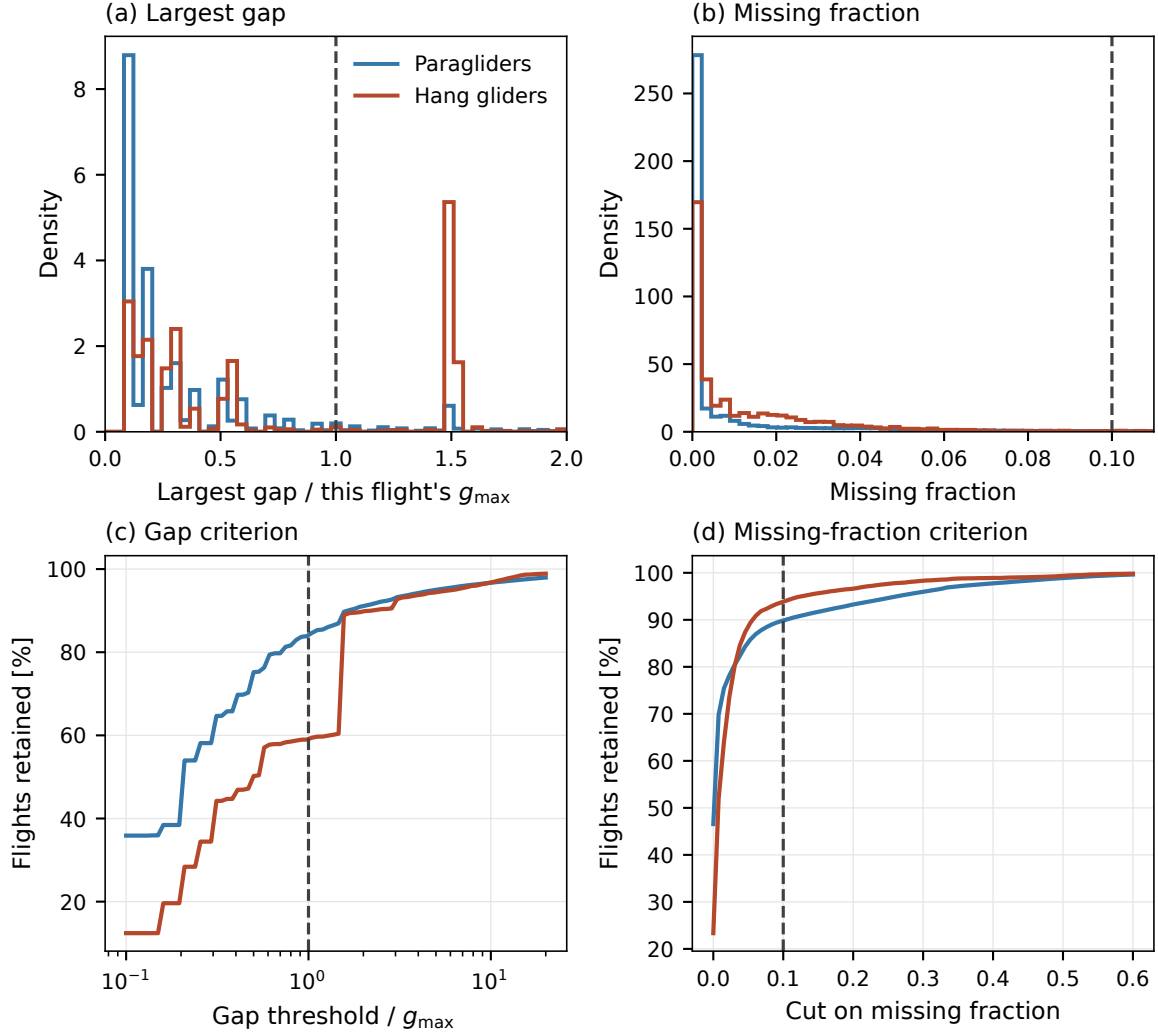


Figure 2.16. Sampling gaps in the raw archive, before cleaning and splitting. (a)/(b): distributions of the largest gap relative to $g_{\max}$ and of the missing fraction on a native-rate grid. (c)/(d): fractions of flights satisfying each criterion separately as its threshold varies. These are diagnostics of whole raw flights; the operational missing-fraction gate acts on segments after splitting, so these curves are not final flight-retention rates.

2.7.7 Smoothing and differentiation

Position errors are amplified by differentiation. We estimate position, velocity and acceleration from the same local polynomial, fitted independently to E , N and z within each retained segment. For $w = 2m + 1$ equally spaced samples around fix i , the Savitzky–Golay fit [18] is

$$P_i(u) = \sum_{k=0}^p a_k(i) u^k, \quad u = -m, \dots, m, \quad (2.24)$$

$$\mathbf{a}(i) = \arg \min_{\mathbf{a}} \sum_{u=-m}^m [x_{i+u} - P_i(u)]^2. \quad (2.25)$$

Here u is a sample offset. The physical quantities at the window centre are

$$\hat{x}_i = a_0(i), \quad \hat{v}_i = \frac{a_1(i)}{\Delta t}, \quad \hat{a}_i = \frac{2a_2(i)}{\Delta t^2}. \quad (2.26)$$

The weights. With $V_{u,k} = u^k$ and $\mathbf{x}_i = (x_{i-m}, \dots, x_{i+m})^\top$, the mathematical least-squares solution is

$$\mathbf{a}(i) = (V^\top V)^{-1} V^\top \mathbf{x}_i. \quad (2.27)$$

The inverse is a formula for the weights, not a recommendation to form a matrix inverse numerically. The implementation uses SciPy’s Savitzky–Golay coefficients. For fixed (w, p) , row k defines weights $c_k(u)$ with

$$a_k(i) = \sum_{u=-m}^m c_k(u) x_{i+u}. \quad (2.28)$$

This forward-offset convention is a correlation; convolution uses the reversed weights. For an odd derivative the reversal changes their signs. Table 2.6 gives the forward-offset weights, before conversion to seconds.

Table 2.6. Interior Savitzky–Golay weights for $w = 5$, $p = 3$. Velocity and acceleration weights must be divided by Δt and Δt^2 , respectively.

	sample offset u				
	−2	−1	0	1	2
c_0	−3/35	12/35	17/35	12/35	−3/35
c_1	1/12	−8/12	0	8/12	−1/12
$2c_2$	2/7	−1/7	−2/7	−1/7	2/7

The order $p = 3$ allows nonzero acceleration and locally varying acceleration. A quadratic already estimates velocity and acceleration; a cubic is a modelling choice, not a mathematical necessity. On a symmetric window, adding a cubic term changes the first-derivative weights but leaves the value and second-derivative weights of the quadratic fit unchanged.

Window length and cadence. A raw spectral diagnostic suggests a change in slope around 0.2 Hz (Fig. 2.18). It motivates the working timescale

$$\tau_c = 1/f_c = 5 \text{ s}, \quad w = \max\left(\left\lceil \frac{\tau_c}{\Delta t} \right\rceil_{\text{odd}}, 5\right). \quad (2.29)$$

Horizontal and vertical timescales are separate configuration entries, currently equal. The code converts this timescale to a *number of samples*: the distance in time between the first and last sample of a window is $(w - 1)\Delta t$. Thus $w = 5$ spans 4 s at a 1 s cadence, and 20 s at a 5 s cadence. Slower recording therefore changes the physical smoothing scale and can remove motion that faster loggers resolve. Comparisons of short-time observables must control for cadence.

The spectral knee is neither a fitted receiver-noise model nor the cutoff frequency of this filter. Rounding a field to steps of size q has variance $q^2/12$ only under a uniform-error approximation; a flat quantization spectrum additionally requires temporal independence. Similar high-frequency spectra do not establish those assumptions or identify the source of the observed knee.

Frequency response. In the segment interior, a Fourier mode $x_i = e^{2\pi i f i \Delta t}$ is multiplied by

$$G(f) = \sum_{u=-m}^m c_0(u) e^{2\pi i f u \Delta t}, \quad S_x(f) = |G(f)|^2 S_x(f). \quad (2.30)$$

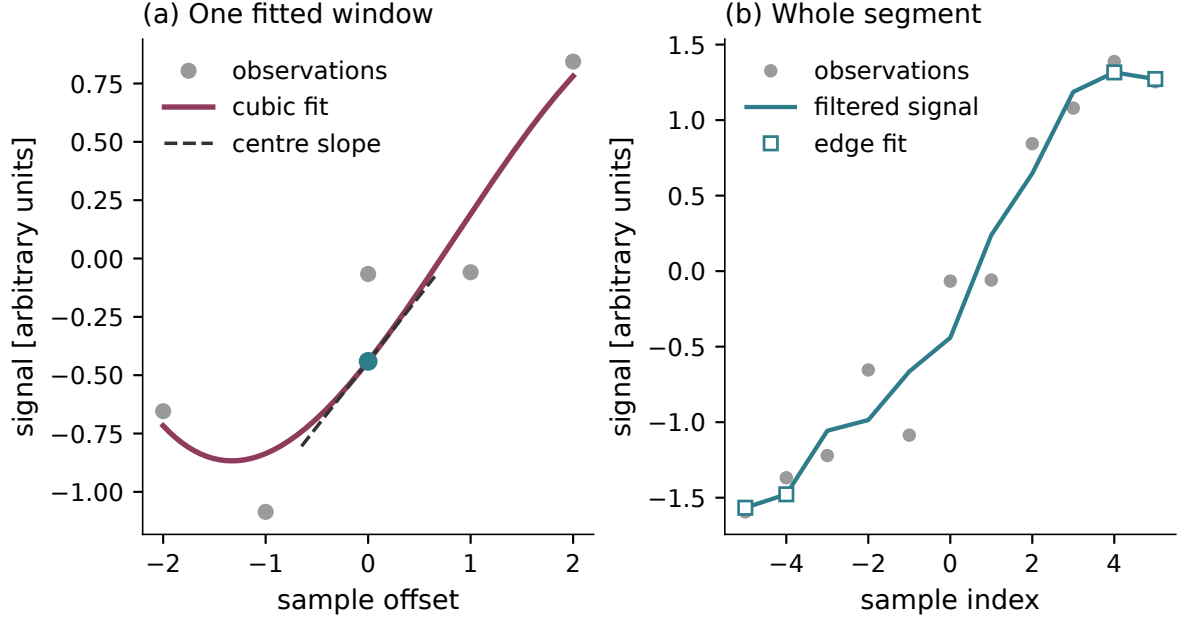


Figure 2.17. A cubic fit to a synthetic signal. Left: one five-sample window, its fit and the fitted value and slope at the centre. Right: the output of the implemented filter over the same series; squares mark the two edge samples at each end. Synthetic values are $\sin(0.28j) + \varepsilon_j$, with independent $\varepsilon_j \sim \mathcal{N}(0, 0.35^2)$.

The response depends on frequency and cadence through

$$\nu = f\Delta t \in [0, \tfrac{1}{2}], \quad (2.31)$$

in cycles per sample, with Nyquist at $\nu = 1/2$. Thus the same $G(\nu)$ applies to every cadence: $\nu = 0.2$ corresponds to 0.2 Hz at $\Delta t = 1$ s and 0.04 Hz at $\Delta t = 5$ s.

Symmetric weights give a real response, $G(\nu) = c_0(0) + 2 \sum_{u=1}^m c_0(u) \cos(2\pi\nu u)$. The phase is unchanged where $G > 0$ and reversed by π where $G < 0$. For the five-sample cubic fit (Table 2.6),

$$G(\nu) = \frac{17 + 24 \cos(2\pi\nu) - 6 \cos(4\pi\nu)}{35}. \quad (2.32)$$

Figure 2.19 shows the resulting power transmission. The filter preserves constants, $G(0) = 1$, and transmits half the power near $\nu = 0.24$. It vanishes at $\nu_0 = \arccos(1 - \sqrt{35/12})/(2\pi) \approx 0.375$, then rises to $|G(1/2)|^2 = (13/35)^2 \approx 0.14$. High frequencies are therefore attenuated unevenly by this short kernel.

Effect on the MSD. Write one recorded coordinate as $\tilde{x}_i = x_i + \varepsilon_i$. For zero-mean errors independent of the true motion and of one another, with $\mathbb{E}[\varepsilon_i \varepsilon_j] = \sigma^2 \delta_{ij}$, the one-coordinate MSD at lag $\ell \geq 1$ is

$$\mathbb{E}[(\tilde{x}_{i+\ell} - \tilde{x}_i)^2] = \mathbb{E}[(x_{i+\ell} - x_i)^2] + \underbrace{\mathbb{E}[(\varepsilon_{i+\ell} - \varepsilon_i)^2]}_{2\sigma^2}. \quad (2.33)$$

The constant noise contribution matters most at short lags, where the true displacement is small.

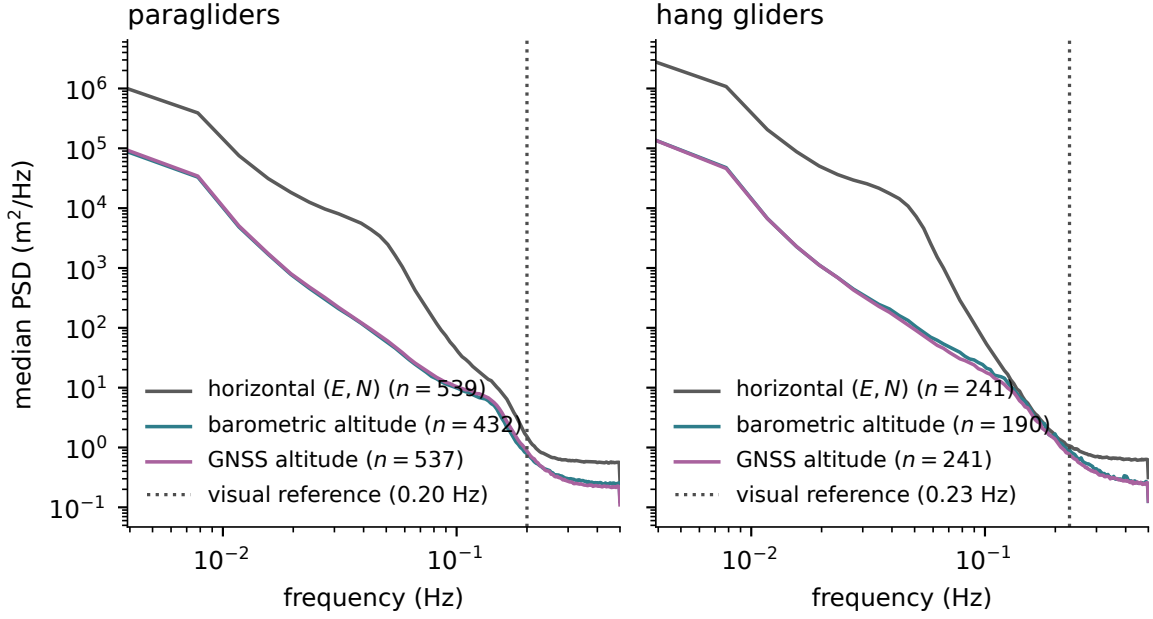


Figure 2.18. Median raw Welch PSDs, by discipline and coordinate channel, on a seeded sample of up to 900 flights per discipline. Spectra use recording intervals near 1 s; legend counts give the usable flights. Horizontal coordinates use a local equirectangular approximation before cleaning, and the curve pools both component spectra from each admitted flight. Channel eligibility is assessed separately, so the three medians need not represent identical flights or intervals. The vertical dotted lines are visual reference frequencies, 0.20 Hz and 0.23 Hz, not estimates with an uncertainty interval. Recording precision, sensor error, interpolation and real motion can all affect these spectra.

Smoothing replaces ε_i by the weighted average $\eta_i = \sum_u c_0(u) \varepsilon_{i+u}$, with c_0 extended by zero outside $|u| \leq m$. Independence of the raw errors leaves only the terms that read the same fix, so with the weight autocorrelation $R(\ell) = \sum_u c_0(u) c_0(u - \ell)$,

$$\mathbb{E}[\eta_i \eta_{i+\ell}] = \sigma^2 R(\ell), \quad (2.34)$$

$$B(\ell) = \mathbb{E}[(\eta_{i+\ell} - \eta_i)^2] = 2\sigma^2 [R(0) - R(\ell)], \quad (2.35)$$

For $|\ell| \geq w$, the two windows share no raw errors and $R(\ell) = 0$.

For $w = 5$, $p = 3$ the weights of Table 2.6 give

$$35^2 R(\ell) = 595, 336, 42, -72, 9, 0 \quad (\ell = 0, \dots, 5), \quad (2.36)$$

Thus $B(\ell)/\sigma^2 \approx 0.42, 0.90, 1.09, 0.96$ at $\ell = 1, \dots, 4$, settling at $34/35 \approx 0.97$ for $\ell \geq 5$. The filter roughly halves the noise contribution beyond its window but makes it lag dependent inside, with an overshoot at $\ell = 3$ (Fig. 2.19). A horizontal MSD adds both coordinate contributions.

This calculation assumes independent errors; it does not calibrate GNSS noise. The filter also changes the true motion unless each window follows a polynomial of degree at most p . A Hurst exponent alone cannot bound that change.

Edges and validation. At either end of a segment, the filter fits the nearest full window and evaluates that polynomial inside its fitted interval. These are off-centre evaluations, not extrapolation beyond the observations. The first and last m samples carry an **edge** flag because their weights and error variance differ from the interior. The filter can overshoot despite earlier cleaning, so speed bounds are not guaranteed by smoothing.

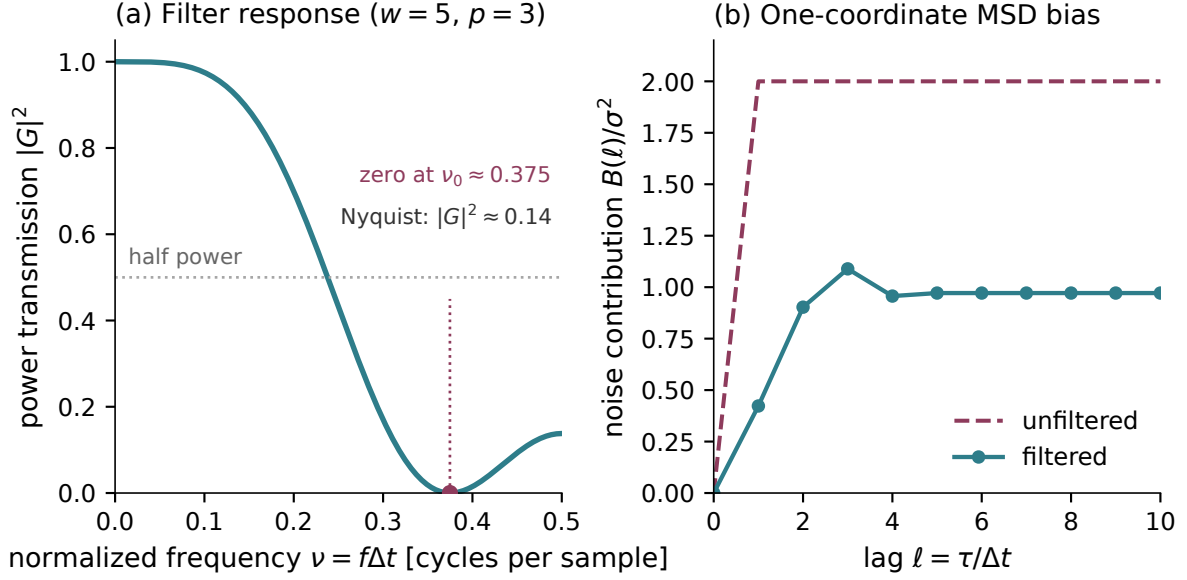


Figure 2.19. Exact interior response of the five-sample cubic filter. Left: power transmission $|G(\nu)|^2$, with $\nu = f\Delta t$ in cycles per sample and Nyquist at $1/2$. Right: additive one-coordinate MSD bias $B(\ell)/\sigma^2$ for independent measurement errors, compared with the unfiltered value 2. Overlapping windows produce the short-lag variation.

The kernel identities and derivative units are checked in automated tests. Archive-level validation remains separate: after rebuilding the cleaned archive, compare the MSD and phase features at neighbouring odd window lengths, within each cadence class, and with edge and reconstructed samples excluded. A flat residual spectrum is not a sufficient criterion: even white input noise has residual PSD $|1 - G(f)|^2 S_\varepsilon(f)$. Until these comparisons are reported, the adopted window is a working choice rather than a certified optimum.

2.7.8 Notation

The coordinates and their derivatives are

$$\begin{aligned} \mathbf{r}(t) &= (E(t), N(t)), & z(t) &= h_{\text{GNSS}}(t), \\ \mathbf{v}(t) &= \dot{\mathbf{r}}(t), & v_z(t) &= \dot{z}(t), \\ v_{xy}(t) &= |\mathbf{v}(t)|, & \Delta \mathbf{r}(t_0, \tau) &= \mathbf{r}(t_0 + \tau) - \mathbf{r}(t_0). \end{aligned}$$

The first trimmed fix sets $t = 0$ and the horizontal origin; smoothing can shift its position slightly. Altitude keeps its logged value. Every segment retains the parent flight's origin and clock. A bare *position* denotes the horizontal vector $\mathbf{r}$; three-dimensional position is stated explicitly.

2.8 Pipeline summary

Figure 2.20 gathers the processing steps of Sec. 2.7 in the order they are applied, from raw IGC tracks to the retained trajectories. Each stage identifies the records it acts on, its effect and the section that explains it. Section 2.9 then describes the resulting dataset.

Each flight is processed independently using its own cadence and geometry, with the shared thresholds in Table 2.8. Adding flights to the archive therefore does not change how an existing record is cleaned.

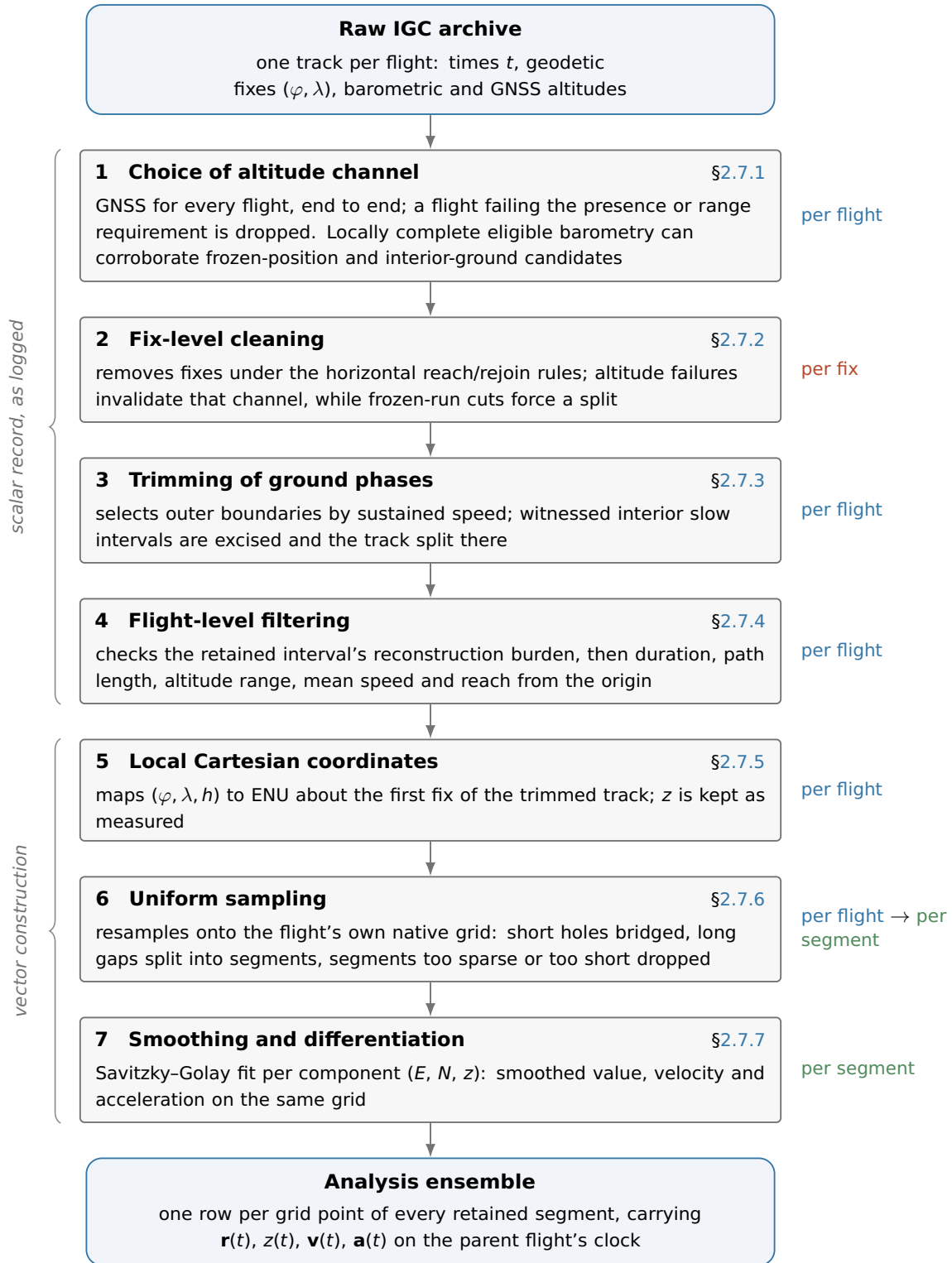


Figure 2.20. The processing chain, from the raw archive to the analysis ensemble. Each stage carries its effect and the section that argues the choice; the badge to the right is the level it acts at (*per fix*, *per flight*, *per segment*). Every check on the scalar record (stages 1–4) precedes the construction of vector quantities (stages 5–7), so no coordinate conversion is computed for a record about to be repaired or dropped. The thresholds each stage applies are listed, with their symbols, in Table 2.8.

What the chain retains. Run over the whole archive, the cascade retains 156,406 of 186,052 paraglider flights (84.1 %) and 6,094 of 6,716 hang-glider flights (90.7 %). Table 2.7 says where the rest went. Counts form a cascade: each rejected flight reached that stage after passing earlier criteria. They differ from the single-criterion raw diagnostics in Secs. 2.7.4 and 2.7.6.

Within the retained flights, cleaning deletes 1.83 % of paraglider fixes and 0.72 % of hang-glider ones, and invalidates an altitude on a further 0.056 % and 0.036 %. The local outlier test flags far more than is deleted, and by design: it never gates, so 0.68 % and 0.96 % of fixes are flagged and *kept*. The flag records local disagreement, while the impossibility gate controls horizontal deletion (Sec. 2.7.2). Resampling reconstructs 0.58 % and 0.30 % of grid points. A split is common but not the rule: 9.9 % of retained paraglider flights and 20.5 % of hang-glider ones carry at least one, the difference following the cadences of Fig. 2.14 as Sec. 2.7.6 anticipated.

Table 2.7. The pre-processing cascade over the full archive, in the order the pipeline applies it. *Standing* counts flights that survived earlier criteria; *share* is the fraction removed from that remaining population. Resampling leaves no retained segment in 20,798 paraglider flights (11.7 % of those reaching that step) and 193 hang-glider flights (3.1 %). These conditional removal fractions differ from the separate raw-population diagnostics in Secs. 2.7.4 and 2.7.6.

Why the flight was dropped	paragliders			
	standing	removed	share (%)	remaining
unreadable track	186,052	27	0.01	186,025
no usable GNSS altitude	186,025	697	0.37	185,328
never airborne	185,328	197	0.11	185,131
integrity gate	185,131	550	0.30	184,581
shorter than the duration cut	184,581	2,084	1.13	182,497
longer than the duration cut	182,497	3	0.00	182,494
path below the floor	182,494	173	0.09	182,321
altitude activity below the floor	182,321	4,899	2.69	177,422
out of reach of its own first fix	177,422	217	0.12	177,205
no segment survived resampling	177,205	20,798	11.74	156,407
no segment carries the smoothing window	156,407	1	0.00	156,406
retained				156,406

Why the flight was dropped	hang gliders			
	standing	removed	share (%)	remaining
no usable GNSS altitude	6,716	135	2.01	6,581
never airborne	6,581	1	0.02	6,580
integrity gate	6,580	53	0.81	6,527
shorter than the duration cut	6,527	79	1.21	6,448
path below the floor	6,448	1	0.02	6,447
altitude activity below the floor	6,447	155	2.40	6,292
out of reach of its own first fix	6,292	5	0.08	6,287
no segment survived resampling	6,287	193	3.07	6,094
retained				6,094

Table 2.8 lists the working thresholds and their symbols by stage.

Table 2.8. Working values of the pre-processing pipeline, by stage. Symbols are those used in the text; configured values are read from `configs/preprocessing.yaml`. The minimum smoothing window is fixed by the polynomial-order rule in Sec. 2.7.7. The listed sections explain each criterion.

Symbol	Value	What it controls
Altitude admission (Sec. 2.7.1)		
—	95 %	minimum fraction of finite, nonzero GNSS readings; a flight below this completeness threshold is excluded
—	30 m	minimum GNSS range among finite, nonzero readings; a flight below it is excluded, without establishing a sensor failure
—	95 %	minimum barometric presence for witness eligibility; each candidate also requires complete local support
—	30 m	minimum barometric range for the same witness eligibility
Fix-level cleaning (Sec. 2.7.2)		
k	5	robust- σ multiplier of the Hampel test, Eq. (2.5)
w	20 s	half-width of the Hampel window; the implementation reuses it as the longest a pending block may stay open before the track is cut (Sec. 2.7.2)
$\varepsilon_{\min}$	15 m	absolute floor of the Hampel test: no flag below this residual, whatever the local scale
—	5 fixes	smallest window population for which the test is evaluated
$v_{xy}^{\max}$	45 m s ⁻¹ (para), 55 m s ⁻¹ (hang)	absolute inter-fix speed bound; the impossibility gate, on its own (Sec. 2.7.2)
$ v_z ^{\max}$	10 m s ⁻¹	cutoff for the two adjacent window statistics in Eq. (2.9); also used by the out-and-back spike rule
w_v	5 s	half-width of that window
—	5 finite steps	smallest window population for which the median is estimable; below it the per-step value stands in
$[h_{\min}, h_{\max}]$	$[-100, 6000]$ m	sensor-plausibility window on the altitude
δ_{xy}	15 m	diameter within which a frozen-lock run must stay
δ_z	5 m	altitude flatness tolerance of the same rule
τ_{freeze}	60 s	minimum duration before a run can be cut by the frozen-position rule
—	10 %	share of fixes above which the cleaning is declared to have failed and the flight is dropped
Ground trimming (Sec. 2.7.3)		
v_0	5 m s ⁻¹	ground-speed threshold used to identify take-off and landing candidates
T_0	30 s	how long $v_{xy} > v_0$ must persist for take-off (and, time-reversed, landing)
T_{ground}	10 min	minimum duration of an interior ground stint before it is excised
—	5 m	altitude flatness tolerance over such a stint
Flight-level filtering (Sec. 2.7.4) — discards whole flights that cannot support the statistics		

continued on the next page

Table 2.8 (continued)

Symbol	Value	What it controls
—	40 min	minimum recorded airborne duration, accumulated within blocks
—	20 km	minimum flown path length
—	16 h	maximum elapsed span, including gaps; records above it are excluded
—	600 m	minimum whole-flight altitude range
—	$v_{xy}^{\max}$ (above)	plausibility bound on the mean ground speed of the cleaned track; reuses the fix-level bound, read over a whole flight (Sec. 2.7.4)
—	$v_{xy}^{\max}$ (above)	reach bound applied at every fix: $d_i \leq v_{xy}^{\max}(t_i - t_0) + 50$ m, where d_i is its distance from the first trimmed fix
Resampling (Sec. 2.7.6) — one uniform grid per flight, holes bridged or split		
$g_{\max}$	$\min(10 \Delta t, \max(20 \text{ s}, 2\Delta t))$	longest gap that is interpolated instead of splitting the flight
—	10 %	largest share of grid points a retained <i>segment</i> may have had to interpolate
—	90 s	minimum retained segment duration; smoothing additionally requires enough grid samples for its window
Smoothing (Sec. 2.7.7) — Savitzky–Golay filter and its derivatives		
p	3	polynomial order of the filter
τ_c	5 s (horizontal), 5 s (vertical)	working timescale setting the window through Eq. (2.29); equal values are adopted for the two channels, without identifying a physical correlation time
—	5 samples	minimum odd window for the adopted cubic fit. The temporal span is $(w - 1)\Delta t$, with w given by Eq. (2.29)

The analysis ensemble. The analysis ensemble comprises the retained segments written to `fixes.parquet`, with one row per grid time carrying (E, N, z) and their first two derivatives. A flight must pass stages 1–4 and retain at least one segment after resampling and smoothing. Segment origins and clocks remain those of the first trimmed fix of the parent flight.

Transport and phase estimators impose their own cadence, lag-support and quality-mask requirements, so their contributing populations can differ. The file schema and estimator inputs are documented in `docs/guide/data-on-disk.md`.

2.9 Preliminary characterization

This section describes the retained flights: their duration, path length, recording interval and geographical coverage. These measurements require no phase segmentation.

Trajectory statistics. Figure 2.21 compares the retained trajectories. Hang-glider flights are typically longer and logged less frequently:

	Paragliders	Hang gliders
Retained flights	156,406	6,094
Median retained span [h]	2.63	3.07
Median path [km]	87	151
Median cadence [s]	1	3
Recorded at 1 Hz [%]	69.9	28.0

Paraglider interquartile ranges are 1.77 h to 3.93 h for duration and 57 km to 134 km for path length. The median record contains 7,132 fixes. The native cadence sets each flight’s smoothing window (Sec. 2.7.7).

The dataset audit counts 0 non-finite coordinates and 0 non-monotone flight clocks in the paraglider audit. Its largest filtered interior step speed is 49 m s^{-1} , with 26 interior steps above the working bound but below the verifier’s gross-anomaly threshold. Across segment boundaries, 30 paraglider and 7 hang-glider flights exceed its separate speed-envelope diagnostic, which compares displacement with elapsed time and the declared coordinate tolerance. These boundary anomalies do not establish a unique sensor-error mechanism. Splitting prevents increment windows from bridging them; it does not independently validate absolute positions after reacquisition.

These checks identify structural failures and large kinematic anomalies. Finite coordinates and bounded speeds do not exclude plausible-looking recorder errors.

Panels (a) and (b) describe retained flights. Here duration is the elapsed span from the first to the last retained fix, including intervening gaps; path length sums only within-segment steps. These quantities can be smaller than those of the trimmed parent flight tested in Sec. 2.7.4, because short, sparse or unreconstructable segments can be discarded. The resulting distributions include 609 paraglider flights (0.39 %) and 6 hang-glider flights below the original 40 min or 20 km threshold. The parent-flight quantities passed those cuts; the cuts are not reapplied to the shorter retained trajectory.

Retention across recording intervals and seasons. The cascade in Table 2.7 retains most attempted flights. Loss of every segment at resampling is the largest conditional cut in both archives:

	Paragliders	Hang gliders
End-to-end retention [%]	84.1	90.7
No segment survives [%]	11.7	3.1
Altitude-range cut [%]	2.7	2.4
Ratio of the two cut rates	4.4	1.3

Each cut rate uses the flights reaching that stage. The 20.798 paraglider losses at resampling combine segments that are too short, too sparse or unreconstructable; they cannot all be attributed to the missing-fraction bound.

Across the six most common known cadences, the paraglider share with no surviving segment ranges from 0.0 % to 22.4 %. This variation calls for sensitivity checks of the missing-fraction and minimum-duration thresholds.

Across the 20 seasons with at least a thousand flights, overall retention ranges from 69.8 % to 91.9 % (Fig. 2.22b). Similar rates can conceal changes in cadence, region and pilot composition.

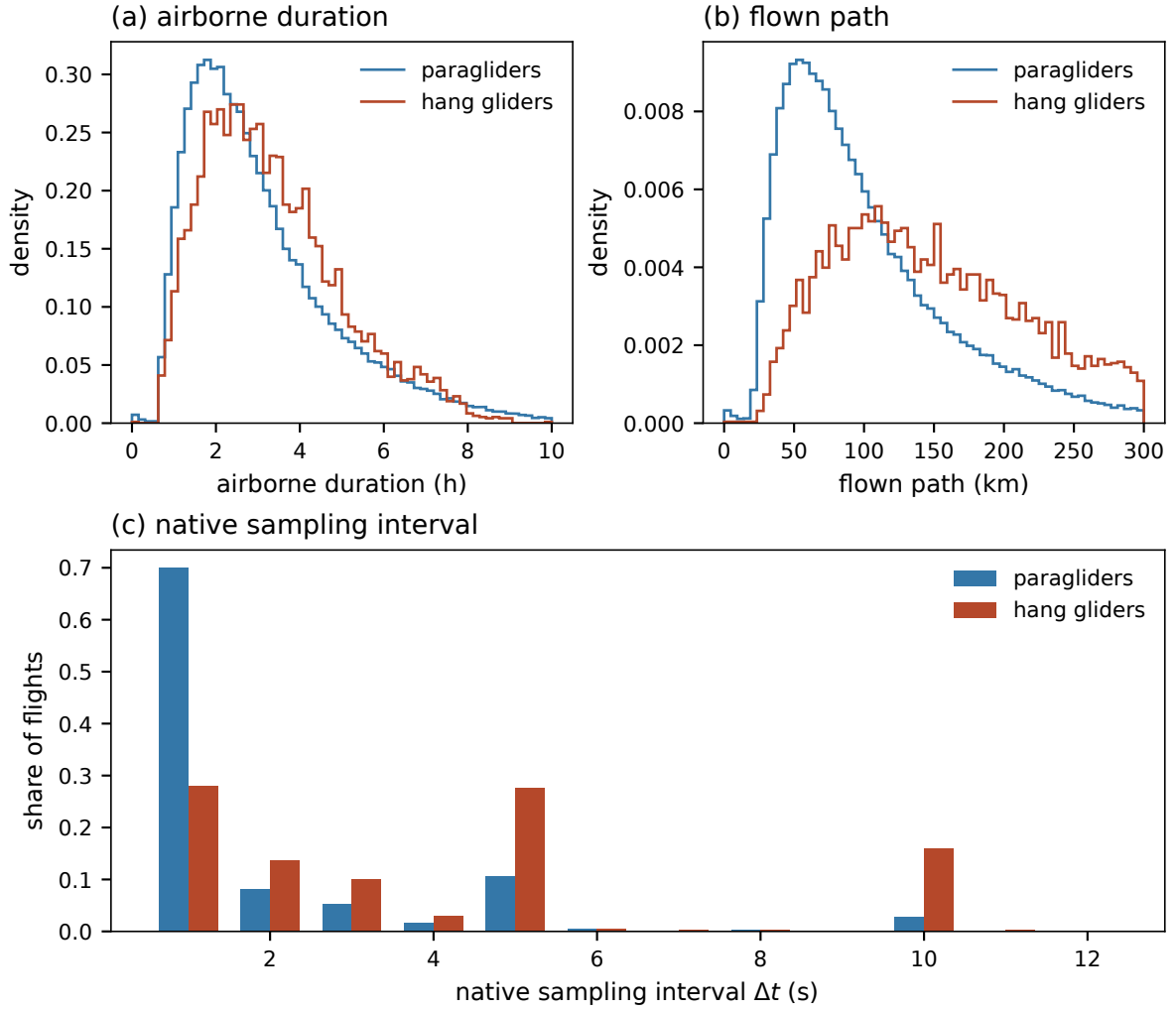


Figure 2.21. The records the pipeline retains. (a) Airborne duration, (b) flown path and (c) native sampling interval Δt , per discipline, over the retained flights rather than over every parsed flight—which is what distinguishes these from Figs. 2.16 and 2.14. The duration labelled in panel (a) is the first-to-last retained-fix span, including gaps, rather than observed airborne time. Path length in (b) sums only within-segment steps. These retained-record quantities differ from the trimmed-parent quantities tested by Sec. 2.7.4; no parent-flight threshold is reapplied after resampling.

Where the flights are. The distribution of take-off sites is presented across three separate panels to account for the geographic coverage of the CFD: Figure 2.23 displays these sites against coastline and national borders.

- **Metropolitan France.** 94.7 % of retained paraglider flights launch here, against 96.7 % of hang-glider ones — the hang gliders are slightly more domestic.
- **La Réunion.** 3.2 % (4,946 flights), the one overseas department with a substantial share.
- **Elsewhere.** 2.2 % (3,379 flights): the Canaries, Andalusia, the Italian Alps, and further out Colombia, Nepal, India, South Africa and Brazil.

Within metropolitan France the ensemble does not sample the country evenly. It occupies 5,341 distinct take-off sites, where distinct sites are launches rounded to a hundredth of a degree (~ 1 km); by share of paraglider launches:

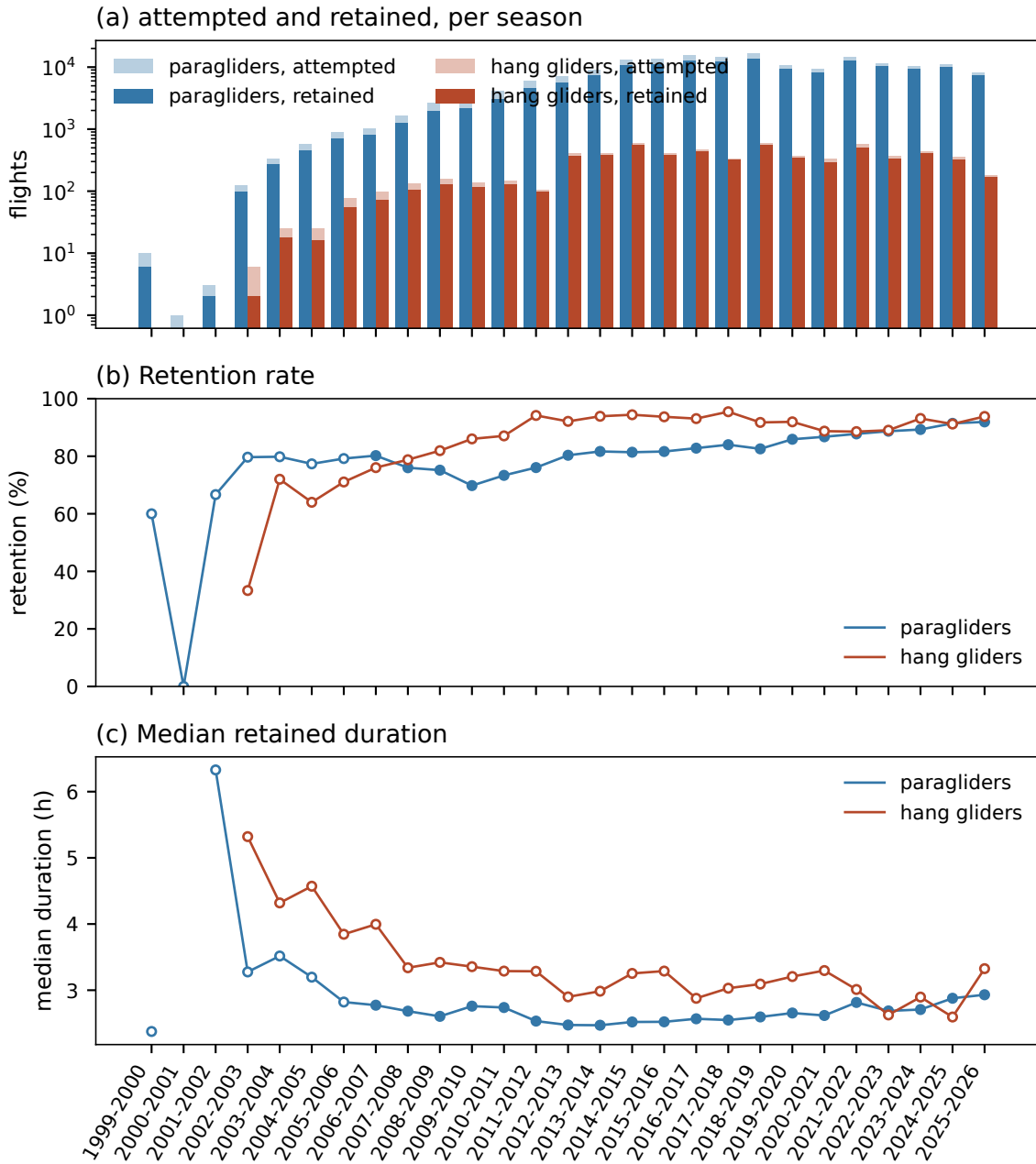


Figure 2.22. The archive season by season. (a) Flights attempted and retained, on a logarithmic scale: the ladder grows by three orders of magnitude over its life. (b) Retention rate and (c) median airborne duration of the survivors; open markers are seasons of fewer than 1,000 flights. The aggregate retention rate is approximately flat across the larger seasons; this does not establish equal selection within cadence or region groups.

- **Alps.** 58.8 %.
- **Pyrenees.** 7.9 %.
- **Massif Central.** 8.2 %.
- **Channel Coast.** 3.2 %: a box over low-relief terrain (Normandy, Picardy, the coastal Nord), not one carved out by excluding the other three.
- **Outside the massifs.** 16.7 %: everywhere else in metropolitan France, a residual defined by exclusion and not a terrain claim.

These groups are longitude and latitude boxes, drawn on Fig. 2.23a; membership is assigned from the first recorded position.

Altitude at the beginning of the record. Figure 2.24 groups flights by their first raw altitude, before trimming: GNSS where available, pressure altitude otherwise. This classifies 99.9 % of retained paraglider flights. Their shares are

Legend label	Logged altitude [m]	Share [%]
Plains	< 300	7.6
Hills	300–800	11.2
Low mountains	800–1500	50.7
High mountains	≥ 1500	30.5

The bands follow Hernández-Aguayo, Cristelli and Benzaquen [19]. They simplify the full classification of Kapos et al. [20]; their historical names do not measure local relief or slope.

The first fix can be away from launch, already airborne, or affected by sensor settling. Datum and atmospheric offsets can move it between bands. This exploratory map therefore describes recorded starting altitude, with pressure fallback confined to this map.

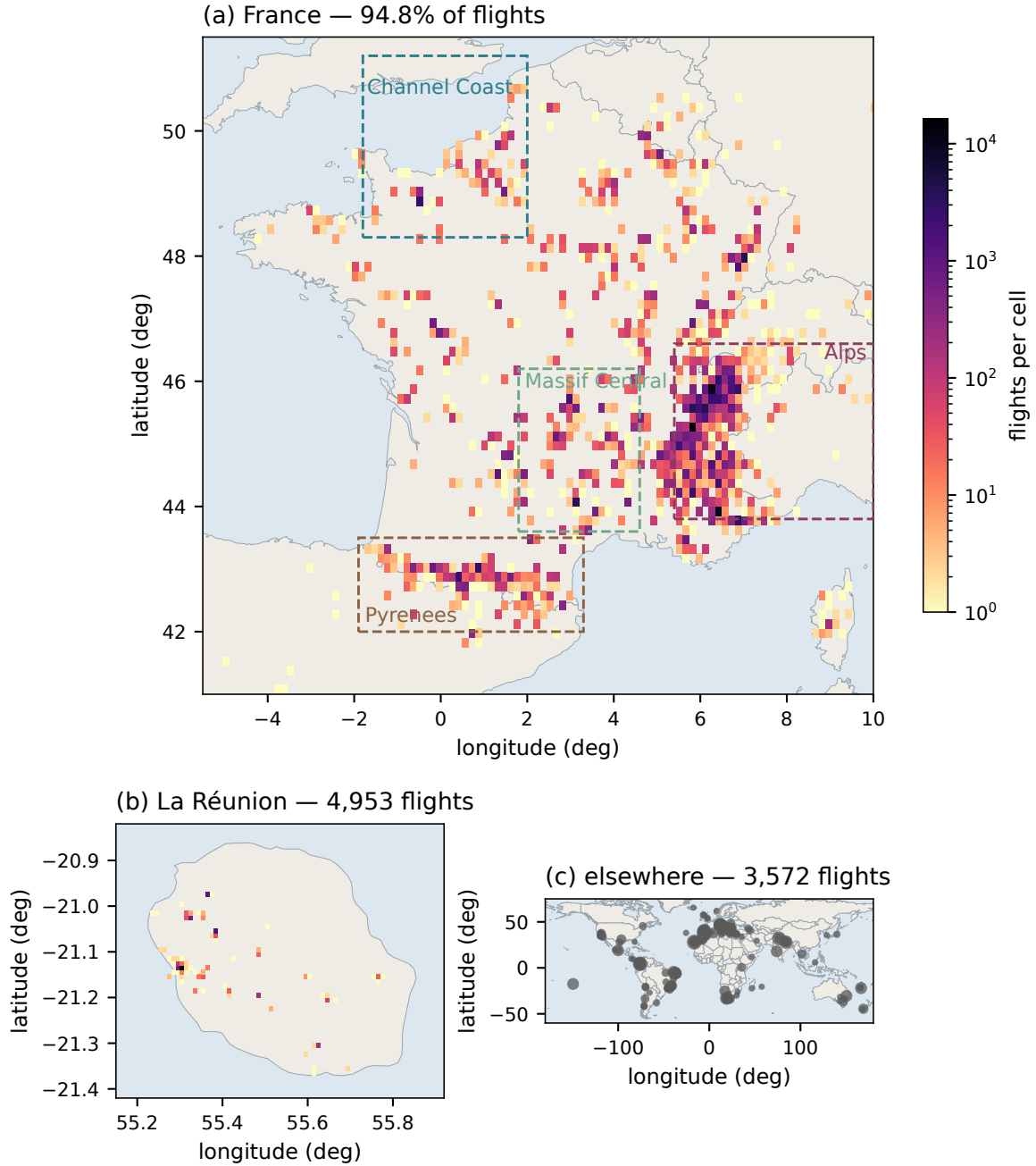


Figure 2.23. Take-off sites of the retained ensemble, on three frames, pooling both disciplines throughout. (a) Metropolitan France, density on a 0.15° cell; the dashed boxes delimit the orographic groups. (b) La Réunion, on a 0.01° cell, the one overseas department with a substantial share. (c) Everything else, one marker per 0.5° cell with the area set by its count.

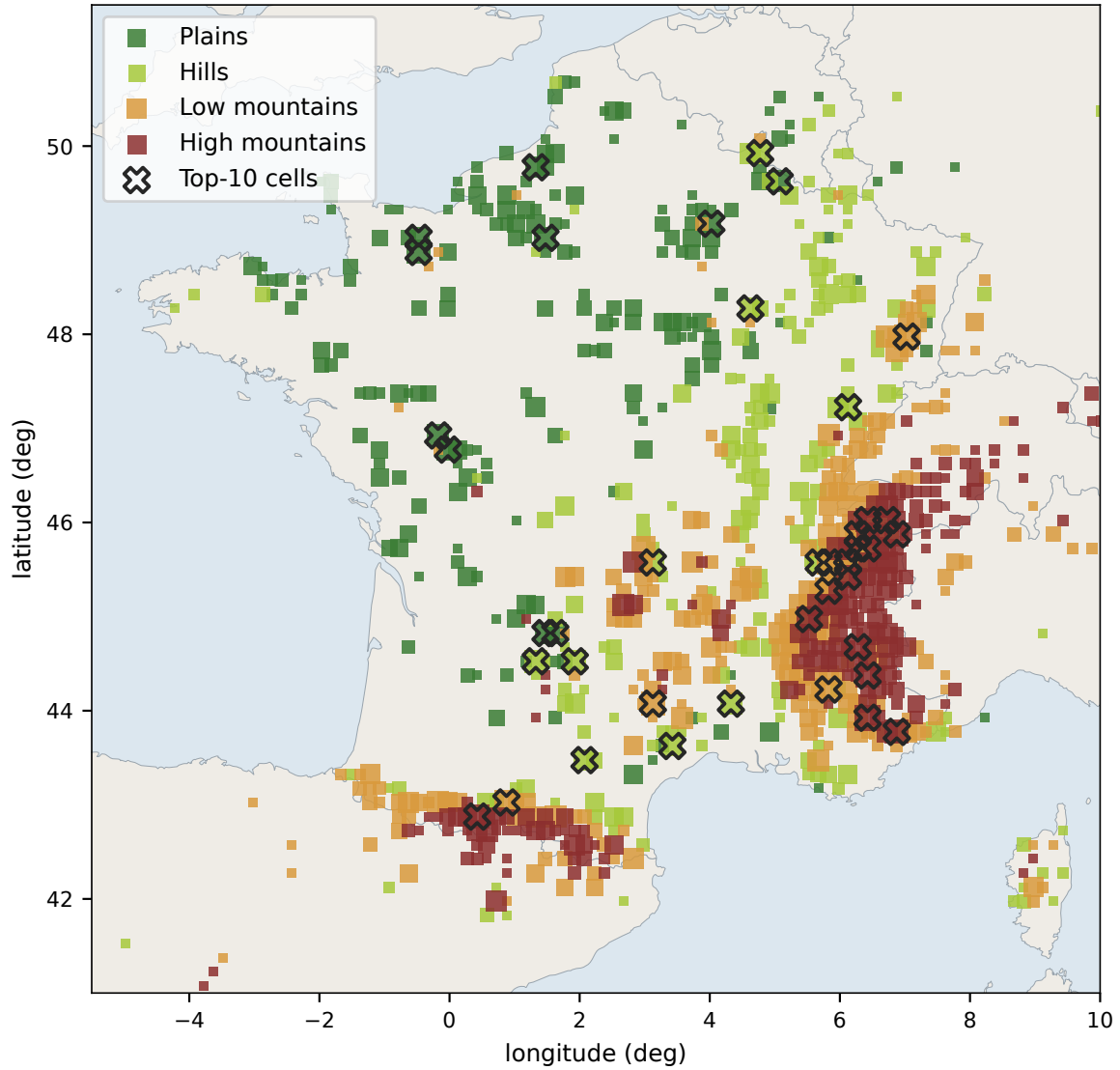


Figure 2.24. First recorded positions of the retained ensemble (both disciplines pooled), grouped by logged altitude using the elevation bands of [19]. These are approximate launch-altitude groups, not measured terrain classes. One marker per occupied 0.15° cell, coloured by its majority class and sized by the logarithm of its flight count; crosses mark the 10 busiest cells of each class.

2.10 Trajectories for the transport analysis

The retained trajectories have a uniform grid within each segment, local horizontal coordinates and explicit reconstruction flags. Phase features can exclude flagged vertical support, although plausible unflagged defects remain possible. Geography, cadence and flight duration also vary across the archive.

Any transport observable computed from these trajectories must specify its weights, lag support and contributing population.

Chapter 3

Horizontal transport and displacement scaling

We measure the mean-square displacement (MSD), its dependence on circuit type and initial altitude, and the scaling of the displacement distribution. Paragliders and hang gliders are analysed separately.

3.1 Mean-square displacement

3.1.1 Estimator and temporal resolution

Let $\mathbf{x}_f(t) = (E_f(t), N_f(t))$ be the cleaned horizontal trajectory of flight f , and let s index its retained continuous segments. Within each segment, define $\Delta \mathbf{x}_{fi}(\tau) = \mathbf{x}_f(t_i + \tau) - \mathbf{x}_f(t_i)$ and $R_{fi}(\tau) = |\Delta \mathbf{x}_{fi}(\tau)|$, defined whenever t_i and $t_i + \tau$ lie in the same segment s ; no increment crosses a segment boundary. Let i index grid origins across the flight, and define the admissible index set

$$\mathcal{I}_f(\tau) = \{i : t_i \text{ and } t_i + \tau \text{ lie in the same retained segment of flight } f\}.$$

Let $n_{f,s}(\tau)$ count the admissible indices in segment s , so that $n_f(\tau) = |\mathcal{I}_f(\tau)| = \sum_s n_{f,s}(\tau)$ is their total across the flight. Set

$$\mathcal{C}(\tau) = \{f : n_f(\tau) > 0\}, \quad N(\tau) = |\mathcal{C}(\tau)|.$$

Let $m_f(\tau)$ be the time-averaged MSD of flight f , and $M_2(\tau)$ the mean of $m_f(\tau)$ over $\mathcal{C}(\tau)$. They are defined by

$$m_f(\tau) = \frac{1}{n_f(\tau)} \sum_{i \in \mathcal{I}_f(\tau)} R_{fi}(\tau)^2, \quad M_2(\tau) = \frac{1}{N(\tau)} \sum_{f \in \mathcal{C}(\tau)} m_f(\tau). \quad (3.1)$$

We first average the squared displacements within each flight, pooling the valid increments from all its retained segments. We then average these flight-level MSDs, giving each contributing flight the same weight $1/N(\tau)$, regardless of its number of increments or segments. This is the *equal-flight* average. Two alternatives were rejected.

- *Equal-increment average*: pooling all increments directly, without first averaging within flights,

$$M_2^{\text{incr}}(\tau) = \frac{1}{\sum_{g \in \mathcal{C}(\tau)} n_g(\tau)} \sum_{f \in \mathcal{C}(\tau)} \sum_{i \in \mathcal{I}_f(\tau)} R_{fi}(\tau)^2 = \sum_{f \in \mathcal{C}(\tau)} \frac{n_f(\tau)}{\sum_{g \in \mathcal{C}(\tau)} n_g(\tau)} m_f(\tau),$$

would give flight f weight $n_f(\tau)/\sum_{g \in \mathcal{C}(\tau)} n_g(\tau)$ in the population TA-MSD: longer flights, which accumulate more valid increments, would dominate the pooled statistic, which would then track long flights rather than represent the population evenly.

- *Equal-segment average*: first computing a TA-MSD for each segment, then averaging over every segment in $\mathcal{C}(\tau)$,

$$m_{f,s}(\tau) = \frac{1}{n_{f,s}(\tau)} \sum_{i \in s} R_{fi}(\tau)^2, \quad M_2^{\text{seg}}(\tau) = \frac{1}{S(\tau)} \sum_{f \in \mathcal{C}(\tau)} \sum_s m_{f,s}(\tau),$$

with $S(\tau)$ the total number of segments with $n_{f,s}(\tau) > 0$ across $\mathcal{C}(\tau)$, would give equal weight to every segment regardless of its length: flights broken into many short segments, typically those with lower signal quality, would then be overrepresented.

For $10 \leq \tau \leq 10000$ s we subsample each segment to a 10 s grid, starting at its first fix. This spacing sets the step between the origins t_i from which the MSD at lag τ is computed within a segment, and it gives every flight the same physical lags to compare. Records with native cadence above 10 s are excluded. Missing positions are linearly interpolated within segments: with an 8 s cadence, for example, the 10 s position lies between the fixes at 8 and 16 s. Table 3.1 counts how often the grid requires this. The 33 evaluated lags are multiples of 10 s, approximately logarithmically spaced and including 100 and 1000 s. Figure markers show the evaluations; lines connect them.

The spacing itself is a trade-off. Finer sampling would raise the computational cost sharply without benefit once it resolves timescales below any actual change in the dynamics; coarser sampling would fail to resolve the fastest manoeuvres, which occur during the search and thermalling phases, where speed, acceleration and heading vary faster than during gliding. Vilpellet et al. [2] report that a single revolution while thermalling takes just under 30 s, so 10 s samples each turn about three times, resolving the fastest dynamics in the archive without oversampling the slower gliding phase.

Discipline	Grid fixes	Coincident	Interpolated	Fraction
Paragliders	85,915,677	81,888,079	4,027,598	4.69%
Hang gliders	4,099,322	3,639,355	459,967	11.22%

Table 3.1. Common-grid positions in the main fixed cohort. Coincidence is checked against actual cleaned timestamps. The last two columns count interpolation added by the 10 s grid, separately from reconstruction during cleaning.

Among the coincident grid positions, 355,170 paraglider fixes and 9,648 hang-glider fixes already carry the cleaning interpolation flag. The short-lag detail at 1–9 s in the general TA-MSD uses only native segments that resolve the requested lag exactly.

3.1.2 General transport and uncertainty

Figure 3.1 compares the TA-MSD, averaged over within-segment origins, with squared distance from the flight’s single post-trim origin (Section 2.7.3). For the latter, t is elapsed time from that origin. Gaps leave missing support without resetting either the clock or the reference position.

A flight contributes at t if a retained segment covers that time and its native sampling interval satisfies $dt \leq t$. Linear interpolation misses any curvature between consecutive fixes and its position error depends on motion within the local interval dt , so it doesn’t improve

as elapsed time t increases. What can decrease is the error *relative to the displacement from the origin*. At a characteristic speed v , unresolved motion has a spatial scale $v dt$, whereas approximately ballistic displacement grows as vt . Their ratio dt/t explains the cutoff: for $t < dt$, interpolation error can be comparable to the displacement; as t grows beyond dt , its relative contribution can decrease.

At each t we report the mean, median and 5–95% range across available flights. The contributing population changes as t grows, since flights whose retained span is shorter than t drop out, and the TA-MSD loses flights the same way as τ grows; Table 3.2 counts the flights that remain. These curves therefore describe a changing mixture of sites, days and flight conditions, which is why the controlled comparisons of Section 3.1.3 instead fix the population once for all lags.

The available-population fits give $H = 0.869[0.864, 0.874]$ for paragliders and $H = 0.841[0.816, 0.855]$ for hang gliders. These summarise growth with changing flight support. The hang-glider fit reaches a tail supported by only 7 groups; its interval uses 998 replicates with complete lag support.

Discipline	Lag (s)	TA-MSD flights	Origin-distance flights
Paragliders	10	155,085	152,771
Paragliders	100	155,044	155,117
Paragliders	1000	154,482	155,250
Paragliders	10000	46,273	72,160
Paragliders	32690	27	1,237
Hang gliders	10	6,060	5,330
Hang gliders	100	6,060	6,059
Hang gliders	1000	6,044	6,070
Hang gliders	10000	2,326	3,480
Hang gliders	43200	1	3

Table 3.2. Number of contributing flights in the general curves. The last row for each discipline is the largest evaluated lag with TA-MSD support. The origin-distance curve uses elapsed time; the TA-MSD requires a within-segment lag no larger than $0.8T_s$. Their support sets therefore differ. Early origin-distance support can increase when elapsed time exceeds an initial excluded interval or resolves a coarse cadence.

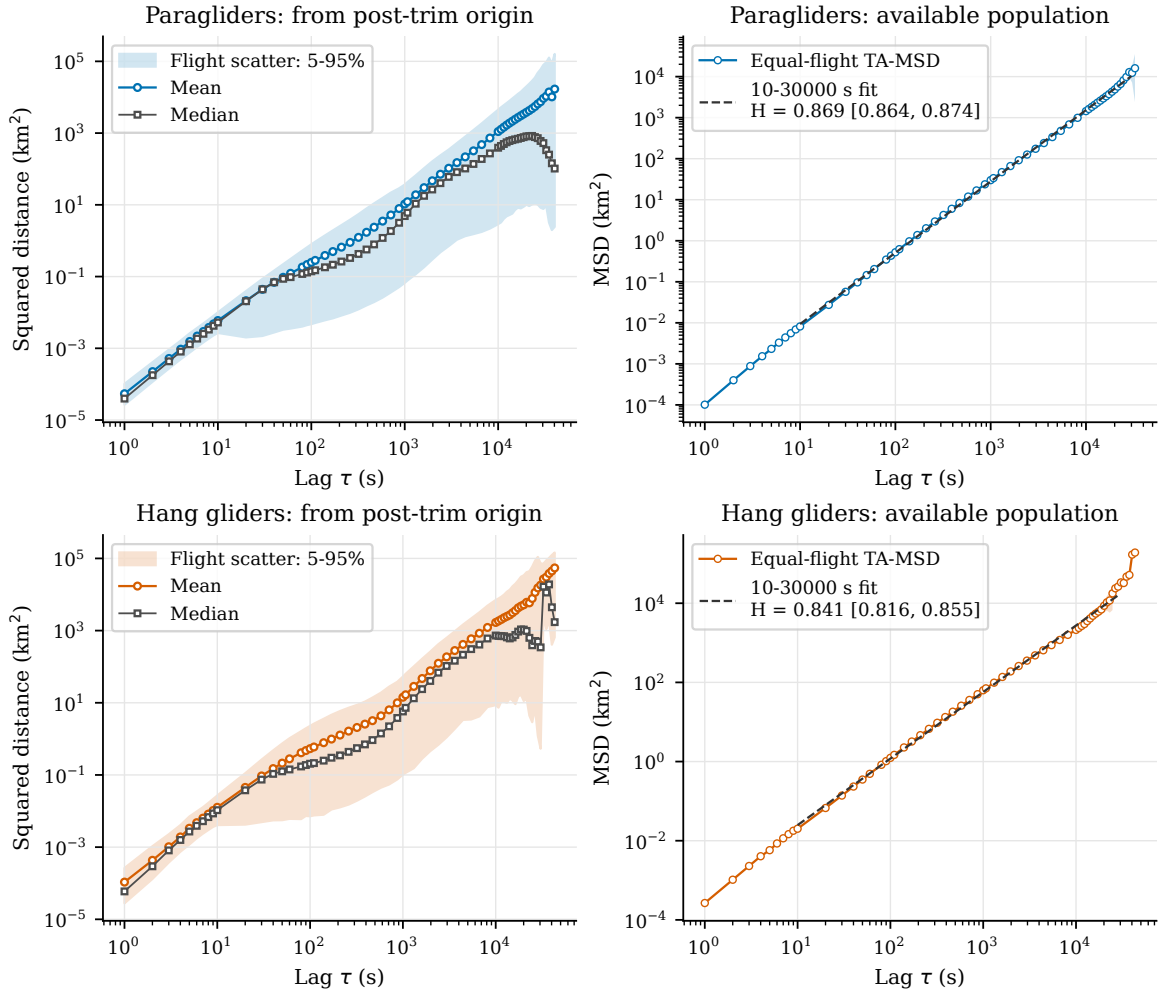


Figure 3.1. General horizontal transport. Left: squared distance from the post-trim origin, with equal-flight mean, median and the empirical 5–95% flight range. Right: equal-flight TA-MSD over the population available at each lag, with pointwise 90% cluster-bootstrap intervals and a log–log fit over 10–30000 s (evaluated lags: 10–28440 s). Native short-lag support differs from common-grid support.

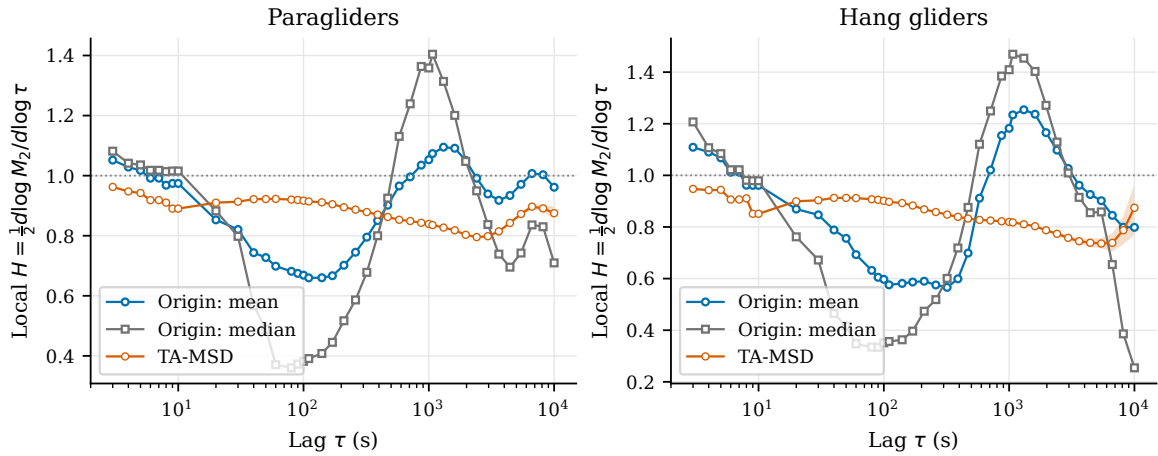


Figure 3.2. Local slopes of the general curves. The origin-distance mean and median have no confidence band here; the TA-MSD band is obtained by refitting bootstrap curves. The dotted reference is ballistic growth, $H = 1$.

For a second-moment curve the local growth exponent is

$$H_{\text{loc}}(\tau) = \frac{1}{2} \frac{d \log M_2(\tau)}{d \log \tau}. \quad (3.2)$$

We estimate this derivative by a local log–log least-squares slope, divided by two, within ± 0.25 decades of each lag and using at least three points. In the general curves, a sparse interior window expands to the three nearest log-lags: the estimate at 20 s uses the values at 10, 20 and 30 s. For a single exponent over a specified interval, write $x_k = \log_{10}(\tau_k/1 \text{ s})$. The equal-weight least-squares slope of a dimensionless logarithmic ordinate z_k is

$$\mathcal{B}[z] = \frac{\sum_k (x_k - \bar{x}) z_k}{\sum_k (x_k - \bar{x})^2}, \quad H = \frac{1}{2} \mathcal{B} \left[\log_{10} \frac{M_2}{1 \text{ m}^2} \right]. \quad (3.3)$$

We fit the controlled analysis over 10–10000 s and refit every bootstrap curve to propagate dependence between lags. The RMS logarithmic residual, in dex, measures the departure from a single power law. Here H denotes an effective growth exponent; its identification with a process’s self-similarity exponent is tested in Section 3.4. Figure 3.2 shows the resulting local slopes of both general curves.

The origin-distance means are approximately ballistic at the earliest resolved lags. Their local slopes reach minima at 110 s ($H_{\text{loc}} = 0.66$) for paragliders and 320 s ($H_{\text{loc}} = 0.57$) for hang gliders within 20–2000 s, then recover towards $H_{\text{loc}} \simeq 1$ around 600–700 s. The reduced growth after the initial ballistic regime is consistent with searching for lift and climbing in a thermal, followed by renewed gliding. Under this interpretation, the minimum marks a rough transition time towards completion of the initial search-and-climb phase. It is distinct from the later recovery towards $H_{\text{loc}} = 1$ and does not estimate a mean duration measured on individual flights. The mean lies above the median, showing the influence of large displacements; the pointwise percentiles do not track the same flights through time.

Uncertainty. The shading around the origin-distance curves of Figure 3.1 is the empirical 5–95% range across flights, a measure of scatter among flights. Every other shaded band in this chapter is a pointwise 90% bootstrap interval. Flights are grouped by catalogue take-off site (the one declared in the FFVL catalogue metadata (Section 2.3, Table 2.1)) and calendar date, retaining dependence within each site–day. Missing keys give singleton groups. From the eligible archive’s G site–day groups, each of $B = 1000$ replicates draws G groups with replacement, then restricts them to the population being analysed. If b_g is the number of times group g is drawn, the weights are

$$\begin{aligned} N_b(\tau) &= \sum_{f \in \mathcal{C}(\tau)} b_{g(f)}, & w_{fi}^{(b)}(\tau) &= \frac{b_{g(f)}}{N_b(\tau) n_f(\tau)}, \\ M_2^{(b)}(\tau) &= \sum_{f \in \mathcal{C}(\tau)} \sum_{i \in \mathcal{I}_f(\tau)} w_{fi}^{(b)}(\tau) R_{fi}(\tau)^2. \end{aligned} \quad (3.4)$$

The same draws are used across lags, observables and cohort comparisons. Quantiles, ratios and fits are recomputed in each replicate. For an estimate $\hat{\theta}$, the 5th and 95th percentiles of the bootstrap replicates $\{\hat{\theta}^{(b)}\}_{b=1}^{1000}$ give a pointwise 90% confidence interval

$$I_{90}(\theta) = \left[\text{perc}_5 \{ \hat{\theta}^{(b)} \}_{b=1}^{1000}, \text{perc}_{95} \{ \hat{\theta}^{(b)} \}_{b=1}^{1000} \right]. \quad (3.5)$$

Percentiles are linearly interpolated between ordered replicates. Furthermore, at a given lag, a replicate whose resampled flights provide no support there yields no estimate and is dropped from that lag’s percentiles.

Cluster bootstrap requires many approximately independent groups and sufficiently stable statistics [21]; flights within a group need not be independent. Table 3.3 describes their spatial and temporal separation. Nearby sites and successive days can share weather, so these distances do not verify independence; the paragraph below checks MSD and fitted-exponent errors using geographic, altitude and seasonal groups. Residual dependence may make the bands too narrow; upload selection also limits inference to the population represented by the archive. General TA-MSD bands are omitted below 20 contributing groups, an operational support threshold rather than a guarantee of coverage.

The statistical validity of these intervals is therefore not established at this stage. Independence between site–day groups is assumed and has never been verified, and the size of the groups makes the assumption doubtful: their median size is one flight, the 95th percentile is five flights for paragliders and three for hang gliders, and even the largest group holds 47 and 13 flights (Table 3.3). A partition this fine absorbs little of the dependence it was introduced to capture. Flights that share one weather episode across neighbouring sites, or across consecutive days at the same site, fall into different groups and are resampled as if they carried independent information. The groups are then probably not independent, and Eq. (3.5) probably understates the true uncertainty, so every band reported in this chapter should be read as a lower bound on the spread. The open question is whether bootstrap groups coarse enough to be approximately independent can be built at all. The paragraph below is a first attempt at it and has not answered it yet, which is why it is typeset in red.

Quantity	Paragliders	Hang gliders
Catalogue sites	3,424	284
Calendar dates	3,642	1,017
Flights per group: median / P95 / max	1 / 5 / 47	1 / 3 / 13
Largest group (% of flights)	0.10	0.56
Nearest same-day site (km)	11.2 [0.04, 173]	24.7 [0.04, 338]
Spatial comparisons	24,509	1,147
Prior observed date at same site: gap (days)	19 [1, 429]	18 [1, 382]
Temporal comparisons	22,258	1,484
Flights with missing site/date keys	0	0

Table 3.3. Site–day diagnostics in the main cohort. Distances refer to the nearest other catalogue site represented on the same date; time gaps refer to the previous observed date at the same site. Distances and time gaps are reported as median [10th, 90th percentile]. Cohort group counts are in Table 3.5.

Reliability of bootstrap uncertainty estimates (work in progress, not working). This paragraph is a working draft and no conclusion in it is settled. The grouping tried here does not yet produce approximately independent bootstrap groups, so its intervals validate nothing and its numbers are reported only to record where the attempt stands.

We repeat the uncertainty check with groups defined by a fixed geographic cell, launch-altitude class and month of the year, pooling all years. Thus July flights from different years share a group when their initial positions lie in the same cell and altitude class. The grid is independent of the observed sites: each initial position belongs to one half-open 50×50 km square in its WGS84 UTM zone, with sides measured in projected metres. Cells are clipped at zone and hemisphere boundaries. The four altitude classes (plains, hills, low and high mountains) retain the existing thresholds of 300, 800 and 1500 m; they are a proxy for launch orography. This construction groups geographical and seasonal context, not necessarily a common weather episode.

All main-cohort flights and their equal-flight weights are retained. We draw 1000 bootstrap

replicates, paired across lags, and repeat with two seeds. The archived site-day bootstrap provides the reference. Table 3.4 compares group composition. Singleton groups remain, but contain only 1.4% of paraglider flights and 5.0% of hang-glider flights. The standard error of H is 3.97–4.05 and 2.00–2.05 times the site-day value, respectively; point estimates are unchanged.

	Site-day	Grid	Median size	Singletons (%)	Max. share (%)
Paragliders	25682	1901	3	33.5	4.4
Hang gliders	1768	267	2	43.4	13.4

Table 3.4. Bootstrap groups of the main cohort before and after cell-altitude-month grouping. “Before” are the site-day groups that resample every other interval in this chapter; “after” are the alternative groups introduced in this check, used only here. Median size is the number of flights per new group. Singletons are expressed as a percentage of new groups; maximum share is the largest group’s percentage of cohort flights. All years are pooled within each month.

We test sensitivity using 10, 25 and 50 km cells, shifting each grid by half a cell along either or both axes. Separately, we merge two or three adjacent months and shift the seasonal boundaries, retaining 50 km cells and the altitude classes (Fig. 3.3). These are alternative resampling partitions, not exclusions of flights.

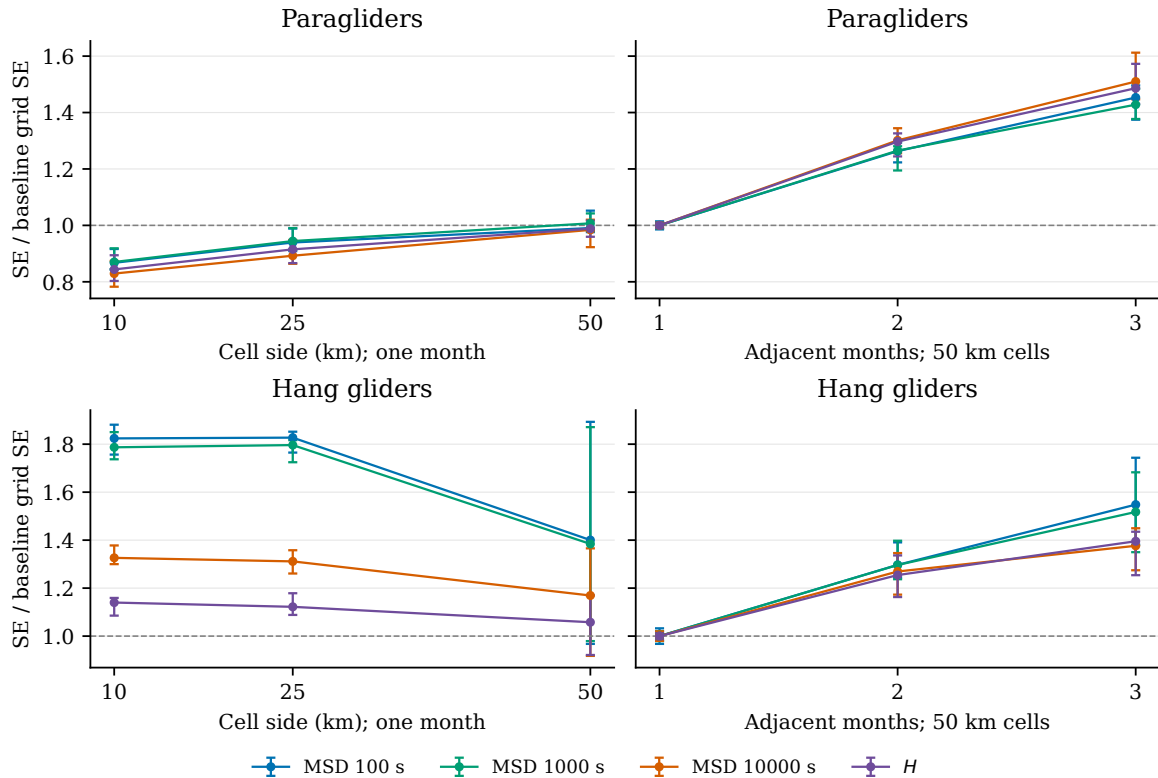


Figure 3.3. Standard errors relative to the unshifted 50 km, one-month grid baseline. Left: cell size and boundary shifts. Right: adjacent-month unions with seasonal boundary shifts. Markers are medians and bars span minima and maxima across partitions and two seeds; they are not confidence intervals. Altitude classes remain separate, and all years are pooled.

The uncertainty has not stabilised across these choices. Merging three months multiplies the median standard error of H by 1.49 for paragliders and 1.40 for hang gliders. For the latter, shifting the 50 km grid gives MSD errors at 1000 s ranging from 0.98–1.87 times the unshifted

value. Larger groups therefore do not by themselves validate independence or interval coverage. Neighbouring cells, months and altitude classes can still share conditions, while persistent geographical differences can also affect the comparison [21]. The new intervals are conditional on this grouping; the archived site-day bands elsewhere remain the reference and are not replaced by this MSD/ H check.

3.1.3 Fixed population and the effect of duration selection

The general TA-MSD of Sec. 3.1.2 averages over whichever flights support each lag, so its population shrinks as τ grows: short flights and short segments drop out one after another. The mix of circuit types and launch sites that remains then changes with τ , and a change in the slope of $M_2(\tau)$ cannot be told apart from a change in the dynamics. Every statistic of the motion in the rest of this chapter is therefore measured on one population that does not depend on τ . Such a population is limited by its shortest member: the longest lag it supports is fixed by how long its segments are. Selecting only long segments gives more lags, and loses flights. This subsection asks whether that selection changes the measured H enough to matter, and chooses the cohort accordingly.

Cohorts. We select the segments once, using

$$\tau_{\max} \leq 0.8T_s, \quad \mathcal{C}_{\tau_{\max}} = \{f : \exists s \in f, T_s \geq \tau_{\max}/0.8\}, \quad (3.6)$$

where T_s is the segment's elapsed duration on the 10 s grid. The condition $\tau_{\max}/T_s \leq 0.8$ leaves $T_s - \tau_{\max} \geq 0.2T_s$, one fifth of the segment, for the origins at the largest lag. The threshold 0.8 fixes how many origins remain there. For the main cohort, $\mathcal{C}_{10000}$, segments last at least 12500 s, so each supplies at least 251 origins on the 10 s grid at $\tau = 10000$ s; requiring only $T_s \geq \tau_{\max}$ would leave one. If a flight has segments of 14000 and 3000 s, only the first contributes, even at $\tau = 10$ s. Flights and selected segments therefore remain the same at every lag.

Weights. For each cohort, $\mathcal{I}_f(\tau)$ and $n_f(\tau)$ refer only to its selected segments. Each flight of $\mathcal{C}_{10000}$ carries the same weight $1/N$ at every lag, because the cohort does not change with τ . The number of admissible origins $n_f(\tau)$ still decreases with lag, so a flight's weight is divided evenly among them. The weight of origin i of flight f at lag τ is

$$w_{fi}(\tau) = \frac{1}{Nn_f(\tau)}, \quad \sum_{i \in \mathcal{I}_f(\tau)} w_{fi}(\tau) = \frac{1}{N}, \quad N = |\mathcal{C}_{10000}|. \quad (3.7)$$

The origins of one flight therefore sum to that flight's share $1/N$, and the origins of all flights sum to one. This is the equal-flight average of Eq. (3.1), applied to a population that does not change with τ : it gives $M_2(\tau) = \sum_f \sum_i w_{fi}(\tau) R_{fi}(\tau)^2$, and the same weights define the mean $\langle A \rangle_\tau = \sum_f \sum_i w_{fi}(\tau) A_{fi}$ of any per-origin quantity A_{fi} , used from Eq. (3.16) onwards.

Effect of the selection on H . We assess duration selection with the nested cohorts $\mathcal{C}_{100}$ and $\mathcal{C}_{1000}$. They contain the main cohort's segments, additional shorter segments of its flights, and additional flights. Selecting long segments also changes who is in the sample. The declared circuit class, open-distance flights against closed routes such as triangles and out-and-return flights (Sec. 3.2), shifts with the cohort: the share of closed routes rises from 56.6% in $\mathcal{C}_{100}$ to 61.0% in $\mathcal{C}_{10000}$ for paragliders, and from 64.0% to 76.5% for hang gliders.

We compare the fitted H of two cohorts over the same lag interval, and the comparison is paired. Every bootstrap replicate draws the same site-day groups (Eq. (3.4)) for all cohorts,

Discipline	$\tau_{\max}$ (s)	Flights	Segments	Groups
Paragliders	100	155,044	188,312	82,650
Paragliders	1000	154,482	167,152	82,390
Paragliders	10000	46,273	46,367	25,682
Hang gliders	100	6,060	7,515	4,264
Hang gliders	1000	6,044	6,610	4,253
Hang gliders	10000	2,326	2,327	1,768

Table 3.5. Nested duration cohorts and their site–day groups, the resampling units of Sec. 3.1.2. Minimum segment durations are 125, 1250 and 12500 s, rounded upward when required by the 10 s grid. Existing cleaning and segment-admission conditions also apply. The table gives only counts. The full membership of each cohort, every flight and the start and stop of each of its segments, is saved in a manifest file with a checksum, so that later analyses can reuse exactly the same population.

and the difference between two exponents is taken replicate by replicate. Sampling noise that the cohorts share cancels in the difference, so what remains isolates the effect of the selection (Figure 3.4, Table 3.6). The fit covers 10–100 s for all three cohorts, and 10–1000 s for $\mathcal{C}_{1000}$ and $\mathcal{C}_{10000}$, because the broader cohorts have no support at longer lags.

Discipline	Fit range (s)	Contrast	ΔH [5–95%]
Paragliders	10–100	$H_{\mathcal{C}_{100}} - H_{\mathcal{C}_{1000}}$	-0.00006 [-0.00009, -0.00004]
Paragliders	10–100	$H_{\mathcal{C}_{100}} - H_{\mathcal{C}_{10000}}$	-0.01687 [-0.01712, -0.01660]
Paragliders	10–100	$H_{\mathcal{C}_{1000}} - H_{\mathcal{C}_{10000}}$	-0.01680 [-0.01705, -0.01654]
Paragliders	10–1000	$H_{\mathcal{C}_{1000}} - H_{\mathcal{C}_{10000}}$	-0.01703 [-0.01733, -0.01674]
Hang gliders	10–100	$H_{\mathcal{C}_{100}} - H_{\mathcal{C}_{1000}}$	-0.00002 [-0.00013, 0.00011]
Hang gliders	10–100	$H_{\mathcal{C}_{100}} - H_{\mathcal{C}_{10000}}$	-0.01646 [-0.01748, -0.01552]
Hang gliders	10–100	$H_{\mathcal{C}_{1000}} - H_{\mathcal{C}_{10000}}$	-0.01644 [-0.01746, -0.01550]
Hang gliders	10–1000	$H_{\mathcal{C}_{1000}} - H_{\mathcal{C}_{10000}}$	-0.01771 [-0.01879, -0.01668]

Table 3.6. Paired differences of the cohort exponents in Fig. 3.4. Each comparison uses identical lags and bootstrap draws. A 90% interval excluding zero resolves a difference under the adopted resampling; inclusion of zero alone would not establish absence of selection bias.

Result. On 10–100 s, $\mathcal{C}_{100}$ and $\mathcal{C}_{1000}$ differ in H by less than 10^{-4} in both disciplines. The paired interval resolves this tiny difference for paragliders; for hang gliders it includes zero. The two cohorts share 99.6% (paragliders) and 99.7% (hang gliders) of their flights, so in each replicate their exponents move together and their difference has a very narrow interval. That is why a difference this small is resolved at all.

The main cohort behaves differently. Its effective exponent is larger than that of every broader cohort, in both fit intervals and both disciplines, and every one of these paired differences excludes zero. The difference between the main cohort and a broader one lies between 0.0164 and 0.0177 in H , which is 1.8–2.0% of the exponent itself.

We accept this bias and adopt $\mathcal{C}_{10000}$ for every analysis below that requires a fixed population. A shift of about 2% in H is small beside what the cohort provides: a population that stays the same over three decades of lag, 10–10000 s, against two for $\mathcal{C}_{1000}$ and one for $\mathcal{C}_{100}$. Two limits apply. The shift can be measured only where the broader cohorts have support, on 10–100 s and 10–1000 s. And every exponent reported below is conditional on flights with an uninterrupted segment of at least 12500 s.

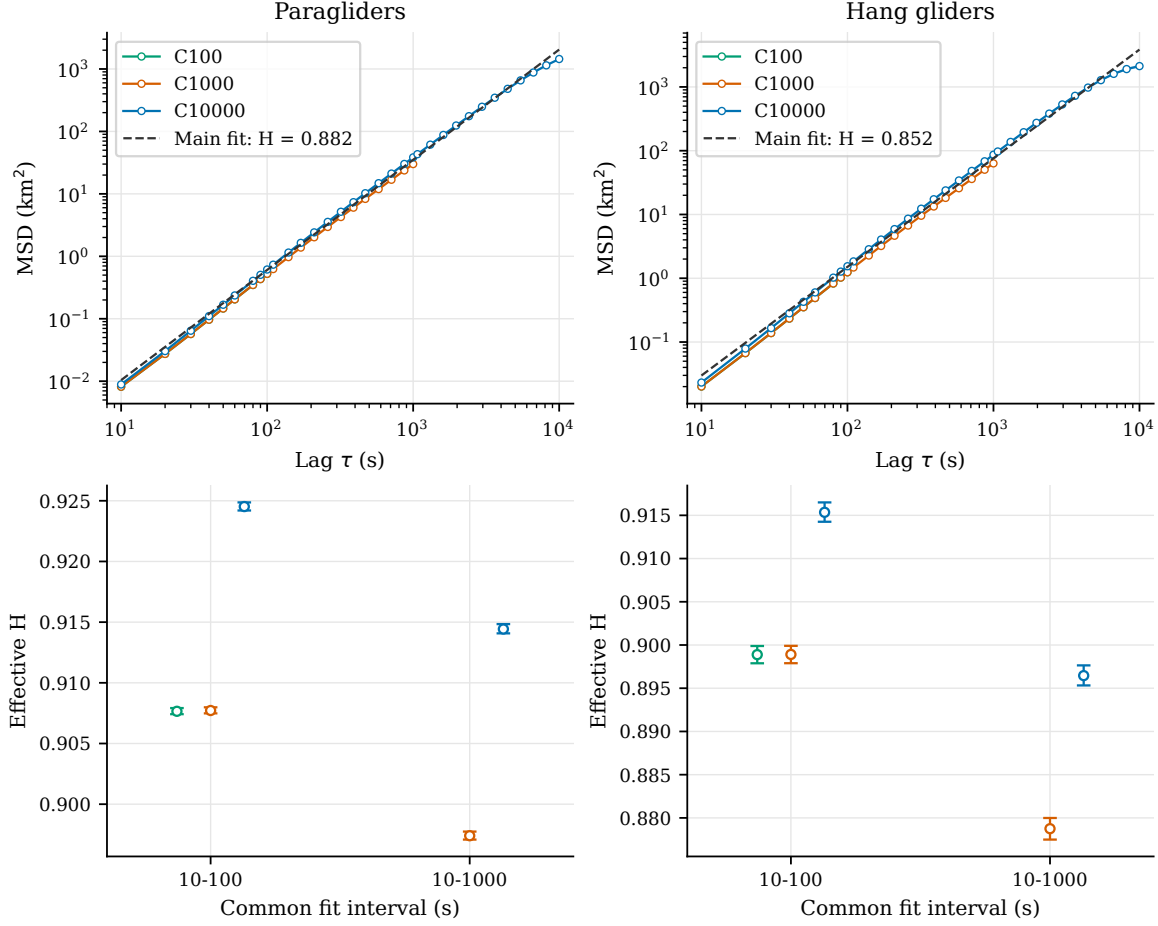


Figure 3.4. Effect of duration selection. Top: MSD in the three nested cohorts; dashed lines fit the main cohort over 10–10000 s. Bottom: effective H with 90% intervals, fitted over 10–100 s for all three cohorts and 10–1000 s for the two longer cohorts. Fits within each comparison use identical lag values.

3.2 Transport by circuit type and initial altitude

The catalogue’s declared circuit class separates open distance flights from closed routes such as triangles and out-and-return flights. We cross this distinction with initial cleaned GNSS altitude: Plains ($z_0 < 300$ m), Hills ($300 \leq z_0 < 800$ m), Low mountains ($800 \leq z_0 < 1500$ m), and High mountains ($z_0 \geq 1500$ m). These are launch-altitude classes; they do not classify the terrain crossed later. This comparison uses paragliders and the same $\mathcal{C}_{10000}$ support rule in all eight cells. Figure 3.5 shows the eight MSD curves; counts and fitted exponents are given in Table 3.7. Eligible flights: 67,003 open (43.2%), 87,807 closed (56.6%), and 275 unclassified.

Main-cohort flights: 17,961 open (38.8%), 28,234 closed (61.0%), and 78 unclassified.

For open routes, Plains and Hills have larger H than either mountain class. Thus the distinction involves the rate of growth as well as the displacement reached at 10000 s. Closed routes have smaller H than open routes in every altitude class. The paired contrasts $H_{\text{open}} - H_{\text{closed}}$ range from 0.023 to 0.129; all four 90% intervals exclude zero. The two lowland classes carry the largest contrast, close to 0.129, and it falls to 0.023 in High mountains. The closed-route exponents also differ between altitude classes: the present intervals do not support a common value across them. Their ordering reverses the open-route one. Paired lowland-minus-mountain contrasts, each lowland class against each mountain class within one circuit, are positive for open routes, between 0.051 and 0.064 in H , and negative for closed

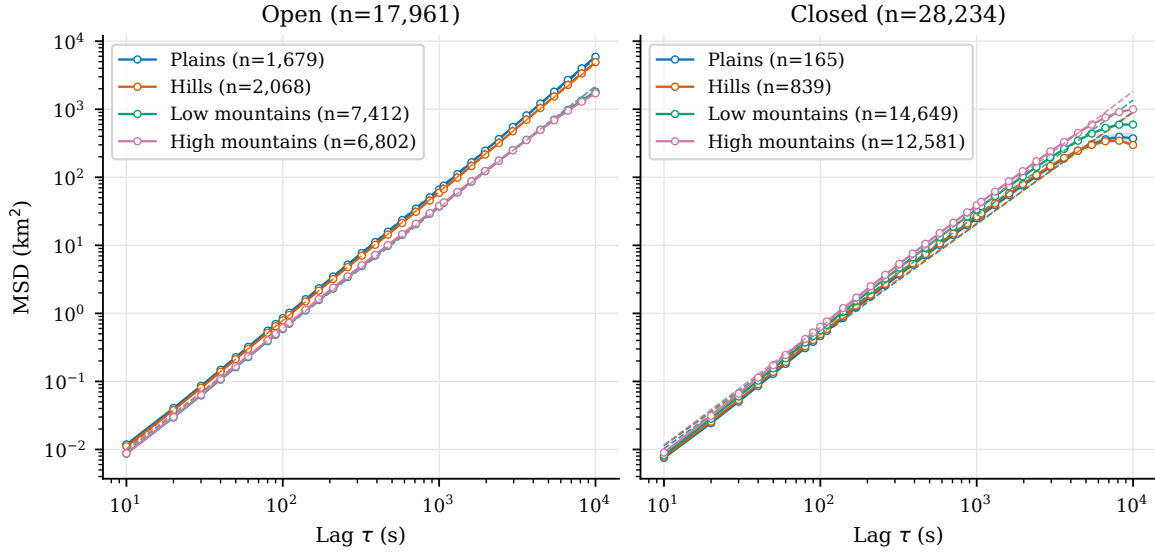


Figure 3.5. Paraglider transport by declared circuit and initial altitude, with fixed flights and segments throughout 10–10000 s. Markers give MSD estimates, shaded bands their pointwise 90% intervals, and dashed lines the global log–log fits. The identical axes permit comparison of displacement amplitude and growth rate.

Circuit	Initial altitude	N	G	H [5–95%]	$M_2(10^4)$ (km ²)
Open	Plains	1,679	1,140	0.953 [0.952, 0.955]	5900
Open	Hills	2,068	1,634	0.944 [0.943, 0.946]	4920
Open	Low mountains	7,412	5,255	0.893 [0.892, 0.895]	1814
Open	High mountains	6,802	4,794	0.889 [0.888, 0.890]	1710
Closed	Plains	165	157	0.825 [0.816, 0.834]	372
Closed	Hills	839	756	0.816 [0.812, 0.819]	299
Closed	Low mountains	14,649	7,954	0.846 [0.845, 0.847]	596
Closed	High mountains	12,581	7,406	0.866 [0.865, 0.867]	1001

Table 3.7. Main-cohort flights N , site-day groups G , global effective exponents and displacement amplitude at 10000 s. Intervals come from the coupled cluster bootstrap. Even the smallest cell retains 165 flights from 157 groups.

routes, with magnitudes between 0.021 and 0.051; all eight intervals exclude zero. Launch altitude therefore enters H with a sign that depends on the declared circuit.

One reading connects the two orderings through a single constraint, how closely the terrain lets a declared route be flown as planned. A closed route commits the pilot to a return: once about half the available time has elapsed, the remainder of the flight carries the glider back towards its start, and displacements measured over lags approaching the circuit’s duration lose the advance accumulated earlier. The suppression is strongest where the return can be flown most directly. Lowland terrain imposes no relief to circumvent and no ridge to follow, so a lowland circuit closes more faithfully, and its net horizontal transport is the more strongly suppressed. Figure 3.5 shows where this acts: over the last decade of lag every closed curve bends below its own fitted line, while the two lowland open curves stay on theirs out to 10000 s. The same freedom acts with the opposite sign on an open route, which asks only for distance: with no relief to follow, the track can stay straighter, and lowland open flights hold a nearly ballistic growth out to 10000 s while the mountain classes lose slope. The orographic setting thus reaches H through the flight plan, and the sign of its effect is set by what the plan asks for.

Three limits attach to this reading. The declared class does not guarantee exact closure of the trimmed trajectory, and the altitude classes describe the launch site rather than the terrain crossed later. The two lowland closed cells are the smallest in Table 3.7 and have the widest intervals, although their contrasts against the mountain cells still exclude zero. And the comparison is observational: it does not isolate orography from weather, route selection, pilots or equipment.

Within the fixed cohort the group fractions $p_g = N_g/N$ are constant. For an exhaustive partition, including unclassified flights,

$$M_2(\tau) = \sum_g p_g M_{2,g}(\tau), \quad H_{\text{loc}}(\tau) = \sum_g \frac{p_g M_{2,g}(\tau)}{M_2(\tau)} H_{\text{loc},g}(\tau). \quad (3.8)$$

The derivative slope weights each group by its contribution to the second moment. Equation (3.8) describes the main-cohort MSD; the plotted local regressions approximate its derivatives. Both circuit classes contribute substantially to this aggregate, with distinct growth rates.

Wind can contribute to both amplitude and apparent scaling. For an additive constant velocity $\mathbf{u}$, write $\Delta \mathbf{x} = \Delta \mathbf{y} + \mathbf{u}\tau$:

$$\langle |\Delta \mathbf{x}|^2 \rangle = \langle |\Delta \mathbf{y}|^2 \rangle + 2\tau \mathbf{u} \cdot \langle \Delta \mathbf{y} \rangle + |\mathbf{u}|^2 \tau^2. \quad (3.9)$$

Even a uniform constant drift can change the slope of this uncentred MSD through its ballistic term. A common deterministic shift cancels from the centred covariance; drifts differing between flights need not cancel from the mixture. A variable wind field can additionally modify velocity correlations. Wind is not separated quantitatively in this comparison.

3.3 Anisotropy and principal directions

Reserved for the analysis of anisotropy and principal directions. The existing PCA results remain in the experiments volume; they are not updated in this chapter.

3.4 Scaling of displacement distributions

The MSD measures one moment of the displacement law. To test whether the whole law rescales with one exponent, we use quantiles, a spectrum of moments and excess kurtosis. All use the main cohort's flights, segments and weights in Eq. (3.7).

3.4.1 Self-similarity and its observable consequences

A process $\mathbf{X}(t)$, with $\mathbf{X}(0) = 0$, is H -self-similar if, for some $H > 0$,

$$(\mathbf{X}(at_1), \dots, \mathbf{X}(at_m)) \stackrel{d}{=} a^H (\mathbf{X}(t_1), \dots, \mathbf{X}(t_m)) \quad (a > 0). \quad (3.10)$$

This identity must hold for every $m \geq 1$ and $t_1, \dots, t_m \geq 0$. Equality in distribution concerns positions at any collection of times, including their dependence [22]. Gaussianity and independence of increments are not required.

Two separate hypotheses stand between this definition and a measurement, and each does different work. Stationary increments make a pooled lag statistic informative about $\mathbf{X}$ itself: with $\mathbf{X}(0) = 0$ they give $\mathbf{X}(\tau) \stackrel{d}{=} \mathbf{X}(t + \tau) - \mathbf{X}(t)$, so displacements collected from many origins estimate the law of the position reached after an elapsed time τ . The archive supplies

no repeated releases from a common initial condition, which leaves this the only route from the increments we observe to the process we describe. Self-similarity then compresses the resulting family of laws: one exponent and one shape determine every lag, including elapsed times the cohort does not cover, and that exponent is the quantity stochastic transport models predict. Without it this chapter reports a separate propagator at each measured lag, with no relation between them. Testing Eq. (3.10) is therefore a test of whether one anomalous exponent describes these flights at all, and the sections below report necessary consequences of that identity rather than a proof of it.

To infer a lag-only scaling law, we additionally assume stationary increments: their joint laws must be unchanged by shifting the origin. Self-similarity alone does not impose this condition. With stationary increments, each amplitude $Y \in \{|\Delta E|, |\Delta N|, R\}$ obeys

$$Y(\tau) \stackrel{d}{=} (\tau/\tau_0)^H Y(\tau_0), \quad \begin{aligned} Q_{Y,p}(\tau) &= (\tau/\tau_0)^H Q_{Y,p}(\tau_0), \\ M_{q,Y}(\tau) &= (\tau/\tau_0)^{qH} M_{q,Y}(\tau_0). \end{aligned} \quad (3.11)$$

Here $Q_{Y,p}$ is a population quantile and $M_{q,Y}$ a finite moment of order q ; below, the same symbols denote their equal-flight estimates. Thus every quantile has the same exponent, quantile ratios are constant, and the moment spectrum is $\zeta_Y(q) = qH$. Signed-component excess kurtosis is also constant when the fourth moment exists and variance is nonzero; it need not vanish. These are necessary consequences for the lag distributions. A finite selection of such statistics cannot establish the joint-law identity in Eq. (3.10).

3.4.2 A common empirical distribution

With the weights in Eq. (3.7), the common empirical distribution is

$$\hat{F}_{Y,\tau}(y) = \frac{1}{N} \sum_{f=1}^N \frac{1}{n_f(\tau)} \sum_{i \in \mathcal{I}_f(\tau)} 1\{Y_{fi}(\tau) \leq y\}, \quad Q_{Y,p}(\tau) = \inf\{y : \hat{F}_{Y,\tau}(y) \geq p\}. \quad (3.12)$$

We average the within-flight fractions below y , then invert the resulting CDF. The quantile is the first observed amplitude at which the cumulative weight reaches p , without interpolation between ranks. This order of operations defines the quantile of the equal-flight mixture.

Absolute components measure amplitudes along east and north. For example, $\Pr(|\Delta E| \leq a) = \Pr(-a \leq \Delta E \leq a)$, so small absolute quantiles measure proximity to zero displacement. The radius $R = (\Delta E^2 + \Delta N^2)^{1/2}$ is invariant under rotation and measures horizontal displacement without assuming isotropy.

3.4.3 Quantiles, growth exponents and relative spread

We evaluate $p = 0.25, 0.50, 0.75, 0.90$ and fit each quantile over the same 33 lags. Using the slope operator in Eq. (3.3),

$$H_{Y,p} = \mathcal{B} \left[\log_{10} \frac{Q_{Y,p}}{1 \text{ m}} \right], \quad \mathcal{B} \left[\log_{10} \frac{Q_{Y,0.25}}{Q_{Y,0.90}} \right] = H_{Y,0.25} - H_{Y,0.90}. \quad (3.13)$$

No factor of two enters a length exponent. The ratio compares small and large amplitudes at each lag, while the fitted exponents summarise their growth across 10–10000 s. The equality follows from linearity of least squares on identical lags; ratio drift and the exponent difference are therefore linked diagnostics. Figures 3.6 and 3.7 give the quantile curves and their ratio for each discipline, and Figure 3.8 collects the fitted quantile exponents.

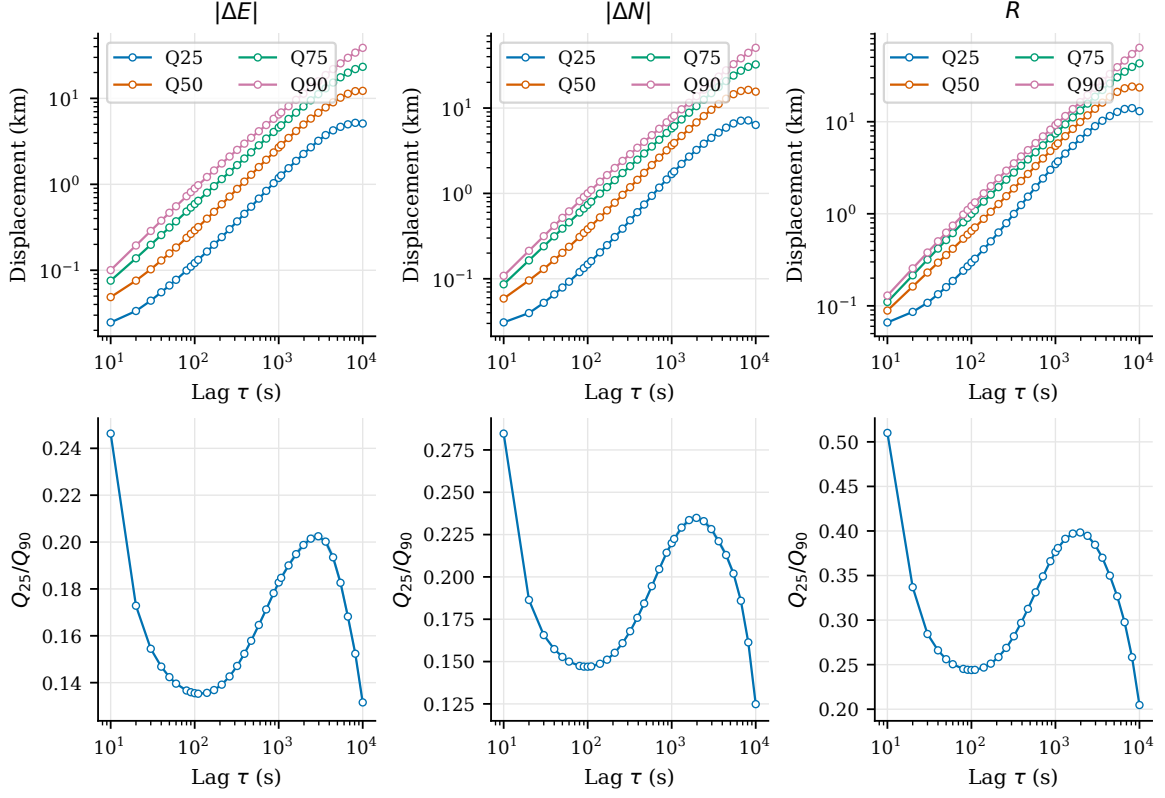


Figure 3.6. Paraglider equal-flight displacement quantiles and their relative spread, on the fixed main cohort. Every panel uses the empirical measure in Eq. (3.12). Shading gives pointwise 90% cluster-bootstrap intervals.

For paragliders, the radial ratio falls from 0.510 at 10 s to 0.244 at 100 s, rises to 0.398 at 1970 s, then falls to 0.205 at 10000 s. Quantile-fit RMS residuals span 0.023–0.084 dex.

For hang gliders, the radial ratio falls from 0.472 at 10 s to 0.192 at 90 s, rises to 0.385 at 1970 s, then falls to 0.233 at 10000 s. Quantile-fit RMS residuals span 0.037–0.101 dex.

A positive global slope of these nonmonotonic ratios can coexist with a net endpoint decrease.

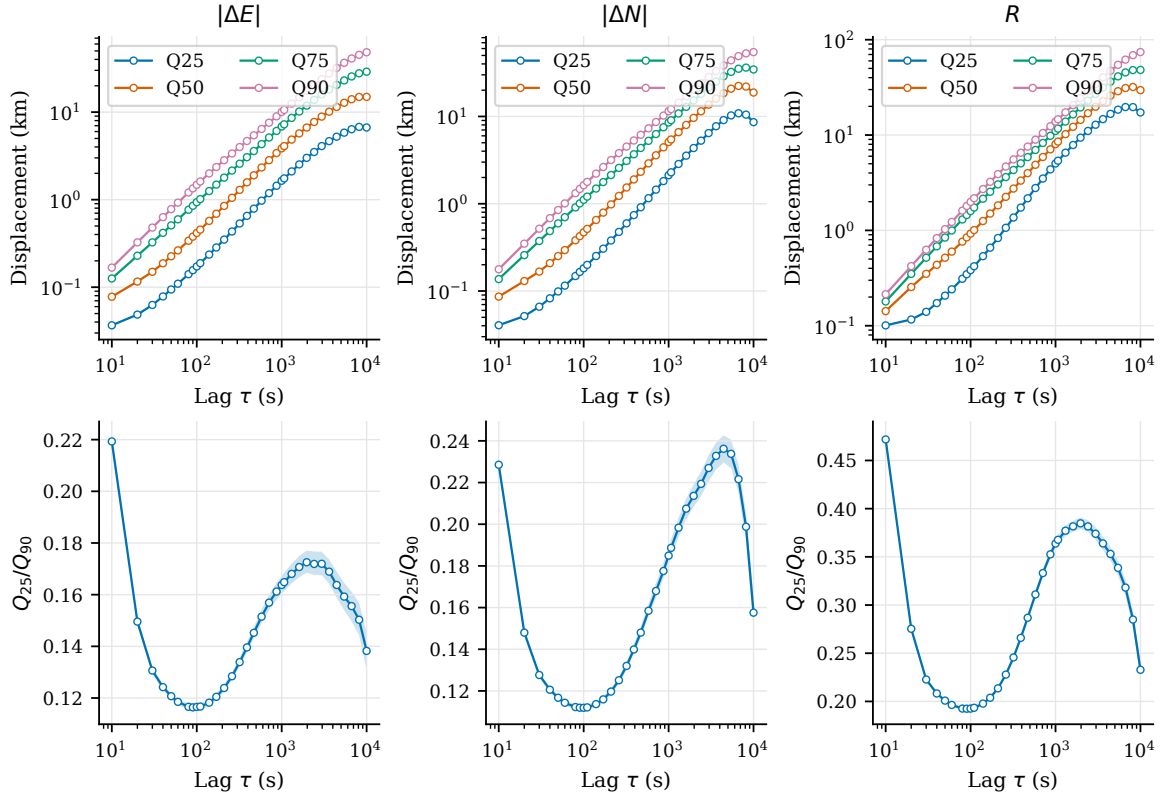


Figure 3.7. The same amplitude quantiles and ratio for hang gliders, with an independent main cohort and the same sampling convention.

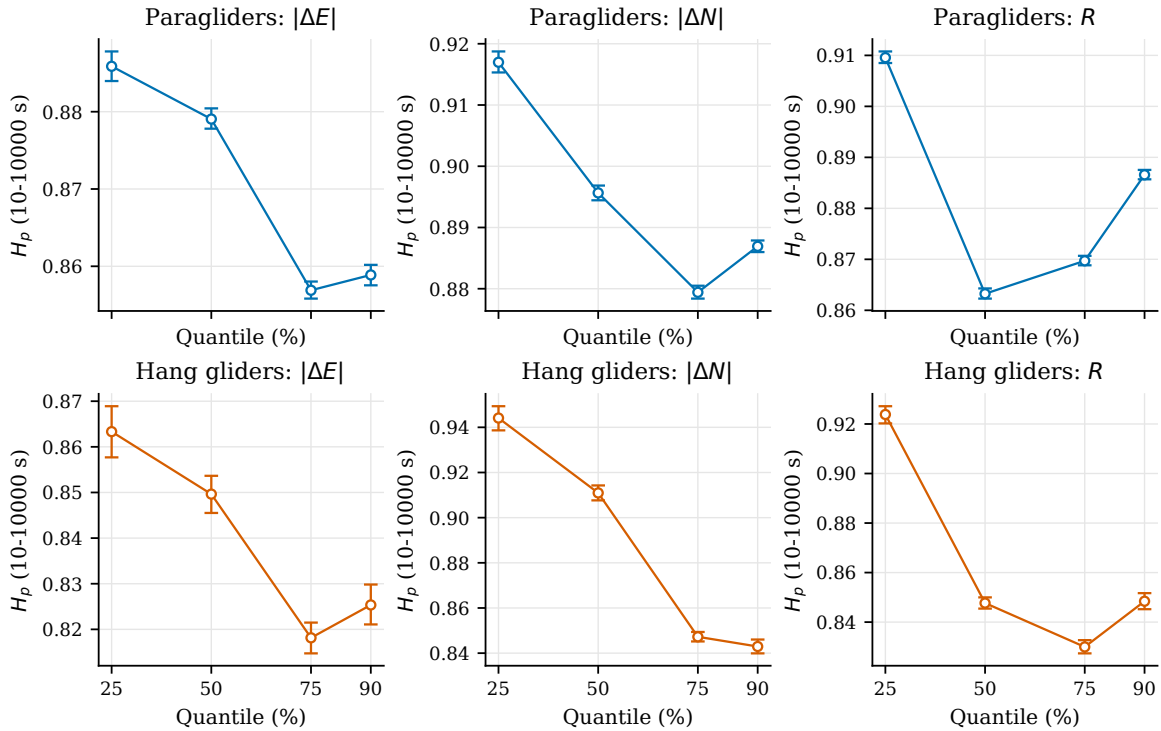


Figure 3.8. Global quantile exponents from 10–10000 s fits. Error bars are the 5th–95th percentiles of the refitted bootstrap slopes. A common scale exponent would give a horizontal sequence within each panel.

3.4.4 Moment spectrum and finite-range scaling

For each amplitude and order we calculate

$$M_{q,Y}(\tau) = \frac{1}{N} \sum_f \frac{1}{n_f(\tau)} \sum_{i \in \mathcal{I}_f(\tau)} Y_{fi}(\tau)^q, \quad \zeta_Y(q) = \mathcal{B} \left[\log_{10} \frac{M_{q,Y}}{(1\text{ m})^q} \right]. \quad (3.14)$$

We use $q \in \{0.25, 0.5, 0.75, 1, 1.5, 2, 2.5, 3, 3.5, 4\}$. Each point of the spectrum is obtained by first computing a moment at all lags, then fitting its log–log slope. The reference $\zeta = qH$ is fitted through the origin with equal weight for the ten orders. Higher orders emphasise large displacements. The radial spectrum contains the MSD, with the identity

$$M_{2,R} = M_{2,|\Delta E|} + M_{2,|\Delta N|} = M_2 \quad (3.15)$$

verified at every lag and in every bootstrap replicate. Table 3.8 reports the main-cohort MSD exponents, Figure 3.9 compares their local and global values, and Figure 3.10 gives the moment spectra. Table 3.9 collects the paired quantile and moment contrasts that test $\zeta = qH$.

Discipline	Fit range (s)	H [5–95%]	RMS (dex)
Paragliders	10–10000	0.882 [0.881, 0.883]	0.046
Paragliders	10–100	0.925 [0.924, 0.925]	0.007
Paragliders	100–1000	0.898 [0.897, 0.898]	0.006
Paragliders	1000–10000	0.804 [0.802, 0.807]	0.026
Hang gliders	10–10000	0.852 [0.849, 0.855]	0.072
Hang gliders	10–100	0.915 [0.914, 0.916]	0.006
Hang gliders	100–1000	0.873 [0.871, 0.874]	0.007
Hang gliders	1000–10000	0.728 [0.720, 0.737]	0.051

Table 3.8. Main-cohort MSD exponents: a single global fit and three common decade intervals. RMS residuals are measured in $\log_{10} M_2$. Each fit retains the same flight and segment population.

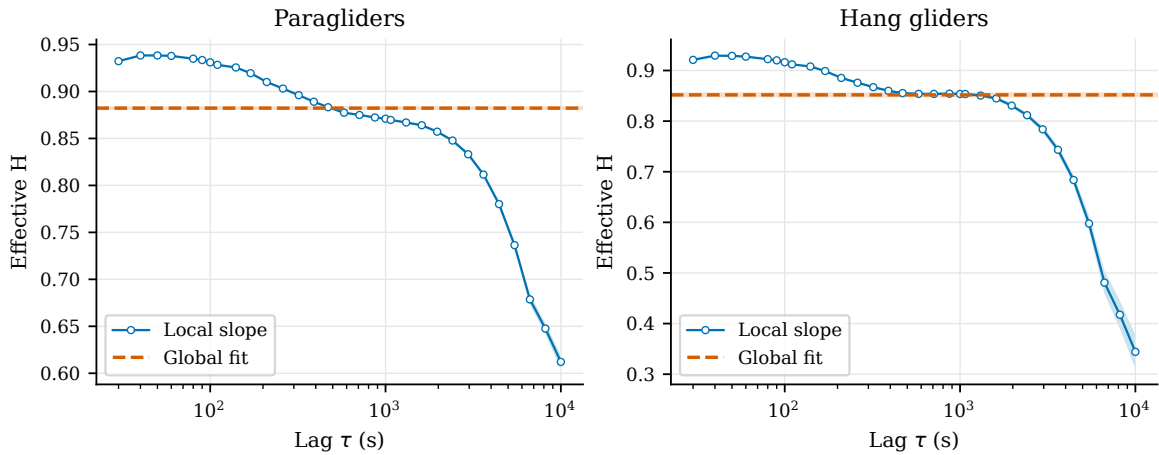


Figure 3.9. Local and global effective MSD exponents on the fixed main cohort. Bands are pointwise 90% bootstrap intervals. The global fit compresses the full curve into one number; the local slopes expose its scale dependence.

For paragliders, the radial contrast $\zeta(4)/4 - \zeta(0.25)/0.25$ is 0.0095 [0.0086, 0.0105]. Moment-fit RMS residuals span 0.010–0.079 dex across all amplitudes and orders.

For hang gliders, the radial contrast $\zeta(4)/4 - \zeta(0.25)/0.25$ is -0.0085 [-0.0112, -0.0057]. Moment-fit RMS residuals span 0.011–0.126 dex across all amplitudes and orders.

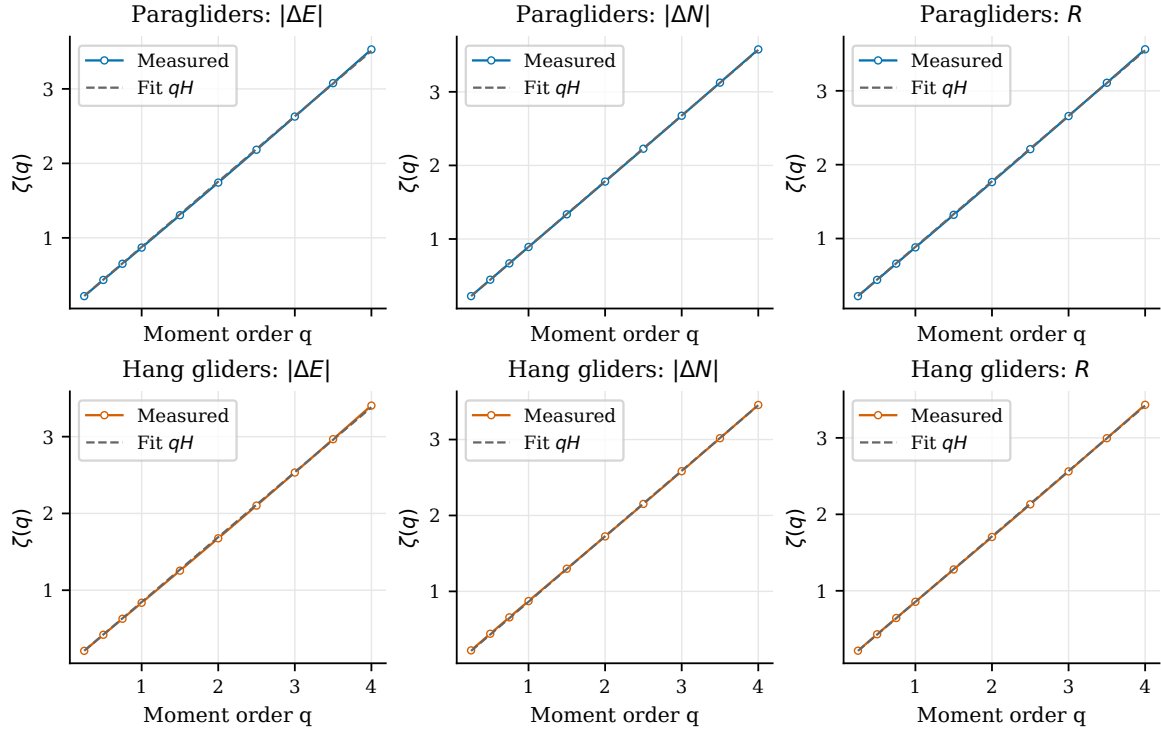


Figure 3.10. Moment spectra for the three displacement amplitudes. Each point is the slope of a 10–10000 s log–log moment fit; shading gives its pointwise 90% interval. Dashed lines are constrained linear fits $\zeta = qH$.

Discipline	Amplitude	$H_{25} - H_{90}$	$\zeta(4)/4 - \zeta(0.25)/0.25$
Paragliders	$ \Delta E $	0.027 [0.025, 0.029]	0.0113 [0.0097, 0.0128]
Paragliders	$ \Delta N $	0.030 [0.028, 0.032]	-0.0004 [-0.0016, 0.0009]
Paragliders	R	0.023 [0.022, 0.024]	0.0095 [0.0086, 0.0105]
Hang gliders	$ \Delta E $	0.038 [0.032, 0.044]	0.0095 [0.0040, 0.0150]
Hang gliders	$ \Delta N $	0.101 [0.096, 0.106]	-0.0269 [-0.0305, -0.0232]
Hang gliders	R	0.075 [0.071, 0.079]	-0.0085 [-0.0112, -0.0057]

Table 3.9. Paired global-slope contrasts, with 5–95% bootstrap intervals. A common scale exponent predicts zero for both the quantile and moment contrasts.

3.4.5 Excess kurtosis of the signed components

Departures from Gaussian shape are assessed on the signed components: taking an absolute value or a radius changes even a Gaussian law. With the equal-flight mean $\langle A \rangle_\tau = \sum_f \sum_{i \in \mathcal{I}_f(\tau)} w_{fi}(\tau) A_{fi}$, define

$$\mu_j(\tau) = \langle \Delta x_j \rangle_\tau, \quad K_j(\tau) = \frac{\langle (\Delta x_j - \mu_j)^4 \rangle_\tau}{\langle (\Delta x_j - \mu_j)^2 \rangle_\tau^2} - 3, \quad j \in \{E, N\}. \quad (3.16)$$

Centring uses the mean of the full mixture at each lag. A Gaussian marginal has $K_j = 0$ [23]; nonzero excess measures a departure in its standardised fourth moment. Zero excess alone does not establish Gaussianity. For self-similarity, the relevant prediction is constancy with lag, regardless of the value. Figure 3.11 shows $K_j(\tau)$ for both components.

For paragliders, (K_E, K_N) changes from $(-0.68, -0.97)$ at 10 s to $(2.77, 0.94)$ at 10000 s.

For hang gliders, (K_E, K_N) changes from $(-0.60, -0.72)$ at 10 s to $(4.03, 2.24)$ at 10000 s.

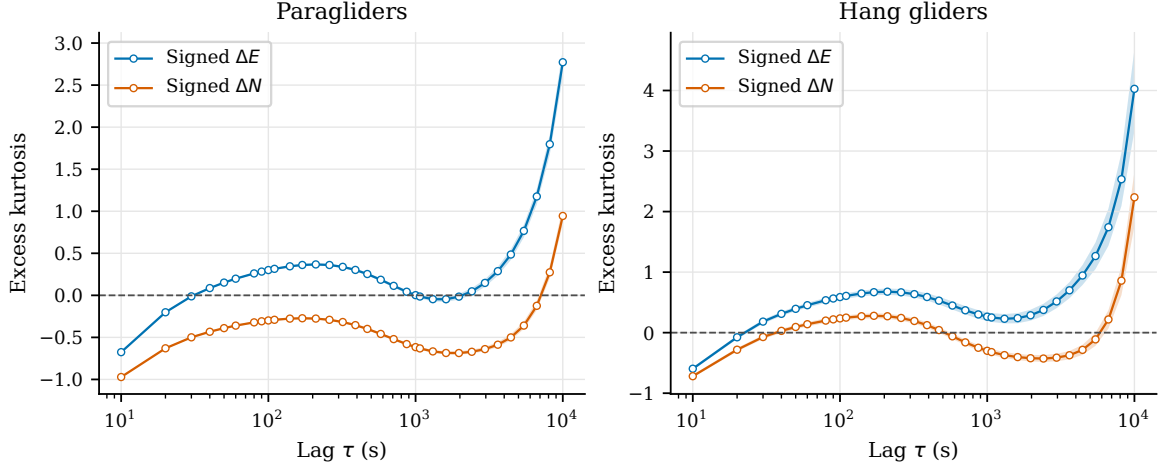


Figure 3.11. Signed-component Fisher excess kurtosis at the same lags, with the same flights, segments and weights as the quantiles and moments. The dashed line is the Gaussian marginal reference. Bands use the coupled cluster bootstrap.

Different flight means and variances can themselves produce nonzero excess in the mixture. The measured kurtosis therefore characterises the pooled displacement law.

3.4.6 Assessment of self-similarity

The quantile curves and the nearly straight moment spectra can be reconciled through

$$M_{q,Y}(\tau) = \int_0^1 Q_{Y,p}(\tau)^q dp. \quad (3.17)$$

Each moment averages over probability levels; the order q controls the emphasis on larger amplitudes. Fitting its slope across three decades further averages over crossovers. Changes in different parts of the distribution can consequently leave the fitted spectrum nearly linear. Moreover, Table 3.9 resolves small departures from $\zeta = qH$; the spectrum is not exactly linear under the adopted bootstrap.

For these data, the lag-dependent quantile ratios provide the more direct evidence against a fixed distributional shape. Their large, nonmonotonic changes are visible without fitting a power law, and their estimation does not require finite high-order moments. Unequal fitted quantile exponents support this reading, but the curved quantile trajectories limit their interpretation as constant exponents. The change in kurtosis supplies complementary evidence. Visual linearity of ten fitted moment slopes is therefore insufficient to outweigh the observed shape changes.

Over 10–10000 s, a single self-similar model with stationary increments does not adequately describe the pooled displacement law. Approximate scaling over narrower intervals remains possible. Fixed flights and segments remove changing membership, and the contrast below measures how much of the remaining structure follows from where within a segment the origins fall. The present analysis does not determine multi-time laws. It therefore does not exclude every nonstationary self-similar process, nor establish an asymptotic multifractal law.

Does the increment law depend on the origin? Equation (3.11) reaches the lag distributions through stationary increments, and the estimators above pool every admissible origin inside a selected segment. That pooling estimates one population law only if the law of $\mathbf{X}(t+\tau) - \mathbf{X}(t)$ does not depend on t . We test this directly. Within each selected segment the admissible origins at lag τ are split into three contiguous blocks of equal size, and the same equal-flight amplitude statistics are measured separately in each. The early and late blocks receive identical origin counts in every segment, so each cohort flight contributes to both and the contrast is paired inside the site-day draws of Sec. 3.1.2. Under stationary increments the late-to-early ratio equals one at every lag. Pooling the three blocks returns the equal-flight moments of Sec. 3.4 exactly, which fixes the diagnostic to the published measurement rather than to a separate estimator.

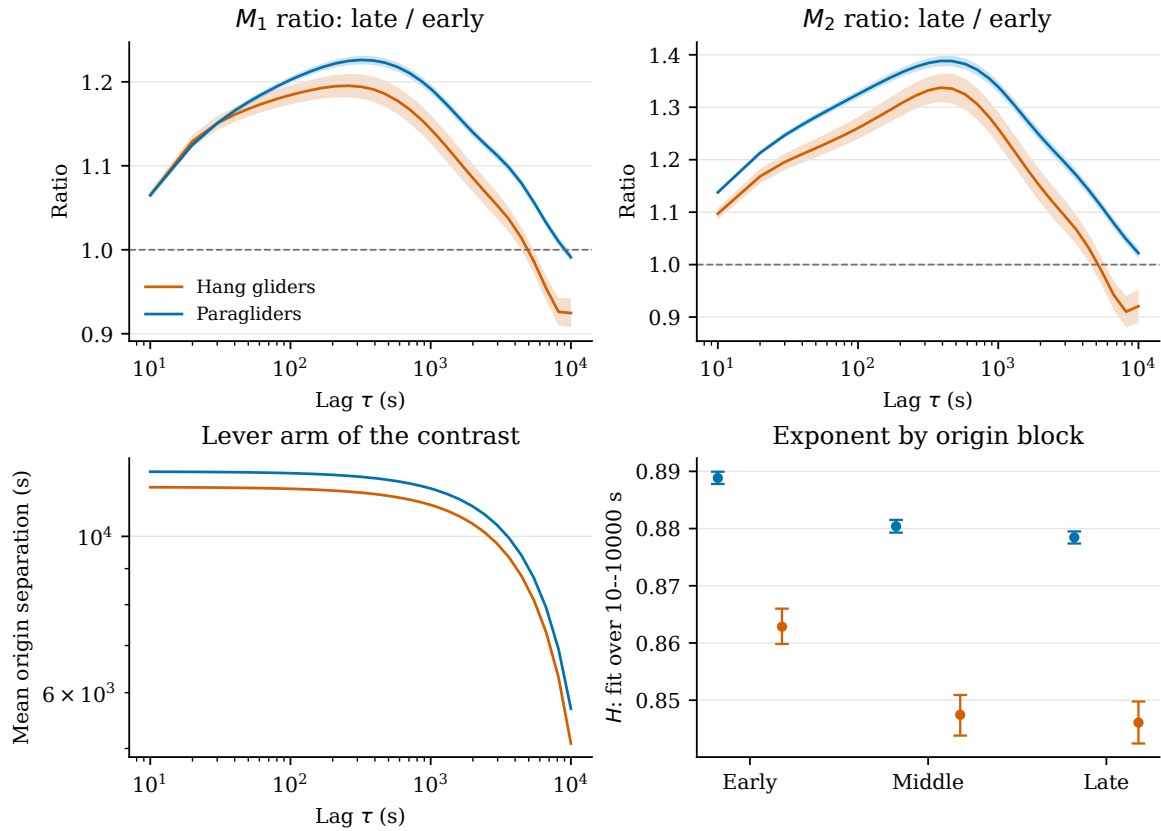


Figure 3.12. Origin dependence of the increment law in the main cohort. Top: the ratio of the late-block to the early-block radial moments, with 95% paired bootstrap bands and the stationary-increment value as a dashed line. Bottom left: the mean separation between the two blocks of origins, which shrinks as the lag consumes the segment. Bottom right: H fitted in each block over 10–10000 s. Bands and bars are pointwise.

For paragliders, the late-origin second moment differs from the early-origin one at 33 of 33 lags. The largest departure is a ratio of 1.388 (1.378–1.398) at $\tau = 390$ s, where the two blocks are separated by 12,100 s on average. Fitting H over 10–10000 s in each block gives a late-minus-early difference of -0.0104 , with a 95% interval from -0.0110 to -0.0098 .

For hang gliders, the late-origin second moment differs from the early-origin one at 32 of 33 lags. The largest departure is a ratio of 1.337 (1.310–1.364) at $\tau = 390$ s, where the two blocks are separated by 11,489 s on average. Fitting H over 10–10000 s in each block gives a late-minus-early difference of -0.0168 , with a 95% interval from -0.0186 to -0.0150 .

The ratio departs from one over the whole lag range in both disciplines, so the increments

of these segments are not stationary. Displacements measured from late origins exceed those from early origins across most of the range, with the largest excess near $\tau \approx 400$ s, and the excess closes at the longest lags. The two blocks are separated by about 12000 s at small lags and by roughly half that at $\tau = 10000$ s, because the admissible origins shrink as the lag consumes the segment; the contrast at the longest lags therefore rests on a shorter lever arm and is not comparable in strength to the rest of the curve.

Two consequences follow for this chapter. First, the amplitude of the displacement law is a property of the flight phase from which origins are drawn, so the pooled one-lag laws of Sec. 3.4 describe a mixture over flight phases rather than one stationary increment law. Second, the fitted exponent is far less sensitive to this mixture than the amplitude: the block-to-block change in H is resolved but small against the amplitude ratios in the same figure. Reading H as a lag exponent of the pooled mixture survives the test; reading it as the self-similarity exponent of a stationary-increment position process does not.

The sign of the effect does not match waiting-time ageing, which suppresses late displacements in a subordinated walk. Launch and initial circling confine the early part of a flight, while route-following occupies its later part, and this ordering alone produces the observed direction. The diagnostic separates origin position from lag; it does not identify the mechanism, and the flight-phase, wind and pilot explanations are not distinguished here.

Bibliography

- [1] Gautam Reddy, Antonio Celani, Terrence J. Sejnowski, and Massimo Vergassola. Learning to soar in turbulent environments. *Proceedings of the National Academy of Sciences*, 113(33):E4877–E4884, 2016.
- [2] Jérémie Vilpelle, Alexandre Darmon, and Michael Benzaquen. From random walks to thermal rides: Universal anomalous transport in soaring flights. arXiv:2601.01293v1 [cond-mat.stat-mech], 2026. Version 1, 3 January 2026.
- [3] Ralf Metzler and Joseph Klafter. The random walk’s guide to anomalous diffusion: a fractional dynamics approach. *Physics Reports*, 339(1):1–77, 2000.
- [4] V. Zaburdaev, S. Denisov, and J. Klafter. Lévy walks. *Reviews of Modern Physics*, 87(2):483–530, 2015.
- [5] Vincent Tejedor and Ralf Metzler. Anomalous diffusion in correlated continuous time random walks. *J. Phys. A: Math. Theor.*, 43(8):082002, 2010.
- [6] Johannes H P Schulz, Aleksei V Chechkin, and Ralf Metzler. Correlated continuous time random walks: combining scale-invariance with long-range memory for spatial and temporal dynamics. *J. Phys. A: Math. Theor.*, 46(47):475001, 2013.
- [7] Fédération Aéronautique Internationale, International Gliding Commission. Technical specification for IGC-approved GNSS flight recorders. Second Edition, with Amendment 8, 2023. Defines the IGC flight-data file format. https://www.fai.org/sites/default/files/igc_fr_specification_with_al8_2023-2-1_0.pdf.
- [8] CEN/TC 136/WG 6. Paragliding equipment — paragliders — part 2: Requirements and test methods for classifying flight safety characteristics. Technical Report Draft of FprEN 926-2, document N 0054, submitted to formal vote, European Committee for Standardization, 2012. Final draft of the standard published as EN 926-2:2013. <http://www.xcmag.com/wp-content/uploads/2012/08/DraftEN926-2.pdf>.
- [9] Fédération Aéronautique Internationale. FAI sporting code, common section 7: Hang gliders and paragliders. 2024 edition, 2024. Defines the FAI hang-glider classes and their sub-class ceilings (§1.4.1). https://www.fai.org/sites/default/files/civl/documents/sporting_code_s7_-_common_2024_v3.pdf.
- [10] Fédération Aéronautique Internationale, CIVL. Sporting code section 7h: Civil flight recorder specification. 2024 edition, stated effective date 1 May 2024, 2024. Section 3.2.1 specifies GNSS altitude relative to the geoid. The available document retains draft annotations. https://fai.org/sites/default/files/civl/documents/sporting_code_s7_h_-_civl_flight_recorder_specification_2024.pdf.

- [11] Omar García Crespillo, Rafael Oliva González, and Stefano Caizzzone. Airborne time-correlated GNSS multipath error modeling of carrier-phase smoothed code. In *Proceedings of the 2024 International Technical Meeting of the Institute of Navigation (ION ITM)*, pages 301–306, 2024. German Aerospace Center (DLR).
- [12] scikit-learn developers. `sklearn.metrics.pairwise.haversine_distances`. scikit-learn API reference. States the great-circle formula used here. https://scikit-learn.org/stable/modules/generated/sklearn.metrics.pairwise.haversine_distances.html.
- [13] Peter J. Rousseeuw and Christophe Croux. Alternatives to the median absolute deviation. *Journal of the American Statistical Association*, 88(424):1273–1283, 1993.
- [14] Laurie Davies and Ursula Gather. The identification of multiple outliers. *Journal of the American Statistical Association*, 88(423):782–792, 1993.
- [15] National Geospatial-Intelligence Agency. Department of defense world geodetic system 1984: Its definition and relationships with local geodetic systems. Technical Report NGA.STND.0036, Version 1.0.0, National Geospatial-Intelligence Agency (NGA), 2014. <https://earth-info.nga.mil/php/download.php?file=coord-wgs84>.
- [16] J. Sanz Subirana, J. M. Juan Zornoza, and M. Hernández-Pajares. Ellipsoidal and Cartesian coordinates conversion. Navipedia, 2011. https://gssc.esa.int/navipedia/index.php/Ellipsoidal_and_Cartesian_Coordinates_Conversion.
- [17] F. N. Fritsch and J. Butland. A method for constructing local monotone piecewise cubic interpolants. *SIAM Journal on Scientific and Statistical Computing*, 5(2):300–304, 1984.
- [18] Abraham Savitzky and M. J. E. Golay. Smoothing and differentiation of data by simplified least squares procedures. *Analytical Chemistry*, 36(8):1627–1639, 1964.
- [19] César Hernández-Aguayo, Matthieu Cristelli, and Michael Benzaquen. From flight logs to atmospheric science: Paragliders as convection sensors for identifying thermal predictors. arXiv:2608.00241v2, 2026. Eq. 1 and Fig. 2 (terrain classification and map); v2, 7 Sep 2026.
- [20] V. Kapos, J. Rhind, M. Edwards, M. F. Price, and C. Ravilious. Developing a map of the world’s mountain forests. In Martin F. Price and Nathalie Butt, editors, *Forests in Sustainable Mountain Development: A State of Knowledge Report for 2000*, pages 4–9. CABI Publishing, 2000.
- [21] A. Colin Cameron and Douglas L. Miller. A practitioner’s guide to cluster-robust inference. *Journal of Human Resources*, 50(2):317–372, 2015.
- [22] Paul Embrechts and Makoto Maejima. An introduction to the theory of selfsimilar stochastic processes. Author manuscript, ETH Zürich.
- [23] NIST/SEMATECH. Measures of skewness and kurtosis. e-Handbook of Statistical Methods, Section 1.3.5.11. Accessed 18 September 2026.
- [24] Peter D. Welch. The use of fast Fourier transform for the estimation of power spectra: a method based on time averaging over short, modified periodograms. *IEEE Transactions on Audio and Electroacoustics*, 15(2):70–73, 1967.

Appendices

Appendices 2.A and 2.B supply the spectral and geodetic definitions used in Chapter 2.

Appendix 2.A

Power spectral density and Welch's method

The spectra in Chapter 2 compare fluctuations in recorded altitude. They contain both flight dynamics and measurement error. A spectral estimate does not separate those contributions without further assumptions.

2.A.1 From covariance to a spectrum

Let $x_n = z_n - \mathbb{E}[z_n]$ be a real, uniformly sampled, second-order stationary process, with sampling interval $\Delta t = 1/f_s$ and covariance $C_k = \mathbb{E}[x_n x_{n+k}]$. When the covariance is summable, its two-sided spectral density is

$$S_2(f) = \Delta t \sum_{k=-\infty}^{\infty} C_k e^{-2\pi i f k \Delta t}, \quad |f| \leq f_s/2. \quad (2.A.1)$$

For real signals S_2 is even. The one-sided density doubles the positive-frequency contribution, giving

$$\int_0^{f_s/2} S_1(f) df = C_0 = \text{Var}(x_n). \quad (2.A.2)$$

Its units are $\text{m}^2 \text{Hz}^{-1}$ for an altitude signal. The Nyquist frequency $f_s/2$ limits distinguishable sampled frequencies: physical motion above that frequency may be aliased into the observed band.

A full flight altitude series need not be stationary. Equation (2.A.2) therefore gives the stationary reference interpretation; it is not a claim that the variance of an entire climbing and descending flight equals the integral of its detrended Welch estimate. Here the spectrum is a finite-record diagnostic, computed after detrending shorter windows.

2.A.2 The estimator used here

For window j of length L , subtract its least-squares linear trend and denote the residuals by $y_{j,n}$. With a periodic Hann taper w_n , define

$$I_j(f_k) = \frac{\left| \sum_{n=0}^{L-1} w_n y_{j,n} e^{-2\pi i k n / L} \right|^2}{f_s \sum_{n=0}^{L-1} w_n^2}, \quad f_k = k f_s / L. \quad (2.A.3)$$

This is the two-sided normalisation. For a one-sided estimate, positive bins are doubled except for the Nyquist bin when L is even; the zero-frequency bin is also left unchanged. Welch's

estimate averages the resulting periodograms over K overlapping windows [24]:

$$\hat{S}_1(f_k) = \frac{1}{K} \sum_{j=1}^K I_{j,1}(f_k). \quad (2.A.4)$$

The implementation uses $L = 256$, half-window overlap, linear detrending and the arithmetic mean within each flight. The frequency spacing is f_s/L ; tapering broadens the effective resolution. Details of the eligible intervals and the separate aggregation across flights are given in [Choice of altitude channel](#).

An individual periodogram has substantial sampling variability. Under suitable stationarity and weak-dependence conditions, averaging independent windows reduces its variance approximately by K . Overlapping windows are correlated, so their effective number is smaller. Increasing L improves frequency resolution but leaves fewer windows to average; keeping L fixed also retains smoothing bias as the record grows. The mean Welch estimator is not, by itself, resistant to arbitrarily large outliers.

Linear detrending suppresses slow variation within each window; it does not remove all physical motion or isolate sensor noise. Likewise, a high-frequency plateau can be compatible with measurement noise without uniquely identifying its cause. Comparisons between GNSS and barometric spectra must account for missing-channel selection, sampling, interpolation and the response of any subsequent filter.

Appendix 2.B

Geodetic transforms: from (φ, λ, h) to ECEF to ENU

This appendix derives the two formulas that Sec. 2.7.5 uses as given: the geodetic-to-ECEF transform, Eq. (2.20), and the ECEF-to-ENU rotation matrix. It closes with the size of the tangent-plane error quoted there. Notation as in the body: WGS84 semi-major axis a , flattening f , $b = a(1 - f)$ the semi-minor axis, $e^2 = f(2 - f) = 1 - b^2/a^2$ the first eccentricity squared [15].

In the exact transform, h denotes ellipsoidal height. The GNSS channel in an arbitrary HPG record need not follow this convention (Sec. 2.7.1). For orthometric height H , the relation is $h = H + \mathcal{N}(\varphi, \lambda)$, where $\mathcal{N}$ is geoid undulation. The pipeline currently inserts the recorded GNSS value as a height proxy; it does not infer a datum or apply a geoid correction. A height error δh perturbs a horizontal displacement from the origin by approximately $|\delta h|d/R_\oplus$ on a short, level route. This is a sensitivity estimate, not an established error bound for every recorder. Moreover, a spatially varying geoid correction does not cancel exactly from altitude increments.

2.B.1 Geodetic coordinates to ECEF

The ECEF frame is Cartesian, centred at the Earth's centre O : the Z axis along the polar axis, the X axis through the intersection of the equator and the prime meridian, Y completing the right-handed triad. By symmetry, a point at longitude λ lies in the meridian plane obtained by rotating the X - Z plane by λ ; writing p for the distance from the polar axis, its ECEF coordinates are

$$X = p \cos \lambda, \quad Y = p \sin \lambda. \quad (2.B.1)$$

The problem therefore reduces to the meridian ellipse: find (p, Z) of a point at geodetic latitude φ and height h .

The surface point. The meridian section of the ellipsoid is the ellipse

$$\frac{p^2}{a^2} + \frac{Z^2}{b^2} = 1. \quad (2.B.2)$$

The geodetic latitude is defined through the surface *normal* (Figure 2.12). The gradient of the left-hand side of (2.B.2) is proportional to $(p/a^2, Z/b^2)$, so the outward normal at a surface point (p_s, Z_s) makes an angle φ with the equatorial plane given by

$$\tan \varphi = \frac{Z_s/b^2}{p_s/a^2} \implies Z_s = \frac{b^2}{a^2} p_s \tan \varphi = (1 - e^2) p_s \tan \varphi. \quad (2.B.3)$$

Substituting (2.B.3) into (2.B.2), with $b^2 = a^2(1 - e^2)$,

$$\frac{p_s^2}{a^2} \left(1 + (1 - e^2) \tan^2 \varphi \right) = 1 \implies p_s^2 = \frac{a^2 \cos^2 \varphi}{\cos^2 \varphi + (1 - e^2) \sin^2 \varphi} = \frac{a^2 \cos^2 \varphi}{1 - e^2 \sin^2 \varphi}, \quad (2.B.4)$$

that is,

$$p_s = N(\varphi) \cos \varphi, \quad N(\varphi) \equiv \frac{a}{\sqrt{1 - e^2 \sin^2 \varphi}}, \quad (2.B.5)$$

and, back through (2.B.3),

$$Z_s = (1 - e^2) N(\varphi) \sin \varphi. \quad (2.B.6)$$

The quantity $N(\varphi)$ so defined is the *prime-vertical radius of curvature*; a computation shows that, away from the poles, it is the distance along the normal from the surface point to the polar axis (the segment $Q-P$ of Figure 2.12): the normal direction in the meridian plane is $(\cos \varphi, \sin \varphi)$, and following it inward from (p_s, Z_s) a length N lands on the axis, $p_s - N \cos \varphi = 0$ by (2.B.5). At the poles the normal lies on the polar axis, so this intersection construction is degenerate; the formula for N remains defined by continuity.

Adding the height. The height h of a fix is measured along the same normal, whose unit vector in ECEF components is

$$\hat{\mathbf{u}} = (\cos \varphi \cos \lambda, \cos \varphi \sin \lambda, \sin \varphi). \quad (2.B.7)$$

Adding $h \hat{\mathbf{u}}$ to the surface point (2.B.5)–(2.B.6) in the meridian plane and restoring the longitude via (2.B.1):

$$X = (N + h) \cos \varphi \cos \lambda, \quad Y = (N + h) \cos \varphi \sin \lambda, \quad Z = (N(1 - e^2) + h) \sin \varphi, \quad (2.B.8)$$

which is Eq. (2.20). The asymmetry between the horizontal factor $(N + h)$ and the vertical one $(N(1 - e^2) + h)$ is the ellipsoid's flattening at work: on a sphere ($e = 0$) both reduce to $R + h$ and the geodetic and geocentric latitudes coincide.

How far the two latitudes differ. The geocentric latitude ψ of a point on the surface is the angle its position vector makes with the equatorial plane, so from (2.B.5)–(2.B.6), at $h = 0$,

$$\tan \psi = \frac{Z_s}{p_s} = \frac{N(1 - e^2) \sin \varphi}{N \cos \varphi} = (1 - e^2) \tan \varphi, \quad (2.B.9)$$

the classic relation: the two agree only where $e = 0$, or at the equator and the poles, where $\tan$ is 0 or infinite. To see the size of the gap, write $\psi = \varphi - \Delta$ and expand for small e^2 . Using $\tan(\varphi - \Delta) \simeq \tan \varphi - \Delta (1 + \tan^2 \varphi)$ in (2.B.9),

$$\tan \varphi - \Delta (1 + \tan^2 \varphi) \simeq (1 - e^2) \tan \varphi \implies \Delta \simeq \frac{e^2 \tan \varphi}{1 + \tan^2 \varphi} = e^2 \sin \varphi \cos \varphi, \quad (2.B.10)$$

that is

$$\varphi - \psi \simeq \frac{e^2}{2} \sin 2\varphi, \quad (2.B.11)$$

which is the expression quoted in Sec. 2.7.5. At this order it is largest at $\varphi = 45^\circ$, where it equals $e^2/2 = 3.347 \times 10^{-3}$ radian, or $0.192^\circ \approx 11.5'$. Mistaking one latitude for the other therefore displaces a fix along the meridian by $R_\oplus (e^2/2) \approx 21$ km at mid-latitudes—the ~ 20 km quoted in the body. The error vanishes at the equator and at the poles; the leading-order approximation is symmetric about 45° .

2.B.2 ECEF to local ENU

The local frame at the origin fix $(\varphi_0, \lambda_0, h_0)$ is spanned by three unit vectors: East, tangent to the parallel; North, tangent to the meridian, pointing toward increasing latitude; Up, along the ellipsoid normal (2.B.7). Away from the poles, East is the direction in which the position (2.B.1) changes under λ alone,

$$\hat{\mathbf{e}}_E = \frac{\partial \mathbf{X} / \partial \lambda}{\|\partial \mathbf{X} / \partial \lambda\|} = (-\sin \lambda_0, \cos \lambda_0, 0), \quad (2.B.12)$$

At a pole, longitude fixes a conventional orientation of the horizontal axes; the displayed vector still defines that convention although the derivative vanishes. Up is $\hat{\mathbf{u}}$ of (2.B.7) evaluated at (φ_0, λ_0) , and North completes the right-handed triad,

$$\hat{\mathbf{e}}_N = \hat{\mathbf{e}}_U \times \hat{\mathbf{e}}_E = (-\sin \varphi_0 \cos \lambda_0, -\sin \varphi_0 \sin \lambda_0, \cos \varphi_0). \quad (2.B.13)$$

The ENU components of a displacement $\Delta \mathbf{X}$ are its projections on the triad, $E = \hat{\mathbf{e}}_E \cdot \Delta \mathbf{X}$ and likewise for N and U ; stacking the three row vectors gives exactly the matrix of Sec. 2.7.5,

$$\begin{pmatrix} E \\ N \\ U \end{pmatrix} = \underbrace{\begin{pmatrix} -\sin \lambda_0 & \cos \lambda_0 & 0 \\ -\sin \varphi_0 \cos \lambda_0 & -\sin \varphi_0 \sin \lambda_0 & \cos \varphi_0 \\ \cos \varphi_0 \cos \lambda_0 & \cos \varphi_0 \sin \lambda_0 & \sin \varphi_0 \end{pmatrix}}_R \Delta \mathbf{X}, \quad (2.B.14)$$

whose rows are orthonormal by construction, so R is a rotation ($R^\top R = 1$) [16]. Two remarks used in the body follow directly. First, the rows depend on the *geodetic* latitude because Up is the ellipsoid normal — the same φ that enters Eq. (2.20), which is why no latitude conversion appears anywhere in the pipeline. Second, the matrix is evaluated once per flight, at the origin fix: the frame is a fixed tangent frame, not one that travels with the wing.

2.B.3 Size of the tangent-plane error

Two different distances must be distinguished. On a sphere, an arc of length d subtends a three-dimensional chord

$$c = 2R_\oplus \sin \frac{d}{2R_\oplus} = d - \frac{d^3}{24 R_\oplus^2} + O(d^5/R_\oplus^4). \quad (2.B.15)$$

The horizontal EN projection from a frame anchored at one endpoint is instead

$$d_{EN} = R_\oplus \sin(d/R_\oplus), \quad d - d_{EN} = \frac{d^3}{6R_\oplus^2} + O(d^5/R_\oplus^4). \quad (2.B.16)$$

This deficit is 0.033 m at 20 km, 0.51 m at 50 km, and about 33 m at 200 km. These estimates assume surface points and an origin at one endpoint. For increments between two remote points in a fixed frame, projection distortion also depends on their distance and bearing from the frame origin. It cannot be bounded from increment length alone. The *vertical* coordinate of a distant surface point is a different matter: the surface falls below the tangent plane by the sagitta $\approx d^2/(2R_\oplus)$, which reaches tens of metres already at $d = 20$ km. This is why the analysis never takes its working altitude from the plane geometry: the vertical coordinate is the adopted altitude channel itself (Sec. 2.7.5, Sec. 2.7.1), and the U component serves only the geometric construction.

2.B.4 Size of the spherical-approximation error

Sec. 2.7.2 computes inter-fix distances by the haversine formula, Eq. (2.2), which treats the Earth as a sphere of fixed radius $R_\oplus = 6,371$ km regardless of latitude or heading. This section bounds what that costs against the WGS84 ellipsoid.

Local metric. For a short step at geodetic latitude φ , the ellipsoid and the sphere give

$$ds_{\text{ell}}^2 = M(\varphi)^2 d\varphi^2 + N(\varphi)^2 \cos^2 \varphi d\lambda^2, \quad ds_{\text{sph}}^2 = R_\oplus^2 (d\varphi^2 + \cos^2 \varphi d\lambda^2). \quad (2.B.17)$$

Define $d\sigma = ds_{\text{sph}}/R_\oplus$ and the bearing α in this spherical metric by $d\varphi = d\sigma \cos \alpha$ and $\cos \varphi d\lambda = d\sigma \sin \alpha$. Then

$$R_{\text{eff}}(\varphi, \alpha) \equiv \frac{ds_{\text{ell}}}{d\sigma} = \sqrt{M(\varphi)^2 \cos^2 \alpha + N(\varphi)^2 \sin^2 \alpha}, \quad (2.B.18)$$

interpolating between the meridian value $M(\varphi)$ ($\alpha = 0$, north–south) and the prime-vertical value $N(\varphi)$ of Eq. (2.B.5) ($\alpha = 90^\circ$, east–west) [15]. $N(\varphi)$ is already in hand; $M(\varphi)$ follows the same route, differentiating the surface point $(p_s(\varphi), Z_s(\varphi))$ of Eqs. (2.B.5)–(2.B.6) with respect to φ and using $ds = M d\varphi$ along the meridian:

$$\frac{dp_s}{d\varphi} = -\frac{N(\varphi)(1-e^2)\sin \varphi}{1-e^2\sin^2 \varphi}, \quad \frac{dZ_s}{d\varphi} = \frac{N(\varphi)(1-e^2)\cos \varphi}{1-e^2\sin^2 \varphi}, \quad (2.B.19)$$

so that $M(\varphi)^2 = (dp_s/d\varphi)^2 + (dZ_s/d\varphi)^2$ collapses, via $\sin^2 \varphi + \cos^2 \varphi = 1$, to

$$M(\varphi) = \frac{N(\varphi)(1-e^2)}{1-e^2\sin^2 \varphi} = \frac{a(1-e^2)}{(1-e^2\sin^2 \varphi)^{3/2}}. \quad (2.B.20)$$

Comparison to the sphere. The reference radius used in Eq. (2.2), $R_\oplus = 6,371$ km, agrees with the ellipsoid’s mean radius $R_m = (2a + b)/3 = a(1 - f/3) = 6,371.009$ km to within 10 m. The fractional difference, expressed as ellipsoidal length relative to spherical length, is approximately $[R_{\text{eff}}(\varphi, \alpha) - R_m]/R_m$, bounded between its values at $\alpha = 0$ and $\alpha = 90^\circ$. Expanding Eqs. (2.B.5) and (2.B.20) to first order in the flattening f (using $e^2 \simeq 2f$),

$$\frac{N(\varphi)}{R_m} - 1 \simeq f \left(\sin^2 \varphi + \frac{1}{3} \right), \quad \frac{M(\varphi)}{R_m} - 1 \simeq f \left(3 \sin^2 \varphi - \frac{5}{3} \right), \quad (2.B.21)$$

both of order the flattening $f = 1/298.257223563 \approx 0.335\%$ at every latitude: from $-5f/3 \approx -0.56\%$ at the equator ($\varphi = 0$, meridian direction) to $+4f/3 \approx 0.45\%$ at the poles ($\varphi = 90^\circ$, where $M = N$ by symmetry). Thus 0.3% is a regional estimate, not a global bound.

At the latitudes flown. Most of the archive launches from the Alps box $\varphi \in [43.8^\circ, 46.6^\circ]$ N used to classify take-off sites (Sec. 2.9, Fig. 2.23). Evaluating Eq. (2.B.21) across that range moves the bound by under 0.02 percentage points, so its midpoint $\varphi = 45^\circ$ is representative:

$$\frac{N(45^\circ)}{R_m} - 1 \simeq \frac{5f}{6} \approx 0.28\%, \quad \frac{M(45^\circ)}{R_m} - 1 \simeq -\frac{f}{6} \approx -0.06\% \quad (2.B.22)$$

(numerically, $N(45^\circ) = 6,388.84$ km and $M(45^\circ) = 6,367.38$ km against $R_m = 6,371.01$ km). At 45° , the sphere underestimates an east–west ellipsoidal step by about 0.28% and overestimates a north–south step by about 0.06%. Across the latitude interval above the absolute discrepancy remains below 0.3%, or roughly 3 cm on a 10 m step. The bound only opens up toward the 0.45–0.56% extremes for the small remainder of flights nearer the equator or at higher latitude (Sec. 2.9, “elsewhere”), corresponding to at most roughly 6 cm on a 10 m step. The uncertainty of a GNSS increment must be assessed separately, since errors at successive fixes can be correlated.